When Is Label-Free Adaptation Knowable?

The Knowability Boundary of Label-Free Adaptation — Manuscript Edition

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Long-form companion to the working paper "When Is Label-Free Adaptation Knowable?" (K-Bound, v0.4). Every theorem, number, table, and figure in this manuscript reproduces from the public repository at `https://github.com/Pratikn03/AutoML_Flagship_V8`*. Nothing has been added beyond the exposition; nothing has been rounded, selected, or otherwise adjusted from the verified artifact manifests.*

Abstract

Test-time adaptation (TTA) improves models on unlabeled target data, yet the very unlabeled objective that recovers accuracy under one distribution shift can silently degrade it under another — and, without labels, a deployed system cannot tell which is happening. Existing methods propose increasingly robust *adaptation mechanisms*; we instead study a prior question: *can a system decide, without labels, whether adaptation should happen at all?*

We formalize adaptation benefit as the excess risk between an adapted candidate and a frozen baseline conditioned on label-free observable evidence, and separate test-time regimes into *knowably helpful*, *knowably harmful*, and *unknowable.* We prove (Theorem 4.1, with an explicit witness) that when two target worlds induce the same observable evidence but opposite benefit, no label-free rule can certify the correct decision in both, so the worst-case-optimal action on such evidence is to abstain. Under an observable risk-alignment assumption we give a finite-sample adapt/freeze/abstain certificate (Theorem 5.1) that controls the false-adapt and false-freeze rates at a chosen level, and we identify a provable positive regime (Theorem 5.3, covariate shift). A quantitative strengthening via the Le Cam two-point lower bound (Theorem 4.1) makes the impossibility result a tight minimax statement: the optimal committal error equals $1 - \mathrm{TV}$ of the evidence distributions, which is 1 in the exact witness and decays to zero only as the evidence separates.

We instantiate the framework (KGA) on a 123-task anomaly-routing benchmark and a controlled mixed-regime study. KGA abstains where the true benefit is near zero, refuses a fusion path that hurts on 80% of tasks (cutting regret-to-oracle $\approx 11\times$ versus always-adapt), and in the mixed regime beats an always-freeze policy by ≈ 0.10 AUROC. In *online non-stationary* CIFAR-10 test-time adaptation, across a helpful-dominated and a harm-dominated stream, no fixed policy is robust: always-adapt collapses in the harm regime (0.339) and always-freeze forgoes gains in the helpful regime — while KGA is near-best in both, giving the highest mean accuracy across regimes (0.490 vs. 0.456 / 0.444). On a *pre-registered* CIFAR-10-C stress grid (severity $\times$ batch $\times$ composition $\times$ aggressiveness; harmful base rate 0.16–0.34), KGA *beats both* trivial policies for Tent and EATA — regret-to-oracle 0.0016 versus always-adapt 0.009 and always-freeze 0.12, at a 0% false-adapt rate — and ties the collapse-resistant SAR.

We are explicit about scope: where harmful adaptation is mild, always-adapt is already strong, so the certificate's distinctive value is bounded by how often *catastrophic, detectable* harm occurs

— itself the quantity the theory characterizes. The surviving open piece is the label-free *bracketing* of the ordinal accuracy comparison on the observable disagreement region in the multiclass and regression cases (Conjecture 5.1); every other element of the characterization is proved.

How to Read This Manuscript

This monograph presents the K-Bound research program at a depth that the 20-page working paper cannot sustain. Theorems and experiments are stated faithfully from the paper; the additional length is entirely expository — motivation, worked examples, proof commentary, and explicit connections between the abstract theory and its empirical realizations. A reader who has the paper in hand can use the chapter headings as an index; a reader coming to K-Bound fresh should read Chapters 1 and 2 first to orient on the problem and its neighbours, then follow Chapters 3 through 5 for the theory, and Chapters 6 through 8 for the instantiation and experiments.

Chapter 1: Introduction. Sets up the TTA double-edge problem and the central question of knowability. Introduces the adapt/freeze/abstain trichotomy, previews all five theoretical results and the headline experimental numbers, and gives an honest statement of scope.

Chapter 2: Background and Related Work. A thorough review of the five lines of work adjacent to K-Bound: TTA mechanisms, TTA failure and collapse studies, guarded/protected adaptation, label-free accuracy estimation, unsupervised risk and identifiability, selective prediction and abstention, conformal and anytime-valid inference, and covariate-shift theory. Closes with a positioning table and a precise statement of the gap K-Bound fills.

Chapter 3: Problem Formulation. The formal setup: source and target distributions, the adaptation benefit $\Delta = R_T(f_0) - R_T(f_a)$, observable evidence Z, the three regimes, and the key definitions of risk alignment and knowability. Includes the "label-free" clarification box and the knowability phase diagram.

Chapter 4: Impossibility and the Unknowable Regime. The first two theoretical results. Theorem 4.1 (non-identifiability with explicit witness) and its Le Cam quantitative strengthening establish the unknowable regime as fundamental. Theorem 4.3 gives the Bayes-optimal gate and the plug-in regret decomposition, which justifies abstaining in the low-margin band.

Chapter 5: Certificates and Positive Regimes. The remaining three theoretical results. Theorem 5.1 is the finite-sample adapt/freeze/abstain certificate with a controlled false-adapt rate; Corollaries give forced abstention and sample complexity. Theorem 5.3 proves a knowable positive regime under covariate shift. Theorems 5.4, 5.5, and 5.6 reduce sign-of-difference identifiability to an ordinal accuracy comparison on the observable disagreement region for binary, multiclass, and regression losses. Conjecture 5.1 states the remaining open piece.

Chapter 6: The KGA Algorithm. The KGA (Knowability-Guided Adaptation) wrapper: evidence extraction, benefit estimation, conformal radius, and the three-way decision. Assumptions, computational cost, and the relationship to any underlying TTA candidate.

Chapter 7: Experiments I — Anomaly Routing. The 123-task clean suite, the harmful-regime refusal experiment, the controlled mixed-regime study, and the non-identifiability witness (both empirical and exact). Evidence ablations and the α-sweep safety/coverage tradeoff.

Chapter 8: Experiments II — Online TTA and CIFAR-10-C. The online non-stationary CIFAR-10 cross-regime test, the per-corruption CIFAR-10-C helpful-dominated protocol, and the pre-registered CIFAR-10-C stress grid. Breadth results on 62 additional harmful-dominated label-free tasks.

Chapter 9: The ELARA/RGA Instantiation. Reliability-gated multimodal fusion as the worked, code-validated special case: the ten-result ELARA theorem stack (T1–T9, GDR) and its mapping to the general K-Bound theorems.

Chapter 10: Discussion, Limitations, and Conclusion. When to deploy K-Bound; honest scope; limitations; and the research frontier (ImageNet-C / ViT scale, natural-shift benchmarks, the open conjecture).

Appendix A: Reproducibility. One-command reproduction; data provenance and checksums; environment and determinism notes.

Appendix B: Extended Tables. The full 65-cell CIFAR-10-C per-corruption-by-severity breakdown and auxiliary result tables referenced from the main chapters.

Appendix C: Notation Reference. A single-page glossary of all symbols used in the monograph.

Readers interested primarily in the *theory* can read Chapters 3–5 as a self-contained unit. Readers interested primarily in the *practical algorithm* can read Chapter 6 with only the informal summaries of Theorems 4.1 and 5.1. Both paths converge at Chapter 10, which synthesizes the implications for deployment.

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Chapter 1

Introduction

1.1 The Deployment Problem

A model trained on one distribution is deployed into a world that has moved. The training images came from one hospital's scanner; the deployed scanner is newer, slightly sharper, with different noise characteristics. The training network saw balanced classes; the test stream is skewed toward one pathology. The training corpus covered one domain; the production queries drift slowly toward a neighbouring domain as user behaviour changes. This pattern — a mismatch between the distribution under which a model was optimized and the distribution under which it operates — is the central operational reality of deployed machine learning.

Test-time adaptation (TTA) is the field's answer. Rather than retraining from scratch whenever the deployment distribution shifts, TTA updates the model on the unlabeled target data that arrives at inference time. The appeal is immediate: labels are expensive or unavailable in deployment, yet the raw inputs carry information about how the distribution has moved. Entropy minimization over batch-normalization parameters [52], self-supervised auxiliary objectives that continue training at test time [47], source-free adaptation by information maximization, and importance-weighted empirical risk minimization [45] are all instances of the same basic idea: use the unlabeled stream to close the performance gap.

1.2 The Double-Edged Nature of Test-Time Adaptation

Adaptation is, however, double-edged in a precise sense. The same unlabeled objective that recovers accuracy under one kind of distribution shift can degrade it under another — and, critically, without labels the deployed system cannot in general determine which situation it is in.

Consider entropy minimization. Under mild covariate shift — the scanner statistics change, but the decision boundary between pathologies does not — minimizing the entropy of predictions sharpens the boundary toward where the new data lives. The adaptation is helpful. But now suppose the distribution shift is a label shift: the prevalence of one class has collapsed to near zero in the target stream. A model that minimizes entropy on this stream will be rewarded for making all-or-nearly-all predictions of the dominant class; it converges to a degenerate solution. The adaptation is harmful, and in the worst case the model collapses entirely. The recent literature on TTA failure modes [58, 38, 37] documents exactly these collapse phenomena: Tent's continual entropy minimization catastrophically fails on adversarial or highly non-stationary streams; EATA's Fisher penalty slows but does not eliminate accumulation of damage; SAR's sharpness-aware resets abort incipient collapses but cannot prevent them in every regime.

The field's response has been to propose sturdier mechanisms. If entropy minimization is fragile, add sample selection to filter high-entropy inputs [36]. If selection is not enough, add sharpness-aware updates that are less likely to collapse [37]. If batch normalization updates are the failure mode, use a fully parameter-free approach [6]. If a single method is fragile, reset after detecting instability [37] or continually re-anchor to the source [53].

Each of these improvements is genuine within its scope. But they all share a common architecture: they commit to adapting, and they try to make that commitment safer. None of them poses the prior question: *given the available evidence, should the system adapt at all?*

1.3 The Central Question

This monograph studies that prior question.

> *Can observable test-time evidence separate helpful, harmful, and unknowable regimes of label-free adaptation, yielding a computable rule for when a system should adapt, freeze, or abstain?*

The question is not rhetorical. It has a substantive, partly negative answer that we make precise. In some regimes — covariate shift, certain kinds of label shift, situations where the frozen and adapted models disagree on inputs whose label orientation is detectable from label-free signals — the answer is yes. In others, the same observable evidence is compatible with both helpful and harmful adaptation, and no label-free rule can certify the correct choice in both worlds. The boundary between these cases is what we call the *knowability boundary* of label-free adaptation, and the central mathematical object is a trichotomy: at any given evidence value, adaptation is either *knowably helpful*, *knowably harmful*, or *unknowable*.

Why does this matter for deployment? Consider a safety-critical setting: an autonomous vehicle's perception stack or a medical imaging pipeline. In such settings, the cost of silent degradation is high and the cost of abstaining — falling back to the frozen model and flagging for human review — is manageable. A system that can identify when it is in an unknowable regime and refuse to adapt is strictly safer than one that always adapts or always freezes, even if its average-case performance is identical. The value of the knowability boundary is not primarily about improving mean accuracy; it is about converting the adapt/freeze decision from an irreducible coin flip into a certified, controlled action.

The same logic applies to model monitoring. A production team that can determine, from unlabeled deployment data alone, whether a proposed update will help or hurt does not need ground-truth labels to make a safe deployment decision. The knowability boundary tells them exactly when that label-free determination is possible and when it is not.

1.4 The Trichotomy and Its Formalization

We formalize the trichotomy as follows. Fix a loss ℓ, a frozen model f_0, and an adapted/gated candidate f_a. The *adaptation benefit* is

$$\Delta \;=\; R_T(f_0) - R_T(f_a),$$

where $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)]$ is the target risk. A positive Δ means adapting helps; a negative Δ means it hurts. *Observable evidence* $Z = \phi(X_{1:m}, f_0, f_a)$ is any statistic computable without target labels: prediction entropy, softmax confidence, score-distribution drift, detector disagreement, predicted-class balance, update norm, or density-ratio diagnostics. The conditional benefit $\Delta(z) = \mathbb{E}[\ell(f_0(X), Y) - \ell(f_a(X), Y) \mid Z = z]$ is the benefit conditioned on evidence $Z = z$.

The regime at evidence value z is:

Knowably helpful if Z certifies $\Delta(z) > 0$; the correct action is ADAPT.

Knowably harmful if Z certifies $\Delta(z) < 0$; the correct action is FREEZE.

Unknowable if the same z is compatible with both $\Delta(z) > 0$ and $\Delta(z) < 0$; the worst-case-optimal action is ABSTAIN.

The key quantity is the *sign* of $\Delta(z)$, not its magnitude. Estimating the sign requires strictly less information than estimating the risk $R_T(f)$ itself, which means the knowability boundary lies strictly outside the identifiability boundary for risk estimation — but it is not everywhere: there exist situations where even the sign is unrecoverable from label-free evidence.

1.5 Contributions

This work makes six contributions, which we enumerate in detail.

Contribution 1: Formalization of the knowability trichotomy. We introduce the adapt/freeze/abstain trichotomy as the natural decision space for label-free TTA, formalize the notion of risk alignment for observable evidence, and define the three regimes precisely (Chapter 3). The formalization connects the TTA literature to the identifiability literature: TTA methods have focused on designing the adaptation mechanism f_a; we instead ask when the sign of $R_T(f_0) - R_T(f_a)$ is recoverable from label-free observations.

Contribution 2: Impossibility theorem with explicit witness (Theorem 4.1). We prove that for any (possibly randomized) label-free decision rule $g(Z)$, there exist two target worlds P_T^1 and P_T^2 with identical observable-evidence laws and opposite-sign benefits. Since the rule cannot distinguish the worlds, its actions are identically distributed in both, and the worst-case-optimal action is to abstain. The construction is non-vacuous: we exhibit it explicitly as $X \sim \mathcal{N}(0, 1)$ under both worlds, $f_0(x) = \mathbf{1}[x > 0]$, $f_a(x) = \mathbf{1}[x < 0]$, and labels flipped between the worlds. A Le Cam two-point quantitative strengthening (Chapter 4) shows that the optimal committal error equals $1 - \mathrm{TV}$ of the evidence laws, which is 1 in the witness and decays to zero as the worlds become separable. Numerically, the KGA validator confirms 100% abstention on the witness and exact tracking of the Le Cam floor across the total-variation sweep.

Contribution 3: Finite-sample adapt/freeze/abstain certificate with false-adapt control (Theorem 5.1). Under a label-free estimator $\widehat{\Delta}(z)$ with a calibrated radius $\varepsilon(z)$ satisfying $\mathbb{P}[|\widehat{\Delta}(z) - \Delta(z)| \leq \varepsilon(z)] \geq 1 - \alpha$, the rule

$$\textsc{adapt if } \widehat{\Delta} - \varepsilon > 0, \quad \textsc{freeze if } \widehat{\Delta} + \varepsilon < 0, \quad \textsc{abstain otherwise}$$

controls the false-adapt and false-freeze rates at α. A corollary shows that in the unknowable two-point regime the certificate *must* abstain with probability at least $1 - 2\alpha$, making abstention mandatory rather than merely conservative. A sample-complexity corollary shows the certificate commits once $n \gtrsim (\sigma^2/\Delta^2)\log(1/\alpha)$ paired benefit samples are available, with an empirical-Bernstein discharge for bounded benefits. An anytime-valid e-process variant (Appendix A) extends the certificate to streaming TTA with no multiplicity correction.

Contribution 4: Covariate-shift positive regime and the sign-of-difference characterization (Theorems 5.3, 5.4–5.6). Under covariate shift ($P_S(Y \mid X) = P_T(Y \mid X)$), importance weighting of the source labels yields a consistent estimator of the target risk, so $\operatorname{sign}\Delta$ is identifiable. Separately, restricting to the *observable disagreement region* $D = \{x : f_0(x) \neq f_a(x)\}$ — observable without labels, since f_0 and f_a are known maps of X — we prove that $\operatorname{sign}\Delta$ equals an ordinal accuracy comparison on D: adapting helps if and only if the adapted model exceeds chance accuracy on D. This characterization holds for binary, multiclass (0/1 loss), and regression (squared loss) in three theorems. Ablations confirm that detector disagreement is the single most important evidence feature, directly supporting the theoretical prediction. Conjecture 5.1 states the remaining open piece: the label-free *bracketing* of the ordinal comparison in the multiclass and regression cases.

Contribution 5: KGA and open-source implementation. We instantiate the framework as KGA (Knowability-Guided Adaptation), a decision wrapper that can be placed around any TTA candidate without modifying it. The algorithm extracts label-free evidence Z, estimates the benefit $\widehat{\Delta}$ with a gradient-boosted regressor trained leave-one-task-out, forms a split-conformal radius ε, and returns the three-way decision. KGA is released as a pure-numpy package (`kga/`) importable as `from kga import KGA` and served at a `POST /decide` endpoint; all code, configs, and result artifacts are public at https://github.com/Pratikn03/AutoML_Flagship_V8 [25]. Chapters 6 and 7–8 present the algorithmic details and the complete experimental evaluation.

Contribution 6: Experiments. We evaluate KGA on nine real experiments spanning multiple regimes.

1. *123-task anomaly-routing benchmark* (ADBench tabular, image-OOD, text, cyber, fraud): KGA abstains where the true benefit is near zero (mean $|\Delta| = 0.021$ on abstained vs. 0.132 on acted tasks) and correctly identifies that the suite is helpful-dominated (adapt precision 0.90, coverage 17%).

2. *Harmful-regime refusal*: with a fusion candidate that hurts on 80% of tasks, KGA refuses it almost everywhere, cutting regret-to-oracle by $\approx 11\times$ (from 0.0214 to 0.0019) while matching the safe baseline AUROC (0.748).

3. *Controlled mixed-regime study* (369 instances, three conditions): KGA achieves adapt precision 0.935 and beats always-freeze by 0.10 AUROC while tying always-adapt in a mild-harm domain.

4. *Clean non-identifiability witness*: the exact Theorem 4.1 construction; KGA abstains on 100% of instances while a forced-commit rule pays regret 0.51.

5. *Statistical rigor over 8 seeds*: multi-seed mean AUROC 0.686 ± 0.003 with paired t-tests confirming KGA is significantly above always-freeze ($p = 7.3 \times 10^{-11}$) and significantly below always-adapt ($p = 5.1 \times 10^{-5}$) in the mild-harm domain.

6. *Regression under covariate shift*: synthetic regression with importance weighting hurting on 79% of instances; KGA refuses it entirely and matches the oracle MSE (0.251) vs. always-adapt (0.338).

7. *Online non-stationary CIFAR-10 TTA*: across a helpful-dominated and a harm-dominated continual stream, KGA achieves the highest mean accuracy across regimes (0.490 vs. always-freeze 0.456 vs. always-adapt 0.444), with always-adapt collapsing to 0.339 in the harm regime.

8. *Per-corruption CIFAR-10-C helpful-dominated control* (65 cells, 13 corruptions $\times$ 5 severities): adaptation is overwhelmingly helpful (false-adapt rate 0.015); KGA matches always-adapt at equal safety, confirming that the certificate does not impose unnecessary cost in benign regimes.

9. *Pre-registered CIFAR-10-C stress grid* (432 conditions/method, harmful base rate 0.16–0.34): KGA beats both trivial policies for Tent and EATA (regret 0.0016 vs. 0.009 / 0.12) at 0% false-adapt, and ties collapse-resistant SAR.

1.6 Honest Scope

We are explicit about the limits of this work in both theory and experiment.

On the theory side, the impossibility result (Theorem 4.1) is close to known non-identifiability of target risk without labels; its purpose here is to *justify the abstain action* and to make the boundary quantitative via the Le Cam two-point lower bound. The certificate (Theorem 5.1) is proved conditional on a valid label-free interval estimator; the empirical-Bernstein discharge requires exchangeability of the paired benefit sample. The sign-of-difference characterization is complete for binary, multiclass, and regression losses; the label-free bracketing of the ordinal comparison in the multiclass and regression cases remains an open conjecture.

On the experimental side, the stress-grid knockout (beating both trivial policies at 0% false-adapt) holds for Tent and EATA on the CIFAR-10-C pre-registered grid; KGA ties collapse-resistant SAR rather than beating it. In per-corruption CIFAR-10-C (helpful-dominated by construction) and within a single-stream online regime, KGA ties the better fixed policy rather than strictly beating it. The decisive advantage is *robustness*: it is the only policy that is near-oracle in both the helpful and harm regimes, which is exactly what the trichotomy predicts. Larger-scale corroboration on CIFAR-100-C, ImageNet-C, and natural-shift benchmarks such as WILDS [26] is future work.

The practical implication is direct. KGA's value scales with how often catastrophic, detectable harm occurs in the deployment stream — precisely the quantity the theory characterizes. In genuinely benign domains where adaptation almost always helps, the certificate is safety insurance at controlled cost. In collapse-prone regimes — small or class-imbalanced batches, single-class label-shift streams, aggressive update schedules, long non-stationary runs — the certificate delivers a real win.

1.7 Chapter Roadmap

Chapter 2 reviews the five adjacent lines of work and positions K-Bound precisely against each. Chapter 3 develops the formal setup: distribution families, the adaptation-benefit definition, observable evidence, and the three regime definitions. Chapters 4 and 5 present the five core theoretical results. Chapter 6 describes the KGA algorithm, its assumptions, and its cost structure. Chapters 7 and 8 present the experimental evaluation across anomaly-routing and online TTA benchmarks. Chapter 9 presents the ELARA/RGA reliability-gated fusion instantiation as a fully code-validated worked example of the general framework. Chapter 10 synthesizes the deployment implications, states the limitations, and outlines future work. Appendix A gives the full reproducibility specification. Appendix B contains extended tables. Appendix C is a notation reference.

Chapter 2

Background and Related Work

We position K-Bound as a precise delta over five adjacent lines of work. In each case the distinction is the one the theory formalizes: between *improving* an unlabeled objective and *certifying the sign* of its benefit before acting. We survey each line in enough depth to make the gap concrete, then close with a positioning table and a synthetic statement of what K-Bound adds.

2.1 Test-Time Adaptation: Mechanisms and Methods

Test-time adaptation originated as a practical answer to the distribution-shift problem: rather than retraining, use the unlabeled target stream to adjust the deployed model. The field has developed along several distinct axes, each representing a different hypothesis about what makes TTA fragile and how to make it robust.

Entropy minimization. **Tent** [52] is the canonical entropy-minimization method. At each test batch, Tent minimizes the Shannon entropy of the model's softmax predictions over only the batch-normalization affine parameters, keeping the rest of the network frozen. The intuition is that predictions on target data should be confident (low entropy), and adjusting normalization statistics accomplishes this with minimal risk of catastrophic forgetting. On i.i.d. corruption benchmarks such as CIFAR-10-C [21] and ImageNet-C, Tent yields substantial accuracy improvements over the frozen baseline.

The fragility of pure entropy minimization, however, is well documented. Tent applied continually to non-stationary or class-imbalanced streams accumulates normalization-parameter damage; once the predictions collapse to a single class, entropy continues to decrease while accuracy plummets. Our stress-grid experiments (Chapter 8) reproduce this collapse in the harm-dominated online CIFAR-10 regime, where always-adapt with Tent falls to 0.339 accuracy while a frozen model holds 0.440.

Batch-normalization statistics adaptation. **Schneider et al.** [42] showed that a large fraction of the performance gap on corruption benchmarks can be closed simply by re-estimating batch-normalization running statistics from a mixture of source and target data, without any gradient update. This observation motivated a family of statistics-only approaches that avoid the optimization risk of entropy minimization entirely, at the cost of needing a sufficient target batch size to estimate stable statistics. **NOTE** [15] extends this direction to non-i.i.d. streams using class-conditional normalization, addressing the label-shift fragility of global statistics.

Filtering and Fisher regularization. **EATA** [36] addresses Tent's catastrophic-forgetting tendency with two additions: an entropy filter that rejects unreliable (high-entropy) samples from the update, and a Fisher-information-based regularization penalty that penalizes updates proportional to their importance for source-domain performance. The entropy filter prevents the worst collapse triggers from entering the update gradient; the Fisher penalty ensures the normalization parameters do not drift too far from their source-optimal values. EATA retains substantial improvement over the frozen baseline on standard benchmarks while partially protecting against the forgetting that pure Tent suffers.

Sharpness-aware updates and resets. **SAR** [37] adds sharpness-aware entropy minimization combined with an online reset mechanism. Sharpness-aware optimization minimizes an upper bound on the worst-case perturbation of the loss landscape, producing parameter updates that land in flat regions less susceptible to subsequent drift. The reset mechanism monitors a model-divergence criterion and resets the normalization parameters to their source values if divergence exceeds a threshold, aborting incipient collapse. SAR is notably more robust than Tent or EATA on the continual and non-stationary regimes that provoke collapse; our experiments confirm this, with SAR's harmful base rate (16% on the stress grid) substantially lower than Tent's (34%) and EATA's (27%). This robustness is also the reason K-Bound ties rather than beats SAR: when the candidate already self-protects, the certificate's additional value is limited.

Source-free and continual adaptation. **SHOT** trains a source-free adaptation by information maximization and self-supervised pseudo-labeling, enabling adaptation when the source data is unavailable at deployment. **CoTTA** [53] addresses the continual adaptation setting by stochastic restoring: with some probability, parameters are reset to their source-pre-trained values, preventing irreversible forgetting across a long non-stationary stream. **TTT** [47] augments the source training with a self-supervised auxiliary task and continues training that task at test time, adapting the shared feature extractor. **MEMO** [56] takes a different route: for each test point it generates multiple augmented views and minimizes the marginal entropy across augmentations, adapting per-sample rather than per-batch.

Parameter-free approaches. LAME [6] is at the opposite extreme: it requires no parameter update at all, instead post-processing the model's predictions using a Laplacian regularization that encourages smoothness across the target batch. Its robustness to collapse is guaranteed by construction, since there are no parameters to corrupt, but it cannot recover the same accuracy gains as gradient-based methods when the distribution shift is large.

What all TTA methods share. Every method in this section, whether gradient-based or gradient-free, whether it uses resets or regularization or filtering, makes the same architectural commitment: *it adapts*. The question of whether adaptation is the right action on a given deployment batch is not posed; the decision is implicit in the method's deployment. K-Bound is mechanism-agnostic with respect to this entire class of methods. It wraps any such candidate — we evaluate Tent, EATA, and SAR — and makes the adapt/freeze/abstain decision at the certificate level, before the adaptation mechanism executes.

2.2 TTA Failure, Collapse, and Evaluation

The recognition that TTA can fail destructively has generated a parallel line of work on understanding and measuring when failure occurs.

Pitfalls and failure modes. Zhao et al. [58] systematically catalogued the pitfalls of test-time adaptation evaluation: sensitivity to the batch-size regime, stream composition, update budget, and the distinction between i.i.d. and temporally correlated streams. Their analysis showed that many published TTA results are obtained in regimes where adaptation is effectively guaranteed to help (large i.i.d. batches of a single corruption type), and that performance degrades substantially — sometimes below the frozen baseline — in the more realistic regimes of mixed, imbalanced, or small-batch streams. This work directly motivates the design of the K-Bound pre-registered stress grid: we cross severity, batch size, composition, and aggressiveness to construct conditions that span the helpful and harmful regimes.

Robustness benchmarks. RDumb [38] demonstrated that a periodically resetting stupid baseline (no adaptation, periodic reset to the source model) outperforms many published TTA methods on long, non-stationary streams. The implication is that sophisticated adaptation mechanisms may be over-optimized for the benign conditions under which they are evaluated and break down precisely when deployment conditions are most challenging.

Standardized benchmarks. Hendrycks and Dietterich [21] introduced the CIFAR-10-C, CIFAR-100-C, and ImageNet-C benchmarks, which apply 19 algorithmically defined corruptions at five severity levels to standard test sets. These benchmarks have become the standard evaluation platform for TTA methods; they are also the platform for our per-corruption experiment

(Chapter 8). However, as Zhao et al. and Press et al. note, per-corruption evaluation is inherently helpful-dominated: each batch contains a single corruption type at a single severity, giving the adaptation mechanism a maximally informative signal. Our stress grid goes beyond this by mixing compositions.

DomainBed. **Gulrajani and Lopez-Paz** [18] introduced the DomainBed benchmark for domain generalization, emphasizing standardized evaluation protocols and a thorough comparison of algorithms under matched hyperparameter selection. Their finding that many domain generalization methods do not consistently outperform empirical risk minimization under rigorous evaluation is methodologically analogous to our finding about TTA in the benign regime: sophisticated methods add value primarily in the regimes where they are designed to help. **WILDS** [26] extends this to real-world distribution shifts, including medical imaging, satellite imagery, and user-generated content. Extending K-Bound to WILDS benchmarks is an explicit item of future work (Chapter 10).

2.3 Guarded and Protected Adaptation

A line of work distinct from both TTA mechanisms and TTA failure analysis guards adaptation against specific failure modes *during* the update.

Entropy thresholding and filtering. EATA's entropy filter is the most direct example: high-entropy samples are excluded from the update gradient on the hypothesis that they carry corrupted or uninformative signal. This protects against one failure direction (adapting on adversarial inputs) but does not address the three-way decision between helpful, harmful, and unknowable evidence, nor does it provide a formal bound on the false-adapt rate.

Betting-style monitors. Sequential testing literature has developed monitors that halt or dampen a running process when a specific failure mode is detected. In the TTA setting, such a monitor could, for example, halt entropy minimization when the estimated class distribution collapses below a threshold. These approaches protect one direction — typically against collapse — but they do not pose the three-way decision with an explicit unknowable region, and they do not certify *before* acting that adaptation will help.

NOTE's class-conditional normalization. **NOTE** [15] addresses the temporal correlation between samples in a non-stationary stream by estimating class-conditional normalization statistics. This is a form of guarded adaptation in the sense that it conditions the normalization update on the estimated class label of each sample, reducing the sensitivity to class imbalance that causes Tent to collapse. The guard is built into the mechanism, not into a separate decision layer.

The structural gap. All guarded-adaptation approaches share the architecture of operating *during* the update. K-Bound operates *before* the update: it makes the adapt/freeze/abstain decision first, and only then — if the decision is ADAPT — does it invoke the underlying mechanism. This separation is not merely organizational; it is what makes the false-adapt guarantee meaningful. A certificate that fires after an adaptation has already occurred cannot meaningfully bound the false-adapt rate; one that fires before can.

2.4 Label-Free Accuracy Estimation

A third line of work predicts target performance from unlabeled data, without the explicit goal of making an adapt/freeze decision.

ATC. Garg et al. [13] introduced Average Threshold Confidence (ATC), which thresholds source-calibrated softmax confidence to estimate the fraction of correct predictions on a target set. ATC achieves strong empirical accuracy on standard benchmarks and has become a reference method for label-free performance estimation. Its training requires a labeled source set to calibrate the confidence threshold; given that calibration, it requires only unlabeled target data.

Agreement-on-the-Line. Baek et al. [2] exploited the empirical observation that model agreement (the fraction of inputs on which two independently trained models make the same prediction) correlates linearly with accuracy on both in-distribution and out-of-distribution data. The linear relationship can be calibrated from in-distribution data and extrapolated to estimate OOD accuracy from OOD agreement, which is label-free and observable. The method is particularly valuable in the covariate-shift regime where the linear correlation holds reliably, and less so in label-shift regimes where it breaks down.

Predicting with confidence. Guillory et al. [17] studied the problem of estimating model accuracy on unseen distributions with confidence intervals, developing methods based on source/target distribution divergence and ensemble disagreement.

The distinction from K-Bound. All of these methods estimate $R_T(f)$ or $R_T(f_0)$ — the target risk of a single model — or detect that TTA has failed *post hoc*. K-Bound's object of study is strictly different: the *sign* of $R_T(f_0) - R_T(f_a)$. This is a relational quantity involving *two* models, not an absolute risk estimate. Certifying the sign is a strictly weaker requirement than estimating either absolute risk, so the knowable set is strictly larger — but, as Theorem 4.1 establishes, it is not all of evidence space. Furthermore, K-Bound provides a *pre-adaptation* certificate with a controlled false-adapt rate: it fires before f_a is deployed, not after. Post-hoc detection that TTA has hurt (as in AETTA and some ensemble-based monitors) is useful for retrospective analysis but cannot prevent a false adaptation from occurring.

2.5 Unsupervised Risk Estimation and Identifiability

The hardness of K-Bound's problem is inherited from the broader problem of unsupervised risk estimation, and the impossibility results in that literature are directly relevant.

Steinhardt and Liang. Steinhardt and Liang showed that unsupervised risk estimation is possible only under a conditional-independence structure between the features, the label, and the observed predictions. In particular, they required that the label can be recovered from the prediction via a known conditional distribution, which is essentially an assumption of calibration stability. Under this assumption, the target risk is identifiable from unlabeled data; without it, the problem is generally intractable. Their work establishes the structural condition under which label-free risk estimation is possible and motivates our weaker target: the sign rather than the magnitude of a risk difference.

Ben-David et al. **Ben-David et al.** [3] established the canonical learning-theoretic framework for domain adaptation. Their central bound expresses the target risk as the source risk plus an $\mathcal{H}$-divergence between the source and target marginals plus a residual term capturing the irreducible concept-shift component. The $\mathcal{H}$-divergence is computable from unlabeled data; the residual term is not. This bound is tight in the sense that all three terms are necessary; the identifiability result of Theorem 5.3 can be understood as showing that under covariate shift the residual term vanishes and the $\mathcal{H}$-divergence is controlled by the density ratio.

Mansour et al. **Mansour et al.** [33] studied domain adaptation from a learning-bounds perspective, extending the Ben-David et al. framework to mixture-of-source distributions and developing discrepancy-based bounds. Their discrepancy measure generalizes the $\mathcal{H}$-divergence and admits importance-weighting interpretations that connect to our covariate-shift positive result.

Label-shift correction. **Lipton et al.** [31] developed Black Box Shift Estimation (BBSE), which corrects for label shift — a change in the marginal $P(Y)$ with stable class-conditional distributions $P(X \mid Y)$ — using an invertible confusion matrix estimated from labeled source data. BBSE identifies the target label marginal from unlabeled target predictions, enabling unbiased risk estimation under label shift. This is one of the two identifiable special cases in the knowability hierarchy (Table 2.1 in Chapter 3): label shift with an invertible confusion matrix places us in the knowably helpful or knowably harmful regime, not the unknowable one.

Dataset shift taxonomy. **Quiñonero-Candela et al.** [39] provide the standard taxonomy of dataset shift: covariate shift, label shift, concept shift, and their combinations. This taxonomy is the organizational backbone of the knowability hierarchy: covariate shift is provably identifiable (Theorem 5.3), label shift is identifiable under a mild structural assumption (invertible confusion), concept shift re-enters the unknowable regime (Theorem 4.1), and mixed/unknown shift requires certify-or-abstain.

Importance weighting. **Shimodaira** [45] introduced importance weighting (weighted log-likelihood) as the canonical approach to covariate-shift adaptation, showing that reweighting the source loss by the density ratio $p_T(x)/p_S(x)$ yields consistent estimators of target risk. **Sugiyama et al.** [46] developed importance weighted cross-validation, which uses the density ratio to select hyperparameters in a way that is consistent under covariate shift. These works are the direct antecedents of Theorem 5.3: the importance-weighted estimator $\widehat{R}_T(f) = n^{-1}\sum_i r(x_i)\ell(f(x_i), y_i)$ is the operational mechanism behind the covariate-shift positive result.

The K-Bound delta over this line. The identifiability literature establishes when risk is identifiable and when it is not. K-Bound's contribution within this line is twofold. First, we target the *sign* of a risk difference, which is identifiable on a strictly larger set than the risk itself: the unknowable regime in K-Bound is strictly smaller than the non-identifiable regime for risk estimation. Second, we instantiate the identifiability result as a *finite-sample, false-adapt-controlled certificate* (Theorem 5.1), going from an asymptotic identifiability statement to an operational, finite-sample decision procedure.

2.6 Selective Prediction and Abstention

Abstention is not new to machine learning; it has been studied in the context of individual predictions for decades.

Chow's reject option. **Chow** [8] introduced the reject option for classifiers: rather than committing to a prediction for every input, a classifier may abstain when its confidence is below a threshold, trading coverage for accuracy. The optimality theory shows that the Bayes-optimal reject rule thresholds the posterior probability, and the coverage-accuracy tradeoff is characterized by the Bayes optimal.

Selective classification. **El-Yaniv and Wiener** [12] and **Geifman and El-Yaniv** [14] developed the modern theory of selective classification, providing finite-sample guarantees on the accuracy of a classifier conditioned on its decision to predict. SelectiveNet extends selective classification to deep neural networks with an end-to-end trainable coverage mechanism.

Conformal prediction. **Vovk et al.** [51] introduced conformal prediction as a framework for constructing prediction sets with valid marginal coverage. **Angelopoulos and Bates** [1] provide a recent tutorial treatment, including split-conformal prediction (which uses a held-out calibration set to set the threshold) and risk-controlling prediction sets. Split-conformal prediction is the mechanism behind Theorem 5.1: the conformal radius ε is constructed from a held-out set of labeled source/validation benefits, and its coverage guarantee transfers to the certificate.

K-Bound's abstention is over a decision, not a prediction. Selective prediction abstracts on individual *predictions*: given a single test input x, should the model commit to a class label $\hat{y}$ or abstain? K-Bound's abstention is at a different level: given a deployment batch and an observable evidence vector Z, should the system commit to the ADAPT or FREEZE *action*, or abstain from the adaptation decision? The object is not a label for a single point but a binary action for an entire deployment episode. This distinction matters because the cost of a wrong action is not the per-sample misclassification rate but the regret relative to the oracle policy — an expectation over the entire deployed batch.

2.7 Conformal Prediction and Anytime-Valid Inference

The certificate construction connects to several recent developments in sequential and anytime-valid inference.

Game-theoretic probability. **Ville** [50] introduced the concept of e-processes (nonneg. supermartingales) and established Ville's inequality, which gives the anytime-valid analogue of Markov's inequality. **Shafer and Vovk** [44] developed the game-theoretic foundations for probability and statistics, showing that betting-style arguments yield valid inference without relying on the law of large numbers.

Testing by betting and e-values. **Shafer** [44] and **Ramdas et al.** [40] developed the testing-by-betting framework: a hypothesis is rejected when the cumulative product of e-values (factors $1+\lambda_t X_t$ where X_t is a bounded observation) exceeds $1/\alpha$. The product is a nonneg. supermartingale under the null, so its exceedance is controlled at level α uniformly over all stopping times (Ville's inequality), without any correction for repeated looks.

Anytime-valid confidence sequences. **Howard et al.** [24] constructed time-uniform, nonparametric confidence sequences: intervals that are valid at all sample sizes simultaneously. **Waudby-Smith and Ramdas** [55] developed betting-based confidence intervals for bounded random variables, establishing tight bounds via predictable bets. **Grünwald et al.** [16] formalized safe testing, which uses e-values to test composite hypotheses in a way that is valid under optional stopping.

Connection to K-Bound. Theorem 5.1 is a fixed-n certificate; the anytime-valid extension (Appendix A) uses a testing-by-betting e-process. The anytime-valid version is strictly more powerful than any fixed-n certificate in the sequential regime (it is valid at any stopping time with no multiplicity correction), while the fixed-n version is tighter at a known sample size. The empirical-Bernstein bound of **Maurer and Pontil** [34] provides the radius discharge: for bounded paired benefits in $[a, b]$, the empirical variance proxy yields a confidence radius under independence/exchangeability.

Classical concentration. **Hoeffding** [22] established the classic exponential tail bound for bounded independent random variables, and **Le Cam** [28] developed the two-point lower bound technique (used in Theorem 4.1) and the fundamental convergence-of-estimates theory. **Tsybakov** [49] and **Lehmann and Romano** [29] are the standard references for nonparametric estimation theory and testing, respectively.

2.8 Benchmarks, Robustness, and Distribution Shift

Robustness benchmarks. **Hendrycks and Dietterich** [21] introduced the corruption benchmarks that anchor all of our CIFAR-10-C experiments. **Recht et al.** [41] showed that ImageNet classifiers lose substantial accuracy on a re-collected ImageNet test set, motivating the study of natural distribution shift. **Taori et al.** [48] measured robustness to a broad set of natural and synthetic distribution shifts, finding that effective robustness (robustness beyond what is predicted by the in-distribution to OOD correlation) is rare. **Miller et al.** [35] established the accuracy-on-the-line phenomenon: across many models and distribution shifts, in-distribution and OOD accuracy are linearly correlated.

Anomaly detection benchmarks. K-Bound's anomaly-routing experiments use the ADBench benchmark [19], which covers tabular, image, text, cyber, and fraud anomaly detection across 57 datasets and multiple detector families. **MVTec AD** [4] and **MVTec 3D-AD** [5] provide industrial visual-inspection benchmarks where distribution shift is induced by modality failure; the ELARA/RGA instantiation (Chapter 9) is evaluated on MVTec 3D-AD. **Real-IAD** [54] is a large-scale real industrial anomaly detection dataset with 30 object categories; Real-IAD-D3 is the ELARA/RGA multimodal reliability experiment. **PyOD** [57] provides the detector library from which most of our detector scores are extracted.

Standard vision. **CIFAR-10** [27], **ImageNet** [9], **ResNet** [20], **ViT** [11], and **Imagenette** [23] are the standard building blocks of our online TTA and CIFAR-10-C experiments.

Label-free performance prediction and accuracy estimation. **Deng and Zheng** [10] studied the problem of evaluating classifier accuracy without labels, developing methods based on the agreement between classifiers and the structure of the confidence distributions. This work is closely related to ATC [13] and Agreement-on-the-Line [2] in its goals and is part of the same literature that K-Bound departs from in its relational target (sign of difference) and pre-adaptation certificate.

Anomaly/outlier detection methods. The anomaly detection backbone of the experiments uses standard methods from the literature: **Isolation Forest** [32], **Local Outlier Factor** [7], **One-Class SVM** [43], **ECOD** [30], and the PyOD suite [57].

2.9 Positioning: The Gap K-Bound Fills

Table 2.1 reproduces and expands the positioning table from the working paper. The six columns characterize each method along the five dimensions that are jointly required for K-Bound's contribution: label-free operation, the distinction between adapting and wrapping, benefit prediction before acting, abstention on the adaptation decision itself, and a formal false-adapt guarantee.

Work	label-free?	adapts?	predicts benefit *before* adapting?	abstains on the decision?	false-adapt guarantee?
Tent [52]	yes	yes	no	no	no
EATA [36]	yes	yes	partial (filtering)	no	no
SAR [37]	yes	yes	stability only	no	no
CoTTA [53]	yes	yes	no	no	no
TTT [47]	yes	yes	no	no	no
MEMO [56]	yes	yes	no	no	no
LAME [6]	yes	yes	no	no	no
NOTE [15]	yes	yes	no	no	no
ATC [13]	yes	monitors	post-hoc estimate	—	no
Agr.-on-Line [2]	yes	monitors	post-hoc estimate	—	no
BBSE [31]	partial	corrects	corrects shift	no	no
KGA (ours)	yes	wrapper	**yes**	**yes**	**yes**

Table 2.1: Positioning of K-Bound against adjacent methods. Prior TTA methods adapt (some with internal guards); performance estimators operate post-hoc. No prior method provides a single pre-adaptation adapt/freeze/abstain certificate with an explicit, proven unknowable region and a false-adapt bound. KGA wraps any TTA candidate without modifying it.

The surviving gap is precise: no prior method provides a single adapt/freeze/abstain certificate with an *explicit, proven* unknowable region and a false-adapt guarantee.

To state this more carefully:

- TTA mechanisms adapt and (some) detect failure, but commitment happens before detection and no false-adapt rate is bounded.
- Performance estimators (ATC, Agreement-on-the-Line) estimate $R_T(f)$ post-hoc; they do not certify sign Δ before the adaptation decision.
- Guarded adaptation (EATA filtering, SAR resets) protects one failure direction but does not pose the three-way decision with an unknowable region.

- Identifiability theory (Ben-David et al., BBSE) shows when risk is identifiable but does not yield a finite-sample, false-adapt-controlled operational certificate.
- Selective prediction (Geifman and El-Yaniv) abstains on individual predictions, not on the adaptation decision as a whole.
- Conformal prediction (Vovk et al., Angelopoulos and Bates) provides the interval mechanism K-Bound uses, but not the framing of the adapt/freeze/abstain decision.

K-Bound combines the identifiability insight (when is sign Δ recoverable?) with the conformal certificate mechanism (how do we control the false-adapt rate given a valid interval?) and the selective-abstention logic (when the interval is too wide to certify either sign, the correct action is to abstain) into a single coherent decision framework. The result is a method that is simultaneously label-free, mechanism-agnostic, pre-adaptive, three-way, and formally guaranteed — a combination no prior work achieves.

Chapter 3

Problem Formulation and the Knowability Trichotomy

This chapter fixes the objects that every later result manipulates: the deployment protocol under which a frozen model meets a shifted, unlabeled target stream; the *adaptation benefit* Δ, which is the quantity a deployed system would like to know; the *label-free evidence* Z, which is the only quantity it can actually observe; and the trichotomy of *knowably helpful*, *knowably harmful*, and *unknowable* regimes that organizes the entire theory. We then formalize the decision problem itself — rules $g(Z) \in \{\textsc{adapt}, \textsc{freeze}, \textsc{abstain}\}$, the oracle, regret-to-oracle, and the false-adapt and false-freeze error rates — and close with three fully worked example environments, a discussion of the standing assumptions, and a reader's map from questions to theorems.

The formal content of this chapter is exactly that of the paper's problem setup and definitions [25]; the worked examples and the assumptions discussion are expository additions that introduce no new claims. A reader who internalizes one idea from this chapter should make it this one: *the constraint that drives everything is not noise but measurability.* The decision rule is a function of Z, and Z is a function of unlabeled data alone. Whether the right decision is *knowable* is a property of the σ-algebra the evidence generates, not of how many samples populate it. Chapter 4 makes that remark a theorem.

3.1 The deployment protocol

The setting is the standard one of test-time adaptation under distribution shift [52, 39], stated so that the information structure — who sees which labels, and when — is completely explicit.

A *source* distribution $P_S(X, Y)$ over inputs $X \in \mathcal{X}$ and labels $Y \in \mathcal{Y}$ is available with labels at training and validation time: the model was fit on labeled source data, and a held-out labeled validation (or probe) set from the source environment may be retained. A *target* distribution $P_T(X, Y)$ governs deployment, and it exposes only its covariate marginal $P_T(X)$ at test time: the deployed system receives a stream (or batch) of unlabeled target inputs $X_1, \ldots, X_m \sim P_T(X)$

and never observes a target label. The two distributions may differ in any way — covariate shift, label-prior shift, concept shift, or arbitrary mixtures [39, 45, 31]; part of the point of the theory is to classify which kinds of difference leave the adaptation decision decidable.

Three further objects are fixed. A loss $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ scores predictions (the paper's results are stated for 0/1 loss, multiclass 0/1 loss, and squared error at the points where the loss matters; until then ℓ is generic). A *frozen model* f_0 is the system as deployed: source-trained, untouched by the target stream. An *adapted or gated candidate* f_a is the alternative the system could deploy instead: the output of a test-time adaptation procedure run on the unlabeled batch (entropy minimization in the style of Tent [52], a reliability-gated fusion, a recalibrated threshold, an importance-weighted refit — the framework is agnostic to the mechanism). What matters is only that at decision time f_0 and f_a are both concrete, known, fixed maps from inputs to predictions: the decision under study is *which of two specific models to deploy*, not how to construct a better candidate. Target risk is

$$R_T(f) \;=\; \mathbb{E}_{P_T}\big[\ell(f(X), Y)\big],$$

the population average loss of a model f on the deployment distribution. Note carefully where the inaccessibility lies: $R_T(f)$ averages over the *joint* law $P_T(X, Y)$, while the system observes draws from $P_T(X)$ only. Both $R_T(f_0)$ and $R_T(f_a)$ are therefore well-defined but unobserved quantities, and every question in this monograph is a question about what can be inferred about them — more precisely, about their difference — through the keyhole of unlabeled data.

3.2 The adaptation benefit

The decision-relevant quantity is not either risk in isolation but their difference.

Definition 3.1 (Adaptation benefit)**.** Fix the loss ℓ, the frozen model f_0, and the adapted/gated candidate f_a. The *adaptation benefit* is

$$\Delta \;=\; R_T(f_0) - R_T(f_a), \qquad \Delta > 0 \iff \text{adapting helps.} \tag{3.1}$$

The sign convention reads naturally: Δ is the risk *removed* by switching from the frozen model to the candidate. When $\Delta > 0$ the candidate is strictly better on the target and the correct action is to ADAPT; when $\Delta < 0$ adaptation is harmful and the correct action is to FREEZE; $\Delta = 0$ is the indifference point. The benefit also has a per-sample form that recurs throughout the proofs: writing $\delta = \ell(f_0(X), Y) - \ell(f_a(X), Y)$ for the paired per-sample benefit, we have $\Delta = \mathbb{E}_{P_T}[\delta]$, and the paired structure — both losses evaluated on the *same* draw (X, Y) — is what later makes finite-sample certification statistically cheap relative to estimating either risk on its own (Chapter 5).

Because the decision will be made on observed evidence, we also need the benefit *conditioned* on that evidence. With Z the observable evidence of the next section, the *conditional benefit* is

$$\Delta(z) \;=\; \mathbb{E}\big[\ell(f_0(X), Y) - \ell(f_a(X), Y) \mid Z = z\big], \tag{3.2}$$

the expected gain from adapting given that the evidence took the value z. The unconditional benefit is recovered by the tower property, $\Delta = \mathbb{E}[\Delta(Z)]$. The paper's central question, stated once and answered over two chapters, is:

When is $\operatorname{sign}\Delta(z)$ *recoverable from* Z *alone?*

Note the deliberate modesty of the target. We do not ask to estimate $R_T(f_0)$ or $R_T(f_a)$ (a cardinal estimation problem, known to be non-identifiable without structural assumptions [3, 31]), nor even to estimate Δ to additive accuracy. We ask only for the *sign* of a difference — a single bit. Part of the content of the theory is that even this single bit is sometimes information-theoretically out of reach (Chapter 4), and that, conversely, the sign is recoverable on a strictly larger set of shifts than either risk is (Theorem 5.4 in Chapter 5).

3.3 Label-free observable evidence

Definition 3.2 (Observable evidence)**.** *Observable evidence* $Z = \phi(X_{1:m}, f_0, f_a)$ is any statistic computable *without* target labels: examples include entropy, confidence, score-distribution drift, detector disagreement, predicted-class balance, update norm, and density-ratio diagnostics.

The definition is intentionally maximal. The evidence map ϕ may be any measurable function of the unlabeled batch $X_{1:m}$ and of the two fixed models; in particular it may use the models' outputs, internal activations, confidence scores, and any comparison between them. Concretely, the evidence panels used in the experiments of Chapters 7 and 8 include: prediction-entropy statistics of f_0 and f_a on the batch; confidence and margin distributions; drift statistics comparing the batch's score distribution to the source validation distribution (for example Kolmogorov–Smirnov distances); the rate and pattern of *disagreement* between f_0 and f_a; the predicted-class balance and its drift from the source balance; the norm of the adaptation update; and density-ratio diagnostics. All of these are functions of covariates and known maps only; none touches a target label. Label-free accuracy-estimation statistics in the style of average thresholded confidence [13] are admissible evidence features under this definition — the framework subsumes them as particular choices of ϕ.

Two structural remarks will carry weight later. First, since f_0 and f_a are fixed known maps, the *disagreement region*

$$D \;=\; \{x : f_0(x) \neq f_a(x)\}$$

is itself observable: membership of each batch point in D is computable without labels, and statistics of the batch restricted to D are legitimate evidence. This observable region turns out to carry the entire decision (Theorem 5.4). Second — and this is the pivot of Chapter 4 — whatever ϕ one chooses, the distribution of Z under a target world P_T depends on that world *only through* $P_T(X)$, because Z is a function of covariates and fixed maps alone. Two target worlds that share a covariate marginal but differ in $P_T(Y \mid X)$ induce *identical* evidence laws for every conceivable evidence map. The impossibility theorem is, at bottom, this observation plus a careful accounting of what it costs.

3.4 The knowability trichotomy

The paper's two definitions classify evidence values by what they permit.

Definition 3.3 (Observable risk alignment)**.** Z is *risk-aligned* if it identifies or bounds $\operatorname{sign} \Delta(z)$ (it need not reveal the label, only whether adaptation helps).

Definition 3.4 (Regimes)**.** At evidence z: *knowably helpful* if Z certifies $\Delta(z) > 0$ (**adapt**); *knowably harmful* if Z certifies $\Delta(z) < 0$ (**freeze**); *unknowable* if the same Z is compatible with both signs (**abstain**).

Definition 3.3 isolates the property that makes label-free decisions possible at all. Risk alignment does *not* ask the evidence to reveal labels, risks, or even the magnitude of the benefit; it asks only that the evidence constrain the benefit's sign. It is a property of the pair (evidence map, family of plausible target worlds): the same ϕ can be risk-aligned against one family of shifts and blind against another. Definition 3.4 then partitions evidence values accordingly. The word "certifies" is doing real work: in the knowable regimes, every target world consistent with the observed evidence has the same benefit sign, so committing is safe; in the unknowable regime there exist worlds consistent with the same evidence whose benefits have opposite signs, and Chapter 4 proves that no rule — however clever, and at any unlabeled sample size — can then commit correctly in all of them. The trichotomy is thus not a design choice but a theorem-shaped fact: the three actions ADAPT, FREEZE, ABSTAIN are exactly the actions matched to the three regimes.

Because "label-free" is used in subtly different senses across the adaptation literature, the paper pins down its meaning in a box that we reproduce here in full, as the protocol is part of the formulation.

Remark 3.1 (What "label-free" means here)*.* *Target-label-free* (our setting): no labels from the target/test stream are ever used. *Validation labels allowed*: labeled source and a held-out labeled validation/probe set may calibrate the benefit estimator $\widehat{\Delta}$ and the radius ε. *Target-unsupervised evidence*: the deployment batch contributes only X, predictions, entropy, confidence, drift and other label-free statistics Z. The online-TTA results that use a clean labelled validation probe (Chapter 8) are target-label-free in this sense — they use no *target* labels — not unsupervised of all labels.

The distinction matters because the impossibility results of Chapter 4 hold against the strongest reading (no target labels, arbitrary evidence maps, source labels freely available), while the positive results of Chapter 5 consume the weaker allowances explicitly: the certificate's radius is calibrated on labeled validation data, and the covariate-shift theorem reweights labeled *source* examples. Nothing anywhere uses a target label.

3.5 Decision rules, the oracle, and error rates

The system's output is a decision, so we now formalize the decision-maker.

Definition 3.5 (Label-free decision rule)**.** A *label-free decision rule* is a (possibly randomized) map

$$g(Z) \;\in\; \{\textsc{adapt},\; \textsc{freeze},\; \textsc{abstain}\},$$

measurable in the evidence Z and in randomness independent of the target world. ADAPT deploys f_a; FREEZE retains f_0; ABSTAIN retains f_0 *and* flags the condition as undecided (declining to certify either committed action). The *coverage* of a rule is $\mathbb{P}[g \neq \textsc{abstain}]$, the fraction of conditions on which it commits.

Two readings of ABSTAIN coexist in the monograph and it is worth separating them now. Operationally, an abstaining system keeps the frozen model running (the conservative default) while routing the condition to fallback handling — human review, data collection, a wider probe. Decision-theoretically, ABSTAIN is a first-class third action whose role is exactly analogous to the reject option in selective prediction [8, 12, 14], with one crucial difference of granularity: selective prediction abstains on individual *predictions*, while our rules abstain on the *adaptation decision* for a deployment condition. The cost accounting of abstention (a coverage cost, distinct from the cost of a wrong commitment) is made precise inside the proofs of Chapter 4.

Performance is measured against the best decision made with full knowledge.

Definition 3.6 (Oracle and regret-to-oracle)**.** The *per-condition oracle* is $g^\star = \mathbf{1}[\Delta > 0]$: it adapts exactly when the true benefit is positive. *Regret-to-oracle* of a rule g denotes $R(g) - R(g^\star)$, where $R(\cdot)$ is the deployed risk of the rule's chosen actions.

For committal gates the deployed risk takes the explicit form analyzed in Chapter 4: with the action-conditional means $a_{\text{FRZ}}(z) = \mathbb{E}[\ell(f_0(X), Y) \mid Z{=}z]$ and $a_{\text{ADP}}(z) = \mathbb{E}[\ell(f_a(X), Y) \mid Z{=}z]$, a gate g has risk $R(g) = \mathbb{E}[g(Z)\, a_{\text{ADP}}(Z) + (1 - g(Z))\, a_{\text{FRZ}}(Z)]$, and the conditional benefit of Definition 3.1 is recovered as $\Delta(z) = a_{\text{FRZ}}(z) - a_{\text{ADP}}(z)$. Theorem 4.3 will show that the regret-to-oracle of any plug-in gate is *exactly* the benefit it forfeits on the evidence values where it picks the wrong sign.

Finally, the two failure modes of a three-way rule are named and will be the quantities the certificate of Chapter 5 controls.

Definition 3.7 (False-adapt and false-freeze rates)**.** For a rule g,

$$\mathrm{FA}(g) \;=\; \mathbb{P}[g = \textsc{adapt} \;\wedge\; \Delta \le 0], \qquad \mathrm{FF}(g) \;=\; \mathbb{P}[g = \textsc{freeze} \;\wedge\; \Delta \ge 0],$$

the probabilities of committing to adaptation when it does not help and of refusing adaptation when it would not hurt, respectively.

The asymmetry of consequences in practice — a false adapt deploys a model that silently degrades, a false freeze merely forgoes an improvement — motivates treating FA as the safety-critical rate; the certificate (Theorem 5.1) bounds both at a user-chosen level α, and the experiments report both, together with coverage and regret-to-oracle. Throughout, $\widehat{\Delta}$ denotes a label-free estimator of Δ with confidence radius ε at miscoverage level α; the operational rule built from $(\widehat{\Delta}, \varepsilon)$ is the subject of Chapters 5 and 6.

3.6 Three worked environments

The definitions above are abstract by design. The following three example environments — expository additions to the paper, chosen so that every quantity is computable in closed form — exhibit one regime each: a knowable covariate shift, an unknowable posterior flip in which the evidence is provably blind, and a low-margin band in which the evidence is informative in principle but commitment is not worth its price. Together they walk the phase diagram of Figure 3.1: far right, far left, and the bottom wedge.

Example 3.1 (A knowable Gaussian location shift)**.** Let the label be a fixed deterministic function of the input, $Y = \mathbf{1}[X > \frac{1}{2}]$, in source and target alike, so that $P_S(Y \mid X) = P_T(Y \mid X)$ and any shift is pure covariate shift [45]. The source is $X \sim \mathcal{N}(0,1)$; the target is $X \sim \mathcal{N}(\mu, 1)$ with μ unknown. The frozen model is $f_0(x) = \mathbf{1}[x > 0]$ (its threshold sits below the Bayes boundary $\frac{1}{2}$), and the adaptation procedure proposes the candidate $f_a(x) = \mathbf{1}[x > 1]$ (a threshold recalibration overshooting the boundary from the other side); under 0/1 loss, with Φ the standard normal distribution function, the risks are the masses of the two mismatch intervals,

$$R_T(f_0) = \mathbb{P}[0 < X \le \tfrac{1}{2}] = \Phi(\tfrac{1}{2} - \mu) - \Phi(-\mu), \qquad R_T(f_a) = \mathbb{P}[\tfrac{1}{2} < X \le 1] = \Phi(1 - \mu) - \Phi(\tfrac{1}{2} - \mu),$$

so the benefit is the explicit function

$$\Delta(\mu) \;=\; 2\,\Phi(\tfrac{1}{2} - \mu) - \Phi(-\mu) - \Phi(1 - \mu).$$

The two error intervals $(0, \frac{1}{2}]$ and $(\frac{1}{2}, 1]$ have equal length and sit symmetrically about the Bayes boundary; since the unimodal target density places more mass on whichever interval is nearer its mode μ, we get $\operatorname{sign}\Delta(\mu) = \operatorname{sign}(\frac{1}{2} - \mu)$, with $\Delta(\mu) = 0$ exactly at $\mu = \frac{1}{2}$ and the antisymmetry $\Delta(\frac{1}{2} + t) = -\Delta(\frac{1}{2} - t)$. Numerically, $\Delta(-1) \approx +0.048$ (adapt), $\Delta(0) \approx +0.042$ (adapt), and $\Delta(2) \approx -0.048$ (freeze). Now take the simplest evidence map: $Z = \bar{X}_m = \frac{1}{m}\sum_{i \le m} X_i$, the batch mean, which is label-free. Then $Z \sim \mathcal{N}(\mu, 1/m)$: the evidence identifies μ consistently, and since $\operatorname{sign}\Delta$ is a known function of μ, the evidence is *risk-aligned* (Definition 3.3) — indeed two target worlds $\mu_1 < \frac{1}{2} < \mu_2$ with opposite benefit have evidence laws separating at total-variation rate

$2\Phi(\sqrt{m}\,|\mu_1-\mu_2|/2)-1 \to 1$. Every $\mu \neq \frac{1}{2}$ is knowably helpful or knowably harmful; the decision is computable from unlabeled data alone.

Example 3.2 (A posterior flip to which all evidence is blind)**.** Now keep the covariate marginal fixed and flip the labels' association with it. In both worlds, $Y \sim \text{Bernoulli}(\frac{1}{2})$ and the covariate is drawn from a symmetric two-component mixture: in world W^+, $X \mid Y{=}1 \sim \mathcal{N}(+1,1)$ and $X \mid Y{=}0 \sim \mathcal{N}(-1,1)$; in world W^- the two class-conditionals are swapped. Both worlds have the *identical* covariate marginal $p(x)=\frac{1}{2}\varphi(x-1)+\frac{1}{2}\varphi(x+1)$, where φ is the standard normal density; what differs is the posterior, which flips from $\mathbb{P}_{W^+}(Y{=}1 \mid x)=1/(1+e^{-2x})$ to $\mathbb{P}_{W^-}(Y{=}1 \mid x)=1/(1+e^{+2x})$. Take $f_0(x)=\mathbf{1}[x>0]$ and $f_a(x)=\mathbf{1}[x<0]$ with 0/1 loss. In world W^+, $R(f_0)=\Phi(-1)\approx 0.159$ and $R(f_a)=\Phi(1)\approx 0.841$, so $\Delta^+=-(2\Phi(1)-1)\approx -0.683$: freezing is correct, by a wide margin. In world W^- the computation mirrors and $\Delta^-=+(2\Phi(1)-1)\approx +0.683$: adapting is correct, by the same margin. Yet *every* evidence statistic $Z=\phi(X_{1:m},f_0,f_a)$ — not just a particular panel, but any measurable label-free map — has identical law in the two worlds, because it is a function of $X_{1:m}$ and the fixed models only and $\text{Law}(X_{1:m})$ is shared. The evidence is blind to a benefit of magnitude 0.683. This is the smoothed sibling of the exact witness that proves Theorem 4.1 (where labels are deterministic and $|\Delta|=1$); the overlap between the classes here shows that the phenomenon is not an artifact of noise-free labels.

It is instructive to contrast this with *genuine label shift*, where the class priors move but the class-conditionals stay fixed [31]. There the covariate marginal does shift (a prior flip $\pi \leftrightarrow 1-\pi$ reflects the marginal, $m_{1-\pi}(x)=m_\pi(-x)$), so some statistics can detect that *something* changed: sign-asymmetric features such as the predicted-class balance of f_0 move with π, while sign-symmetric panels — confidence $|x|$-margins, disagreement indicators, entropy — remain exactly blind. But detecting motion is not certifying a sign: the same movement of a class-balance statistic can be produced by a covariate shift under which the benefit keeps its sign, so converting the detection into a decision requires label-shift structure (stable class-conditionals and an invertible confusion matrix, as in black-box shift estimation [31]) — structure that is itself untestable from unlabeled target data. The knowability hierarchy of Chapter 5 makes this precise shift class by shift class.

Example 3.3 (The low-margin band)**.** Return to Example 3.1 and let the target location approach the crossover: $\mu=\frac{1}{2}+t$ with t small. A first-order expansion gives $\Delta(\frac{1}{2}+t)\approx -0.094\,t$ (the derivative of Δ at the crossover is $2\varphi(\frac{1}{2})-2\varphi(0)\approx -0.094$), so the benefit margin $|\Delta|$ collapses linearly as the condition nears the boundary. The evidence remains risk-aligned in the limit — with infinite data the sign is identifiable for every $t\neq 0$ — but a deployment batch of size m resolves μ only to precision $O(1/\sqrt{m})$, so conditions with $|t|\lesssim 1/\sqrt{m}$ are *effectively* undecidable at the operating sample size, and the benefit at stake there is only $|\Delta|\lesssim 0.1/\sqrt{m}$. This is the regime in which abstention is cheap by design: Theorem 4.3 shows that committal regret on the band $\{|\Delta|<\varepsilon\}$ is at most ε, and the finite-sample certificate of Chapter 5 abstains there automatically because its confidence interval $\widehat{\Delta}\pm\varepsilon$ straddles zero. In the phase diagram this is the lower wedge:

not blind, but not worth the risk of commitment either.

The three examples mark the three corners of the problem. In Example 3.1 the evidence separates opposite-benefit worlds and the margin is healthy: commit. In Example 3.2 no evidence map can separate them at any sample size: abstain, necessarily. In Example 3.3 separation is possible but the margin is too small to matter at the achievable resolution: abstain, economically. The theory of the next two chapters is the general statement of exactly these three verdicts.

3.7 Standing assumptions, and what each one buys

We collect the standing conventions of the framework as explicit assumptions. These are not new postulates — each is stated or used in the paper — but displaying them separately clarifies which results lean on which, and what breaks when each fails.

Assumption 3.1 (Fixed decision triple)**.** The loss ℓ, the frozen model f_0, and the candidate f_a are fixed, known, measurable objects at decision time; the risks $R_T(f_0)$ and $R_T(f_a)$ exist and are finite.

Assumption 3.2 (Label-free protocol)**.** No target labels are observed at any stage of the decision. Labeled source and held-out labeled validation data may be used to calibrate estimators and radii (Remark 3.1); the deployment batch contributes covariates and model outputs only.

Assumption 3.3 (Evidence-measurable rules)**.** Decision rules are measurable functions of the evidence Z and of randomness independent of the target world.

Assumption 3.1 is what makes the question well-posed as a *decision between two named alternatives*. If the candidate were allowed to change after the decision (for instance, if adaptation continued to update f_a on data the certificate never saw), the certified quantity Δ would no longer describe the deployed comparison; in the online experiments of Chapter 8 this is respected by re-running the decision at each adaptation step. The fixedness of (f_0, f_a) is also what makes the disagreement region D observable, on which the sign-of-difference theory of Chapter 5 lives. Boundedness of per-sample benefits — needed for the empirical-Bernstein radius [34, 22] inside the certificate — is a local hypothesis of Theorem 5.1, not a standing one.

Assumption 3.2 is the problem statement itself: with target labels the benefit could be estimated directly and the entire subject would collapse into ordinary validation. Its bite is asymmetric across the theory. The impossibility results of Chapter 4 hold *against* every rule satisfying it, so they are strongest when the allowance for source labels is most generous; the positive results consume the allowance explicitly — the certificate's radius is calibrated on a labeled validation sample assumed exchangeable with deployment conditions (a hypothesis stated and discharged in Chapter 5, in the conformal tradition [51]), and the covariate-shift theorem (Theorem 5.3) reweights labeled source points.

Question	Result	Chapter
Can any label-free rule decide correctly in all worlds?	Theorem 4.1 (with witness; Le Cam bound)	4
What does a forced commitment cost, exactly?	Theorem 4.3 (regret identity; minimax floor)	4
When can a rule commit with finite-sample guarantees?	Theorem 5.1 (certificate; forced abstention)	5
Can decisions stay valid on a stream with repeated looks?	Theorem 5.2 (anytime-valid e-process)	5
Is any shift class provably knowable?	Theorem 5.3 (covariate shift)	5
What single quantity determines sign Δ?	Theorem 5.4 (disagreement region)	5
What remains open?	Conjecture 5.1 (label-free bracketing)	5

Table 3.1: The reader's map from the questions raised by this chapter's formulation to the results that answer them. The first two rows draw the boundary of the possible (Chapter 4); the remaining rows populate the possible side of it (Chapter 5). The KGA algorithm (Chapter 6) operationalizes the certificate row, and the experiments (Chapters 7–8) measure every quantity defined here — benefit, coverage, false-adapt rate, regret-to-oracle — on real benchmarks.

Assumption 3.3 looks like bookkeeping but is the load-bearing wall of Chapter 4: a rule that is a function of Z inherits the law of Z, so two worlds with identical evidence laws force identical behavior — in distribution, exactly — whatever the rule's internal sophistication. Independent randomization buys nothing, because it is world-independent by construction.

Finally, *risk alignment* (Definition 3.3) is deliberately *not* a standing assumption. It is the property whose presence defines the knowable regimes and whose failure defines the unknowable one; assuming it globally would assume away the central question. The KGA algorithm of Chapter 6 needs it only on the conditions where it commits — where it fails, the certificate cannot fire and the rule abstains, which Chapter 5 shows is then mandatory (any rule with valid false-adapt and false-freeze control must abstain with probability at least $1 - 2\alpha$ on evidence shared by opposite-benefit worlds).

3.8 A reader's map from questions to results

The formulation poses a short list of natural questions; each is answered by exactly one result in the next two chapters. Table 3.1 is the itinerary.

One closing orientation. The order of the results is the logic of the subject: first the negative boundary, because without it every positive claim would be suspected of hiding an assumption; then the exact price of ignoring the boundary; then the certificates and positive regimes that live within it. Chapter 4 now takes up the first two rows of the map — the impossibility theorem with its explicit witness, its quantitative Le Cam strengthening, and the plug-in regret identity that prices every wrong commitment.

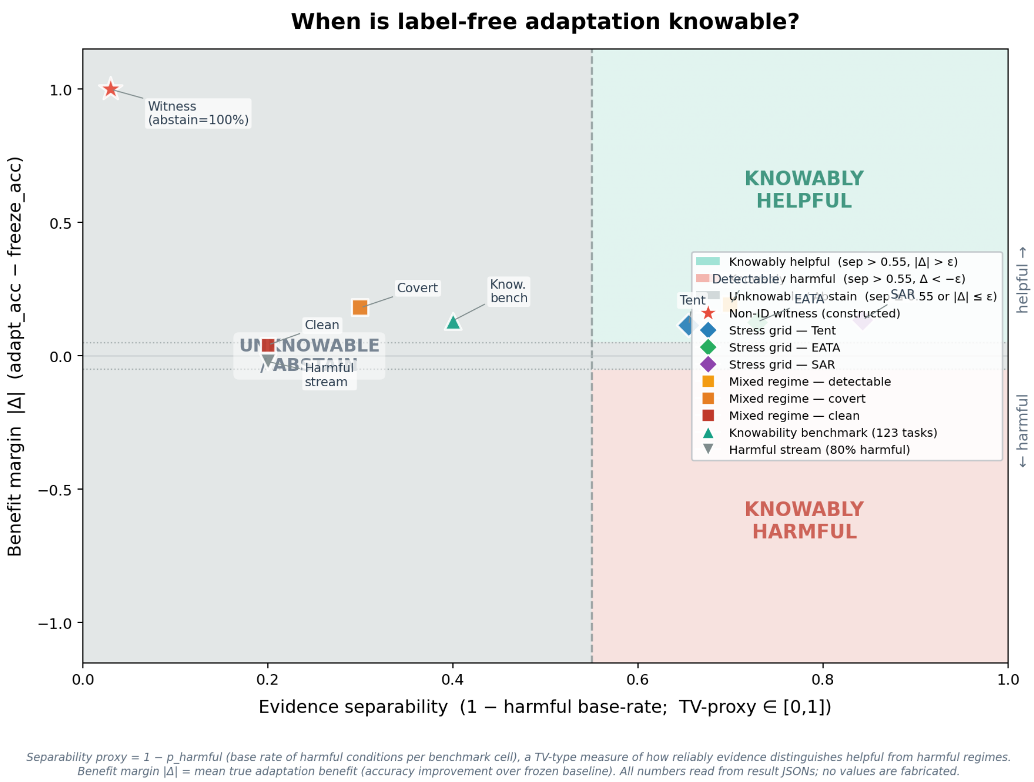

Figure 3.1: **The knowability phase diagram.** Each test-time condition lands at a point set by the evidence separability between opposite-benefit worlds (x-axis; a total-variation/detectability proxy) and the benefit margin $|\Delta|$ (y-axis). KGA commits (ADAPT/FREEZE) only outside the central *unknowable* wedge, where the certificate radius exceeds the estimated benefit ($\varepsilon > |\widehat{\Delta}|$); inside it the worst-case-optimal action is ABSTAIN (Theorem 4.1). Benchmark points — the CIFAR-10-C stress grid (Tent/EATA/SAR), the controlled mixed regime, and the clean non-identifiability witness (Chapters 7 and 8) — are overlaid from their measured values. The two axes of this diagram are exactly the two hypotheses of the impossibility theorem: the x-axis measures how far the evidence laws of opposite-benefit worlds separate (Chapter 4's total-variation quantity), and the y-axis measures how much a wrong commitment costs (the margin $|\Delta|$ in the regret identity of Theorem 4.3).

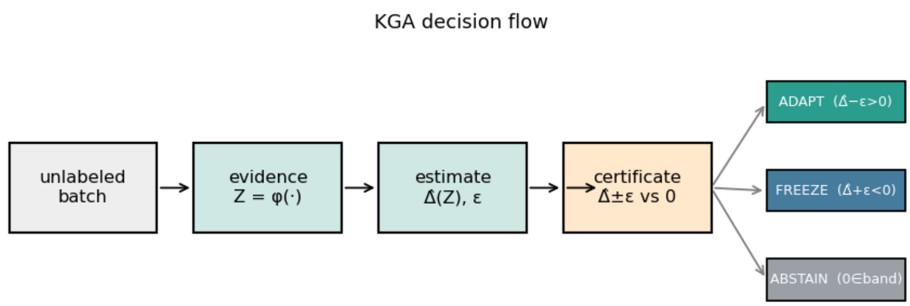

Figure 3.2: The KGA decision flow realizing the formulation of this chapter: label-free evidence $Z = \phi(X_{1:m}, f_0, f_a)$ is extracted from the unlabeled batch and the two fixed models; a calibrated benefit estimate $\widehat{\Delta}(Z)$ with conformal radius ε is formed; and the rule commits to ADAPT only if $\widehat{\Delta} - \varepsilon > 0$, to FREEZE only if $\widehat{\Delta} + \varepsilon < 0$, and otherwise ABSTAINs (keeping f_0 and flagging the condition). The three exits are the three regimes of Definition 3.4; the guarantees attached to each exit are proved in Chapters 4 and 5, and the algorithm itself is the subject of Chapter 6.

Chapter 4

The Unknowable Regime: Non-Identifiability and the Minimax Floor

The first half of the theory draws the boundary of the possible. This chapter proves two results. The first (Theorem 4.1, with its quantitative Le Cam strengthening, Theorem 4.2) shows that the unknowable regime of Definition 3.4 is not an artifact of weak evidence features or small batches: there exist pairs of target worlds whose label-free evidence laws coincide *exactly* — for every evidence map and every unlabeled sample size — while their adaptation benefits have opposite signs, and on such pairs the optimal committal error admits a closed-form floor, $1 - \mathrm{TV}$, which equals 1 at the witness. The second (Theorem 4.3) prices every violation of the boundary: the regret-to-oracle of any plug-in gate equals, *exactly*, the expected benefit forfeited on the evidence values where the estimated sign is wrong, which yields the matching minimax floor $\mathbb{E}[|\Delta(Z)|]/2$ for the unknowable regime and justifies abstention on the low-margin band.

The paper's running example sets the stakes. Two hospitals' scanners induce identical confidence and entropy statistics in a deployed classifier, yet in one hospital adapting helps and in the other — a flipped label frequency — it badly hurts; no label-free rule can separate them, so the safe action is to abstain. The mathematics below shows that this story is not a corner case but a face of the problem: whenever the posterior $P_T(Y \mid X)$ can move while the covariate law $P_T(X)$ does not, the evidence σ-algebra simply does not contain the answer. Everything in this chapter is proved against the strongest reading of the protocol of Chapter 3: arbitrary evidence maps, arbitrary (randomized) rules, unlimited unlabeled data, and free access to labeled source data.

We emphasize the chapter's intellectual posture at the outset, because the paper does. Non-identifiability of unlabeled target *risk* is, by itself, close to folklore — it is the reason the unsupervised risk estimation, domain adaptation, and label-shift literatures all carry structural assumptions [3, 33, 31]. What this chapter contributes is (i) the statement at the level of the *decision*, where it justifies a specific third action; (ii) an explicit, maximally separated witness pair on which every later experiment can be run; (iii) the exact minimax form of the floor via the Le Cam two-point method [28, 49]; and (iv) the exact regret identity that converts the floor into a deployment-relevant quantity. Each component is validated numerically by a machine-checked script in the public repository [25], and we quote those checks where they occur.

4.1 The non-identifiability theorem

We begin with the qualitative statement, copied verbatim from the paper.

Theorem 4.1 (Non-identifiability; with witness)**.** *Let $g(Z) \in \{\textsc{adapt}, \textsc{freeze}, \textsc{abstain}\}$ be any (possibly randomized) rule depending on label-free evidence Z. There exist two target worlds P_T^1, P_T^2 with* (i) $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$ *and* (ii) $\operatorname{sign} \Delta^1 = -\operatorname{sign} \Delta^2 \neq 0$. *For any such pair,* $\mathrm{Law}(g)$ *is identical across the two worlds; hence no label-free rule commits to the benefit-maximizing action in both, and the action minimizing the worst-case committal regret over $\{P_T^1, P_T^2\}$ is* ABSTAIN.

Before the proof, three reading notes. First, the theorem has an existence half and a universal half: it *exhibits* one pair of worlds satisfying (i)–(ii) (the witness), and then shows that for *every* pair satisfying (i)–(ii) — witness or otherwise — every rule is stuck. Second, the conclusion is about laws, not samples: $\mathrm{Law}(g)$ identical across worlds means the rule's behavior is indistinguishable *in distribution*, so no accounting trick (more batches, more features, more randomization) recovers a difference. Third, the optimality of ABSTAIN is a genuine minimax statement over a two-point family, in the classical mold of two-point lower bounds [28, 49]; the next section upgrades it to an exact quantitative floor.

Proof. The proof has four steps: construct the witness; verify properties (i) and (ii); show every label-free rule has world-independent law; and solve the resulting worst-case decision problem.

Step 1 (witness construction; non-vacuous). Let $X \sim \mathcal{N}(0, 1)$ under *both* worlds, so $P_T^1(X) = P_T^2(X)$. Take the fixed predictors $f_0(x) = \mathbf{1}[x > 0]$ and $f_a(x) = \mathbf{1}[x < 0]$, and 0/1 loss. Define the two conditional label laws degenerately:

$$P_T^1(Y \mid X)\colon\ Y = \mathbf{1}[X > 0], \qquad P_T^2(Y \mid X)\colon\ Y = \mathbf{1}[X < 0].$$

In world 1, $f_0(X) = \mathbf{1}[X > 0] = Y$ holds almost surely, so $R_T^1(f_0) = \mathbb{P}[f_0(X) \neq Y] = 0$; and since $\mathbb{P}[X = 0] = 0$, we have $f_a(X) = \mathbf{1}[X < 0] = 1 - Y$ almost surely, so $R_T^1(f_a) = 1$. Hence

$$\Delta^1 = R_T^1(f_0) - R_T^1(f_a) = 0 - 1 = -1 < 0,$$

and freezing is the benefit-maximizing action in world 1. World 2 is the mirror image: $f_a(X) = Y$ and $f_0(X) = 1 - Y$ almost surely, so $R_T^2(f_0) = 1$, $R_T^2(f_a) = 0$, and $\Delta^2 = +1 > 0$: adapting is correct. The benefit gap is maximal ($|\Delta^j| = 1$, the full range of 0/1 risk), so the construction is not a degenerate tie.

Step 2 (identical evidence laws). The evidence $Z = \phi(X_{1:m}, f_0, f_a)$ is, for any evidence map ϕ and any batch size m, a function of $X_{1:m}$ and of the fixed maps f_0, f_a only: writing $\psi(\cdot) := \phi(\cdot, f_0, f_a)$, we have $Z = \psi(X_{1:m})$ with ψ measurable and world-independent. The law of $X_{1:m}$ is $\mathcal{N}(0,1)^{\otimes m}$ in both worlds, so the pushforward laws coincide: $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(X_{1:m}) \circ \psi^{-1} = \mathrm{Law}(Z \mid P_T^2)$. This is property (i); property (ii) was verified in Step 1 ($\operatorname{sign} \Delta^1 = -1 = -\operatorname{sign} \Delta^2 \neq 0$). Thus (i) and (ii) hold simultaneously.

Step 3 (the rule's law is world-independent). A label-free rule is a measurable map of the evidence and of independent randomness: $g = \Gamma(Z, U)$, where U is a randomization variable whose law does not depend on the target world (it is generated by the system, not by the data), and Γ is measurable. Under world j the law of g is the pushforward of $\mathrm{Law}(Z \mid P_T^j) \otimes \mathrm{Law}(U)$ under Γ. By (i) the two product laws coincide, hence so do the laws of g. In particular the three numbers

$$q := \mathbb{P}[g = \textsc{adapt}], \qquad r := \mathbb{P}[g = \textsc{freeze}], \qquad 1 - q - r = \mathbb{P}[g = \textsc{abstain}]$$

are the *same* in both worlds.

Step 4 (no simultaneous correctness, and the worst-case-optimal action). The benefit-maximizing action is FREEZE in world 1 and ADAPT in world 2. A rule committing to the correct action with probability one in world 1 needs $r = 1$; in world 2 it needs $q = 1$; both cannot hold since $q + r \leq 1$. Hence no label-free rule commits to the benefit-maximizing action in both worlds. To identify the minimax action, measure committal regret as the excess 0/1 risk of a committed action versus the better committed action in that world, and assign abstention a separate coverage cost (it is a different kind of outcome — a flagged non-decision — and is priced separately; the quantitative theorem below makes this explicit). In world 1, committing to ADAPT costs $R_T^1(f_a) - R_T^1(f_0) = |\Delta^1| = 1$ and committing to FREEZE costs 0; so world 1 charges expected committal regret $q \cdot |\Delta^1| = q$. Symmetrically world 2 charges $r \cdot |\Delta^2| = r$. The worst case over the two-point family is

$$W(q, r) \;=\; \max(q,\, r),$$

to be minimized over the simplex $\{q, r \geq 0,\; q + r \leq 1\}$. Since $W(q, r) \geq 0$ with equality if and only if $q = r = 0$, the unique minimizer is $q = r = 0$, i.e. the rule that plays ABSTAIN with probability one. $\square$

The paper attaches an explicit calibration of the theorem's weight, which we reproduce because it governs how the result should be cited.

Remark 4.1 (Weight of the result). Theorem 4.1 is close to known non-identifiability of target risk without labels; its purpose here is to *justify the abstain action*, not to claim impossibility as novel. The quantitative strengthening below (Theorem 4.2) makes it a tight minimax statement.

Remark 4.2 (Both hypotheses are necessary). The two-point structure is sharp in both coordinates. If (i) fails — the evidence laws differ by total variation $\tau > 0$ — then Theorem 4.2 shows the optimal committal error drops to $1 - \tau < 1$, and a certificate may commit on the distinguishable part (Chapter 5 builds exactly that). If (ii) fails — the two worlds' benefits share a sign — then committing to the common benefit-maximizing action is correct in both worlds even though the evidence cannot tell them apart; indistinguishability is harmless when the decision does not depend on the distinction. The impossibility requires the conjunction: *the evidence cannot separate the worlds, and the worlds disagree about what to do.* The unknowable regime of Definition 3.4 is precisely the set of evidence values where some such pair lives.

4.2 The explicit witness, in full

Because the witness carries both the theorem's existence half and the paper's cleanest experiment, we devote a section to it: first the algebra in complete detail, then its interpretation, then the empirical realization.

4.2.1 Construction and verification

The witness consists of: covariate law $X \sim \mathcal{N}(0,1)$, shared by both worlds; predictors $f_0(x) = \mathbf{1}[x > 0]$ and $f_a(x) = \mathbf{1}[x < 0]$, fixed and known; loss 0/1; and the two degenerate posteriors $Y = \mathbf{1}[X > 0]$ (world 1) and $Y = \mathbf{1}[X < 0]$ (world 2). Three features deserve explicit verification beyond the proof above.

The risks, entry by entry. Under world 1, the event $\{f_0(X) \neq Y\} = \{\mathbf{1}[X > 0] \neq \mathbf{1}[X > 0]\} = \emptyset$, so $R^1_T(f_0) = 0$ exactly (not merely almost). The event $\{f_a(X) \neq Y\} = \{\mathbf{1}[X < 0] \neq \mathbf{1}[X > 0]\} = \{X \neq 0\}$ has probability one under the atomless Gaussian, so $R^1_T(f_a) = 1$. The four risks are

$$R^1_T(f_0) = 0, \quad R^1_T(f_a) = 1, \quad R^2_T(f_0) = 1, \quad R^2_T(f_a) = 0,$$

giving $\Delta^1 = -1$ and $\Delta^2 = +1$: the largest benefit gap the 0/1 scale permits, in opposite directions.

Evidence-law equality, for every ϕ and every m. The claim is not that some chosen panel of statistics happens to match across worlds; it is that the entire observable σ-algebra matches. Any label-free evidence is $Z = \psi(X_{1:m})$ for a fixed measurable ψ (the models being fixed known maps, they are absorbed into ψ), and equality of the laws of $X_{1:m}$ forces equality of the laws of Z — a pushforward of equal measures under one map. Consequently *no* statistic computable without target labels — no entropy histogram, no drift score, no disagreement pattern, no density-ratio diagnostic, nor any statistic anyone will ever invent — distinguishes the worlds at any batch size. The quantifier order is the point: the construction defeats all evidence maps simultaneously, not a fixed one.

What kind of shift this is. Between the two worlds the covariate law is untouched and the posterior flips: $P^1_T(Y{=}1 \mid x) = \mathbf{1}[x > 0]$ versus $P^2_T(Y{=}1 \mid x) = \mathbf{1}[x < 0]$. In the taxonomy of shifts this is pure concept shift [39], the row of the knowability hierarchy that Chapter 5 marks non-identifiable; Example 3.2 of Chapter 3 is its smoothed sibling, with overlapping classes and $|\Delta| = 2\Phi(1) - 1 \approx 0.683$ in place of the extremal $|\Delta| = 1$, showing the phenomenon survives label noise. It is also instructive to view the witness through the disagreement-region lens of Theorem 5.4: here the observable disagreement region is $D = \{x : f_0(x) \neq f_a(x)\} = \{x \neq 0\}$, essentially the whole line, and the candidate's accuracy on it is $a_a^D = 0$ in world 1 and $a_a^D = 1$ in world 2 — the identity sign $\Delta = \operatorname{sign}(a_a^D - \frac{1}{2})$ holds in both worlds, and what the witness makes impossible is precisely any label-free *bracketing* of a_a^D around $\frac{1}{2}$. The impossibility and the sign-of-difference characterization are two views of one boundary.

4.2.2 Empirical realization

The witness is constructive, so it can be run. The paper reports a clean witness experiment (the full experimental protocol appears in Chapter 7): two simulated worlds with $X \sim \mathcal{N}(0, 1)$, $f_0 = \mathbf{1}[X>0]$, $f_a = \mathbf{1}[X<0]$, and flipped labels, so the benefit is exactly opposite (world means -1.0 versus $+1.0$) while every evidence feature is a function of X only. Across 300 instances, all five evidence features used by KGA are statistically indistinguishable between the worlds — two-sample Kolmogorov–Smirnov p-values in $[0.16, 0.96]$, all above 0.05 — yet the ground truth is opposite, so Z cannot decide. KGA abstains on 100% of instances, and a rule forced to commit by sign $\widehat{\Delta}$ pays regret 0.51, near chance. This is the clean theory–experiment bridge (Figure 4.1).

The repository's theorem validator (`experiments/kbound/theory_validation/val_thm1_lecam.py`, results in `results_thm1_lecam.json` [25]) re-runs the witness at scale with 20,000 Monte-Carlo batches of $n = 64$ unlabeled samples per world and reports four independent checks. (1) The closed-form benefits $\Delta^1 = -1$, $\Delta^2 = +1$ are reproduced exactly by Monte Carlo. (2) For each of five per-sample evidence features (the raw covariate, the margin $|x|$, the predicted class of f_0, the disagreement indicator, and a smooth confidence transform), the permutation-debiased total-variation estimate between the two worlds' evidence laws is at most 1.9×10^{-3} — zero up to the estimator's own null floor — with two-sample KS p-values between 0.28 and 1.0; and the n-sample aggregated evidence statistic has exact TV 0 for every n in $\{1, 2, 4, \ldots, 256\}$ (debiased estimates $\leq 4.8 \times 10^{-3}$). (3) Optimizing the committal error over a dense family of threshold rules (all thresholds, both polarities) on each evidence statistic yields $\inf_g M(g)$ between 0.991 and 1.000 across the five statistics, against the theoretical floor of 1. (4) The actual KGA conformal three-way rule, run exactly as in the paper with $\alpha = 0.1$, abstains on 4,000 of 4,000 instances (100% abstention; zero false adapts and zero false freezes): its calibrated radius ($\varepsilon \approx 1.15$) straddles zero because the evidence carries no benefit signal, so the certificate cannot fire — the behavior that Chapter 5 proves is mandatory for any rule with valid error control. The paper's summary sentence is exact: in the witness the empirical $\inf_g M$ is ≈ 1 (KGA abstains 100%); in the detectable regime the empirical minimax error tracks the closed-form $1 - \mathrm{TV}$ across all n and shift magnitudes.

4.3 The quantitative floor: a Le Cam two-point bound

Theorem 4.1 says committal rules fail on matched evidence; it does not yet say how badly, nor what happens as the evidence laws begin to separate. The paper's quantitative strengthening

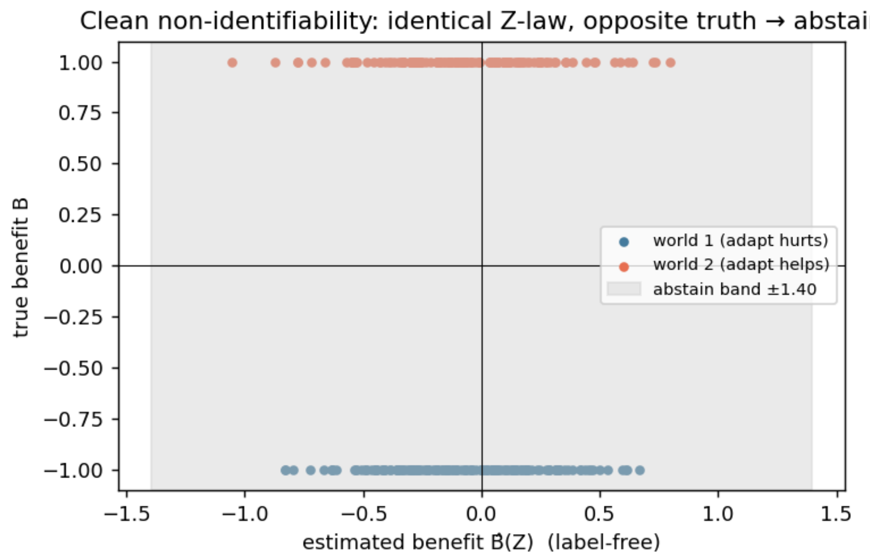

Figure 4.1: The clean non-identifiability witness of Theorem 4.1, realized empirically: identical evidence law across the two worlds (two-sample KS $p > 0.05$ on all five evidence features), exactly opposite true benefit (-1.0 versus $+1.0$), and 100% abstention by the KGA rule; a rule forced to commit pays regret 0.51, near chance. The experiment instantiates hypotheses (i) and (ii) of the theorem exactly rather than approximately, which is what makes it a *witness* rather than a hard benchmark.

answers both with an exact identity, in the two-point tradition of Le Cam [28]; the intuition, in the paper's words: the more alike two such worlds look in the observable evidence (smaller TV distance), the closer any rule forced to commit is pushed toward a coin flip, and the error floor vanishes only as their evidence separates.

We first isolate the two classical ingredients as lemmas, with proofs, since the theorem's proof is a direct composition of them.

Lemma 4.1 (Testing affinity identity)**.** *Let Q_1, Q_2 be probability measures on a common measurable space. Over all randomized tests ψ (measurable functions with values in $[0,1]$),*

$$\inf_{\psi} \{\mathbb{E}_{Q_1}[\psi] + \mathbb{E}_{Q_2}[1-\psi]\} \;=\; 1 - \mathrm{TV}(Q_1, Q_2),$$

and the infimum is attained by the deterministic likelihood-ratio test $\psi^\star = \mathbf{1}[\mathrm{d}Q_2 > \mathrm{d}Q_1]$.

Proof. Let $\nu = Q_1 + Q_2$, which dominates both measures, and let q_1, q_2 be the corresponding densities. For any test ψ,

$$\mathbb{E}_{Q_1}[\psi] + \mathbb{E}_{Q_2}[1-\psi] = \int \psi\, q_1 \,\mathrm{d}\nu + \int (1-\psi)\, q_2 \,\mathrm{d}\nu = 1 + \int \psi\,(q_1 - q_2)\,\mathrm{d}\nu,$$

using $\int q_2 \,\mathrm{d}\nu = 1$. The integrand is minimized pointwise over $\psi(z) \in [0,1]$ by setting $\psi = 1$ where $q_1 - q_2 < 0$, $\psi = 0$ where $q_1 - q_2 > 0$, and arbitrarily on the crossing set $\{q_1 = q_2\}$ (which contributes zero); this is exactly $\psi^\star = \mathbf{1}[q_2 > q_1]$, a deterministic test, so randomization buys nothing. The attained value is

$$1 - \int (q_2 - q_1)_+ \,\mathrm{d}\nu.$$

It remains to identify the integral with total variation. For any measurable A, $Q_2(A) - Q_1(A) = \int_A (q_2 - q_1)\,\mathrm{d}\nu \le \int (q_2 - q_1)_+ \,\mathrm{d}\nu$, with equality at $A^\star = \{q_2 > q_1\}$; hence $\mathrm{TV}(Q_1, Q_2) := \sup_A |Q_1(A) - Q_2(A)| = \int (q_2 - q_1)_+ \,\mathrm{d}\nu$, which also equals $\frac{1}{2}\int |q_1 - q_2|\,\mathrm{d}\nu$ because $\int (q_2 - q_1)\,\mathrm{d}\nu = 0$ forces the positive and negative parts to have equal mass. Substituting gives the identity [28, 49, 29]. □

Lemma 4.2 (Data-processing inequality for total variation)**.** *Let P_1, P_2 be probability measures and T a measurable map. Then $\mathrm{TV}\big(P_1 \circ T^{-1},\, P_2 \circ T^{-1}\big) \le \mathrm{TV}(P_1, P_2)$, with equality whenever T is a sufficient statistic for the pair $\{P_1, P_2\}$.*

Proof. For the inequality, every event of the pushforward laws is the preimage of an event: $\sup_A |P_1(T^{-1}A) - P_2(T^{-1}A)|$ is a supremum over the sub-collection $\{T^{-1}A\}$ of measurable sets, hence at most the supremum over all measurable sets, which is $\mathrm{TV}(P_1, P_2)$. For the equality claim: if T is sufficient for $\{P_1, P_2\}$, the likelihood ratio $\mathrm{d}P_2/\mathrm{d}P_1$ is (P_1-a.s.) a function of T by the factorization theorem [29], so the optimal set $A^\star = \{\mathrm{d}P_2 > \mathrm{d}P_1\}$ realizing the supremum in Lemma 4.1's identification of TV is itself of the form $T^{-1}(B)$, and the restricted supremum already attains the unrestricted one. □

With the two lemmas in hand we state the paper's theorem verbatim.

Theorem 4.2 (Quantitative non-identifiability; Le Cam two-point lower bound)**.** *Consider two target worlds P_T^1, P_T^2 with $\Delta^1 < 0 < \Delta^2$. Let the label-free evidence be $Z = \phi(X_{1:n}, f_0, f_a)$ with n-sample evidence laws $P_{T,Z}^{j,(n)} := \mathrm{Law}(Z \mid P_T^j)$. For any (randomized) committal rule $g(Z) \in \{\textsc{adapt}, \textsc{freeze}\}$ define the mixed committal error $M(g) := \mathbb{P}_{P_T^1}[g = \textsc{adapt}] + \mathbb{P}_{P_T^2}[g = \textsc{freeze}]$. Then*

$$\inf_g M(g) \;=\; 1 - \mathrm{TV}\big(P_{T,Z}^{1,(n)}, P_{T,Z}^{2,(n)}\big) \;\ge\; 1 - \mathrm{TV}\big(P_T^{1,(n)}, P_T^{2,(n)}\big).$$

In the witness above (Z a function of X, $P_T^1(X) = P_T^2(X)$) the evidence TV is 0, so $\inf_g M(g) = 1$ for every *n: no committal rule beats blind guessing in the worst world, and under any positive coverage cost* ABSTAIN *is uniquely worst-case optimal. If the worlds become δ-detectable so the evidence laws separate, $\inf_g M(g) = 1 - \mathrm{TV} \to 0$ at the separation rate (for a location shift summarized by a sufficient statistic, $\mathrm{TV} = 2\Phi(\frac{1}{2}\sqrt{n}\,\delta) - 1$).*

Proof. We expand the paper's proof into five steps.

Step 1 (reduction to binary testing). A randomized committal rule on Z is characterized by its adapt-probability $\psi(z) := \mathbb{P}[g = \textsc{adapt} \mid Z = z] \in [0, 1]$, a randomized test of the two evidence laws. Identify $\{g = \textsc{adapt}\}$ with deciding world 2 (where adapting is correct, since $\Delta^2 > 0$).

Writing $Q_j := P^{j,(n)}_{T,Z}$, the mixed committal error is

$$M(g) = \mathbb{P}_{Q_1}[g = \textsc{adapt}] + \mathbb{P}_{Q_2}[g = \textsc{freeze}] = \mathbb{E}_{Q_1}[\psi(Z)] + \mathbb{E}_{Q_2}[1 - \psi(Z)],$$

the sum of the test's type-I and type-II errors. The infimum of M over committal rules is therefore the infimum of the testing risk over randomized tests.

Step 2 (the equality). Lemma 4.1 applied to Q_1, Q_2 gives $\inf_g M(g) = 1 - \mathrm{TV}(Q_1, Q_2)$, attained by the likelihood-ratio rule that plays ADAPT exactly on $\{\mathrm{d}Q_2 > \mathrm{d}Q_1\}$. Note this is an identity, not merely a bound: the optimal committal rule's mixed error *equals* one minus the evidence separation. This is the tightness direction — no slack is lost between the decision problem and the total-variation geometry, because the optimal test is exhibited.

Step 3 (the inequality to the full data). The evidence is a measurable reduction of the raw unlabeled batch: $Z = T(X_{1:n})$ with $T := \phi(\cdot, f_0, f_a)$. By Lemma 4.2, $\mathrm{TV}(Q_1, Q_2) \le \mathrm{TV}\big(P^{1,(n)}_T, P^{2,(n)}_T\big)$, the total variation between the worlds' n-sample *covariate-data* laws. Negating, $1 - \mathrm{TV}(Q_1, Q_2) \ge 1 - \mathrm{TV}(P^{1,(n)}_T, P^{2,(n)}_T)$: summarizing data can only raise the error floor. Equality holds when Z is sufficient for the two-world testing problem — the evidence then loses nothing.

Step 4 (the witness: floor 1 at every n). In the witness, Z depends on the worlds only through $\mathrm{Law}(X_{1:n})$, and these laws coincide; hence $Q_1 = Q_2$, $\mathrm{TV}(Q_1, Q_2) = 0$, and $\inf_g M(g) = 1$ for every n. Unpacking what $M(g) = 1$ forces: since $M(g) = \mathbb{P}_1[\mathrm{err}] + \mathbb{P}_2[\mathrm{err}] \ge 1$ for every committal g, at least one world sees error probability $\ge \frac{1}{2}$ — no committal rule beats blind guessing in its worst world, and this does not improve with n. Now add a coverage cost $c \in (0, \frac{1}{2})$ for abstention, and recall from the witness that a wrong commitment costs $|\Delta^j| = 1$. A rule with commitment probabilities (q, r) (Step 3 of Theorem 4.1's proof; world-independent by matched evidence) has worst-case expected cost

$$\max\big(q + (1-q-r)c,\ r + (1-q-r)c\big) \;=\; \max(q, r) + (1-q-r)\,c \;\ge\; \tfrac{q+r}{2} + (1-q-r)\,c,$$

which, as a function of the total commitment mass $s = q + r \in [0, 1]$, is $\ge s/2 + (1-s)c$, strictly increasing in s because $c < \frac{1}{2}$. The minimum is at $s = 0$, value c: always-ABSTAIN is the unique worst-case-optimal rule.

Step 5 (the detectable rate). Suppose the worlds are δ-detectable through a real-valued per-sample summary with unit noise scale: the summary is Gaussian with means separated by δ and variance 1 (the canonical location family; in the validator below, the worlds are $P^1(X) = \mathcal{N}(-\delta/2, 1)$ and $P^2(X) = \mathcal{N}(+\delta/2, 1)$). The sample mean $\bar{X}_n$ is sufficient for the location parameter, so by Lemma 4.2 the evidence loses nothing and $\mathrm{TV}(Q_1, Q_2) = \mathrm{TV}\big(\mathcal{N}(-\frac{\delta}{2}, \frac{1}{n}), \mathcal{N}(+\frac{\delta}{2}, \frac{1}{n})\big)$. For two equal-variance Gaussians $\mathcal{N}(\theta_1, \sigma^2)$, $\mathcal{N}(\theta_2, \sigma^2)$ the densities cross exactly at the midpoint $(\theta_1 + \theta_2)/2$, so the positive part integrates to $\mathrm{TV} = \Phi\big(\frac{|\theta_1 - \theta_2|}{2\sigma}\big) - \Phi\big(-\frac{|\theta_1 - \theta_2|}{2\sigma}\big) = 2\Phi\big(\frac{|\theta_1 - \theta_2|}{2\sigma}\big) - 1$. With $|\theta_1 - \theta_2| = \delta$ and $\sigma = 1/\sqrt{n}$ this is $\mathrm{TV} = 2\Phi(\frac{1}{2}\sqrt{n}\,\delta) - 1$, whence $\inf_g M(g) = 1 - \mathrm{TV} = 2\Phi(-\frac{1}{2}\sqrt{n}\,\delta) \to 0$ as $\sqrt{n}\,\delta \to \infty$: the committal-error floor decays at the Gaussian separation rate, and the unknowable regime shrinks exactly as fast as the evidence separates. $\square$

Remark 4.3 (Where the bound is and is not tight). Two distinct tightness claims should be kept apart. The first equality, $\inf_g M = 1 - \mathrm{TV}$ *on the evidence laws*, is always exact (Lemma 4.1 exhibits the optimal test); there is no asymptotic or constant slack anywhere in the chain from decision problem to total-variation geometry. The second relation, the inequality to the *full-data* TV, is an inequality precisely because a poorly chosen evidence map can discard separation that the raw batch contains; it collapses to equality when Z is sufficient for the two-world problem (Lemma 4.2), as in the location family of Step 5. The design lesson for evidence panels is contained in that gap: the floor an operator actually faces is $1 - \mathrm{TV}(\text{evidence})$, which the operator partially controls through ϕ, bounded below by $1 - \mathrm{TV}(\text{full data})$, which nature controls. In the witness nature sets the latter to 1 and the choice of ϕ is irrelevant.

4.3.1 Numerical validation of the floor

The validator `val_thm1_lecam.py` [25] checks the identity $\inf_g M(g) = 1 - \mathrm{TV}$ in both regimes, using 20,000 Monte-Carlo evidence instances per world ($n = 64$ unlabeled samples each, seed 12345) and optimizing committal rules over a dense threshold-and-polarity family on each evidence statistic.

In the *witness* regime the results were quoted in Section 4.2.2: empirical floors 0.991–1.000 against the theoretical 1, exact n-sample evidence TV equal to 0 at every n, and 100% KGA abstention. In the *detectable* regime, $P^1(X) = \mathcal{N}(-\mu, 1)$ and $P^2(X) = \mathcal{N}(+\mu, 1)$ with opposite benefits, the closed form of Step 5 (with $\delta = 2\mu$) is traced across both the shift magnitude and the sample size. Sweeping μ at $n = 64$: at $\mu = 0.1$ the exact floor is $1 - \mathrm{TV} = 0.4237$ and the best empirical committal error is 0.4189; at $\mu = 0.25$, floor 0.0455 versus empirical 0.0451; at $\mu = 0.5$ the floor is 6.3×10^{-5} and the empirical error is 0.0. Sweeping n at fixed $\mu = 0.25$, the (floor, empirical) pairs are (0.8026, 0.8134) at $n = 1$, (0.6171, 0.6055) at $n = 4$, (0.3173, 0.3209) at $n = 16$, (0.0455, 0.0452) at $n = 64$, and (6.3×10^{-5}, 0.0) at $n = 256$ — the empirical optimum tracks the closed form throughout, fluctuating around it in both directions by Monte-Carlo error (the threshold is optimized on the same finite sample, so small dips below the population floor are expected and honest). Meanwhile the KGA three-way rule abstains on 100% of instances at $\mu \in \{0, 0.1\}$, on 0.45% at $\mu = 0.25$, and on 0% at $\mu \geq 0.5$ with zero committal error: the rule stops abstaining exactly as the worlds become detectable, which is the operational content of the phase diagram's x-axis (Figure 3.1).

4.4 The Bayes gate and the plug-in regret identity

The impossibility results bound what *cannot* be decided; the second result of this chapter prices decisions exactly, and in doing so converts the abstract floor into deployed risk. The objects are the action-conditional risks of Chapter 3: with $a_{\mathrm{FRZ}}(z) = \mathbb{E}[\ell(f_0(X), Y) \mid Z{=}z]$ and $a_{\mathrm{ADP}}(z) = \mathbb{E}[\ell(f_a(X), Y) \mid Z{=}z]$, so that $\Delta(z) = a_{\mathrm{FRZ}}(z) - a_{\mathrm{ADP}}(z)$, a gate $g\colon z \mapsto \{0, 1\}$ (encoding 1 = ADAPT,

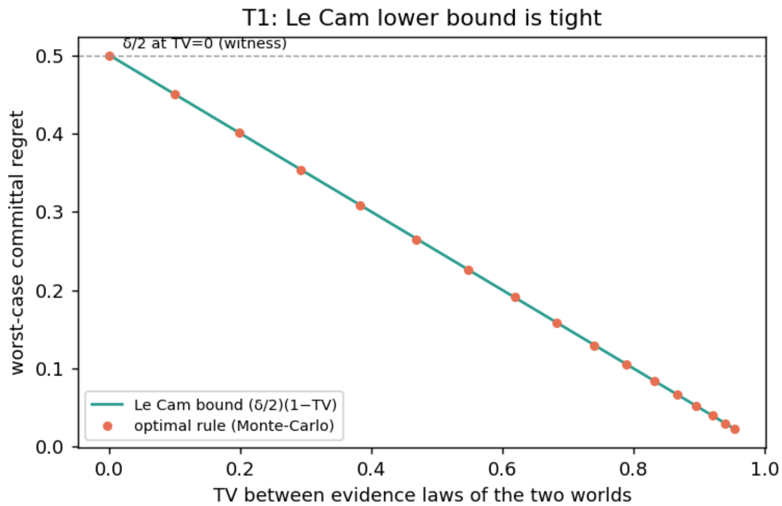

Figure 4.2: Theorem 4.2 in measurement: the optimal committal rule's worst-case regret coincides with the Le Cam floor $\frac{\delta}{2}(1 - \mathrm{TV})$ across the total-variation sweep; at $\mathrm{TV} = 0$ (the witness) it equals $\delta/2$, the minimax floor of Proposition 4.1 with benefit gap δ. Produced by `scripts/theory_extensions_validation.py` [25]. The plot is the bridge between this section and the next: the testing floor $1 - \mathrm{TV}$ (this section) multiplies the benefit margin (next section) to give worst-case regret.

$0 = $ FREEZE) has risk

$$R(g) \;=\; \mathbb{E}\big[g(Z)\, a_{\mathrm{ADP}}(Z) + (1 - g(Z))\, a_{\mathrm{FRZ}}(Z)\big].$$

The Bayes gate is $g^\star(z) = \mathbf{1}[\Delta(z) > 0]$; a plug-in gate built from any label-free benefit estimator $\widehat{\Delta}$ is $\widehat{g}(z) = \mathbf{1}[\widehat{\Delta}(z) > 0]$. The paper's intuition for what follows: a gate's excess risk over the oracle equals the benefit it forfeits exactly where it picks the wrong sign — near-zero-benefit mistakes are cheap, large-benefit ones are not, which is why abstaining in the low-margin band is safe.

Theorem 4.3 (Plug-in regret decomposition)**.** *For every estimator $\widehat{\Delta}$,*

$$R(\widehat{g}) - R(g^\star) \;=\; \mathbb{E}\big[\,|\Delta(Z)|\,\mathbf{1}\{\operatorname{sign}\widehat{\Delta}(Z) \neq \operatorname{sign}\Delta(Z)\}\,\big],$$

where $\operatorname{sign}(t) = \mathbf{1}[t > 0]$*. In particular $g^\star$ minimizes R, regret is zero iff the gate's sign matches a.e., and on the boundary band $\{|\Delta| < \varepsilon\}$ the committal regret is $\leq \varepsilon$—justifying* ABSTAIN *there.*

Proof. We expand the paper's one-paragraph proof into its pointwise case analysis, the integration step, and the three consequences.

Step 1 (the conditional risk is affine in the action). Condition on $Z = z$ and write $\rho(a; z)$ for the conditional risk of playing action $a \in \{0, 1\}$ at z: by definition of the action-conditional means,

$\rho(1;z) = a_{\text{ADP}}(z)$ and $\rho(0;z) = a_{\text{FRZ}}(z)$, so

$$\rho(1;z) - \rho(0;z) \;=\; a_{\text{ADP}}(z) - a_{\text{FRZ}}(z) \;=\; -\Delta(z) :$$

adapting lowers conditional risk by exactly $\Delta(z)$ (raises it when $\Delta(z) < 0$). The total risk of any gate is the average of its pointwise choices, $R(g) = \mathbb{E}_Z[\rho(g(Z);Z)]$, by the tower property.

Step 2 (pointwise excess of the plug-in gate). Fix z and compare $\widehat{g}(z)$ with $g^\star(z) = \mathbf{1}[\Delta(z) > 0]$. There are three cases. If $\widehat{g}(z) = g^\star(z)$, the excess $\rho(\widehat{g}(z);z) - \rho(g^\star(z);z)$ is 0. If $\widehat{g}(z) = 1$ but $g^\star(z) = 0$ — the gate adapts although $\Delta(z) \le 0$ — the excess is $\rho(1;z) - \rho(0;z) = -\Delta(z) = |\Delta(z)|$, using $\Delta(z) \le 0$. If $\widehat{g}(z) = 0$ but $g^\star(z) = 1$ — the gate freezes although $\Delta(z) > 0$ — the excess is $\rho(0;z) - \rho(1;z) = \Delta(z) = |\Delta(z)|$. In all cases,

$$\rho(\widehat{g}(z);z) - \rho(g^\star(z);z) \;=\; |\Delta(z)|\; \mathbf{1}\{\widehat{g}(z) \ne g^\star(z)\},$$

and since $\widehat{g}(z) = \mathbf{1}[\widehat{\Delta}(z) > 0]$ and $g^\star(z) = \mathbf{1}[\Delta(z) > 0]$, the disagreement event $\{\widehat{g}(z) \ne g^\star(z)\}$ is exactly the sign-mismatch event $\{\operatorname{sign}\widehat{\Delta}(z) \ne \operatorname{sign}\Delta(z)\}$ under the stated convention $\operatorname{sign}(t) = \mathbf{1}[t > 0]$. (Ties $\Delta(z) = 0$ are immaterial: whatever the convention assigns them, they enter the right-hand side multiplied by $|\Delta(z)| = 0$.)

Step 3 (integrate, and read off the consequences). Taking $\mathbb{E}_Z$ of the pointwise identity gives the displayed equation exactly — it is an identity, not an inequality, valid for every estimator, every evidence map, and every loss with finite conditional means; no boundedness, continuity, or distributional assumption enters. The three consequences: (a) running the same two steps with an arbitrary gate g in place of $\widehat{g}$ shows $R(g) - R(g^\star) = \mathbb{E}[|\Delta(Z)|\mathbf{1}\{g \ne g^\star\}] \ge 0$, so $g^\star$ minimizes R over all gates; (b) the regret vanishes iff the mismatch indicator vanishes Δ-weighted almost everywhere, i.e. iff the gate's sign matches that of $\Delta(Z)$ a.e. on $\{\Delta(Z) \ne 0\}$; (c) on the boundary band $\{|\Delta(Z)| < \varepsilon\}$, the pointwise excess is $|\Delta(z)|\,\mathbf{1}\{\cdot\} < \varepsilon$, so any committal choice there — right or wrong — forfeits less than ε of risk; abstaining on the band is therefore safe at price at most ε, which is the decision-theoretic justification for the certificate's abstention region in Chapter 5. □

The identity composes with the matched-evidence situation of Theorem 4.1 to give the floor in regret units; this is the paper's proposition, stated verbatim.

Proposition 4.1 (Label-free minimax floor)**.** *In the unknowable two-point regime of Theorem 4.1 (*$\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$*,* $|\Delta^1| = |\Delta^2| = |\Delta|$*, opposite signs), every label-free estimator has worst-case expected regret* $\ge \mathbb{E}[|\Delta(Z)|]/2$*, attained by abstaining.*

Proof. Let $\widehat{\Delta}$ be any label-free estimator, $\widehat{g}$ its plug-in gate, and allow the gate to be randomized: define $q(z) = \mathbb{P}[\widehat{g}(z) = 1 \mid Z = z]$, the adapt-probability at evidence z. Because the estimator and any internal randomization are label-free, and the evidence law is the *same* measure under both worlds, the function q is common to the two worlds — this is Step 3 of Theorem 4.1's proof applied conditionally at each z. Now evaluate the sign-error probability at z in each world: in world 1 the benefit is negative, so the gate errs exactly when it adapts, with probability $q(z)$; in world 2 the benefit is positive, so the gate errs exactly when it freezes, with probability $1 - q(z)$. The two error probabilities sum to 1 at every z, so the larger is $\geq \frac{1}{2}$. By Theorem 4.3, extended to randomized gates by averaging the pointwise identity over the gate's internal randomness, the regret in world j is $\mathrm{reg}_j = \mathbb{E}\big[|\Delta(Z)| \cdot \mathbb{P}(\text{sign error at } Z \text{ in world } j)\big]$; hence

$$\mathrm{reg}_1 + \mathrm{reg}_2 = \mathbb{E}\big[|\Delta(Z)|\,(q(Z) + 1 - q(Z))\big] = \mathbb{E}\big[|\Delta(Z)|\big],$$

and therefore $\max_j \mathrm{reg}_j \geq \frac{1}{2}(\mathrm{reg}_1 + \mathrm{reg}_2) = \mathbb{E}[|\Delta(Z)|]/2$. The floor is attained: the maximally non-committal allocation $q \equiv \frac{1}{2}$ — the gate-level expression of abstention, declining to favor either sign — has $\mathrm{reg}_1 = \mathrm{reg}_2 = \mathbb{E}[|\Delta(Z)|]/2$, so its worst case equals the floor exactly. In the three-action protocol the ABSTAIN action realizes the same worst-case value through its coverage accounting, which is the sense of the statement's final clause. □

Remark 4.4 (Reading the floor). The proposition is the regret-denominated shadow of Theorem 4.2 at $\mathrm{TV} = 0$: the testing floor says the sign error must be $\geq \frac{1}{2}$ in some world, and the regret identity says each unit of sign error costs $|\Delta(Z)|$. At intermediate separations the same composition gives worst-case regret of order $\frac{|\Delta|}{2}(1 - \mathrm{TV})$, which is exactly the curve traced by the validation plot of Figure 4.2. Note also what the floor does *not* say: it does not make abstention free. Abstention forfeits $\mathbb{E}[|\Delta|]/2$ of worst-case benefit too — the floor is a statement that nothing label-free can do better on matched evidence, not that abstaining is costless. The value of abstention is realized only when the system can *also* commit on the conditions where evidence does separate, which is the certificate's job in Chapter 5.

4.4.1 Numerical validation of the identity

Because Theorem 4.3 is an exact identity, its validator (`experiments/kbound/theory_validation/val_thm2_regr` [25]) can demand machine precision rather than statistical agreement, and it does, in two layers. Layer (A), the literal content of the theorem: on $n = 400{,}000$ sampled evidence points with known conditional benefit $\Delta(z) \sim \mathcal{N}(0, 1)$ and a non-trivial action-conditional baseline $a_{\mathrm{ADP}}(z)$ (so the cancellation in $R(\widehat{g}) - R(g^\star)$ is exercised, not assumed), it computes the left side from the conditional action risks and the right side from the mismatch identity, for plug-in estimators $\widehat{\Delta} = \Delta + \text{noise}$ across seven noise levels from 0 to 2.0; the script *asserts* that the maximum absolute gap over all noise levels is below 10^{-9}, and the measured gap is $\sim 10^{-17}$ — pure floating-point rounding, as the paper reports ("the identity holds to $\sim 10^{-17}$"). Layer (B): the same comparison with *realized* per-sample losses drawn around the conditional means, confirming the identity at the accuracy Monte Carlo permits (all cells within 4 standard errors).

The validator also checks each clause of the theorem and the proposition. At zero estimator noise the gates agree everywhere and both sides are exactly 0 (regret zero iff signs match a.e.). On the boundary band $\{|\Delta| < \varepsilon\}$ with $\varepsilon = 0.05$, the maximum per-z regret contribution observed is

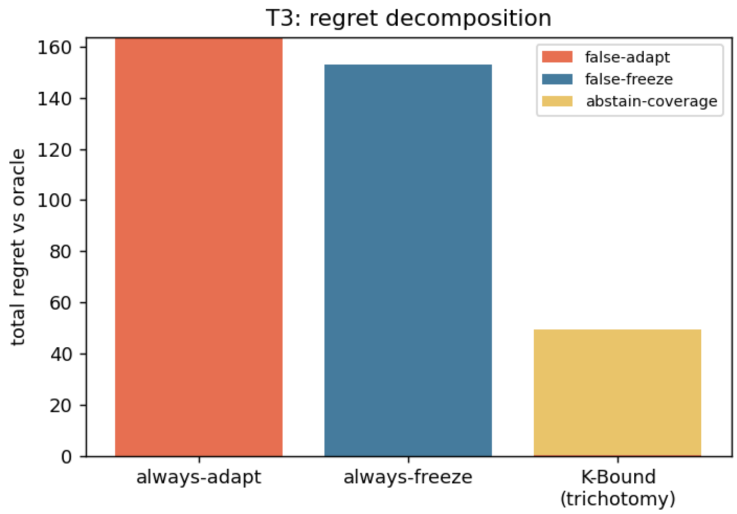

Figure 4.3: Synthetic illustration of the regret split (Theorem 4.3 / Proposition 4.1): total regret decomposes into false-adapt, false-freeze, and abstain-coverage contributions; the trichotomy's total is far below both trivial policies and is dominated by benign abstain-coverage rather than harmful false-adapt. Synthetic data; produced by `scripts/theory_extensions_validation.py` [25]. The decomposition is the deployment-facing restatement of the regret identity: every unit of regret is attributable to a named decision error, weighted by the benefit at stake.

below ε, the "abstaining there costs at most ε" clause. Regret rises monotonically with estimator noise, and large noise approaches — but does not reach — the coin-flip ceiling; the ceiling itself is attained by a fully uninformative estimator (independent of Δ, the empirical surrogate of the matched-evidence regime), whose sign-mismatch rate is $\approx\frac{1}{2}$ and whose regret is 0.3986 against the floor $\mathbb{E}[|\Delta|]/2 = 0.3985$ — Proposition 4.1 realized to three decimal places.

4.5 Remarks: scope, mechanism, and what abstention buys

Remark 4.5 (Relation to selective prediction)*.* The reject option is as old as decision theory: Chow's rule abstains from *classifying* a point when the conditional error probability exceeds a threshold [8], and the selective-prediction literature develops the resulting risk–coverage tradeoff [12, 14]. The present chapter lifts the same logic one level: the object declined is not a prediction but the *adaptation decision* for a deployment condition, the "conditional error" is the probability of picking the wrong sign of Δ, and the analogue of Chow's excess-risk computation is exactly Theorem 4.3, with $|\Delta(z)|$ playing the role of the margin. The analogy is precise enough to transfer intuition — low-margin points are where rejection is cheap in both settings — but the impossibility half has no per-prediction counterpart: a selective classifier can always in principle resolve its uncertainty with more labeled data, whereas Theorem 4.2 exhibits decision uncertainty that no amount of the only available (unlabeled) data resolves.

Remark 4.6 (The impossibility is about evidence-measurability, not sample size). Every quantity in Theorem 4.2 is indexed by the batch size n, and the witness conclusion holds *for every* n: the floor is 1 at $n = 10$ and at $n = 10^9$. This locates the obstruction precisely. Statistical folklore expects uncertainty to shrink at rate $n^{-1/2}$, and indeed the concentration tools that power the certificate of Chapter 5 — Hoeffding and empirical-Bernstein bounds [22, 34] — shrink the *radius* ε at exactly that rate. But a confidence radius shrinks toward the estimand's identifiable value; when the sign of Δ is not a function of the evidence law, there is no identifiable value to shrink toward. Formally, the decision rule is measurable with respect to $\sigma(Z)$ (plus independent randomness), and on the witness pair the two worlds induce the *same* measure on $\sigma(Z)$; the answer is not a random variable on the space the rule lives on. More data refines the empirical estimate of that common measure — it cannot break the tie. This is why the chapter's results are phrased as non-identifiability rather than as a rate: the failure is of the σ-algebra, not of the variance.

Remark 4.7 (What breaks without each hypothesis). It is worth walking the hypotheses of the two theorems once more, now with their proofs in hand. *Matched evidence laws* (hypothesis (i)): drop it and Lemma 4.1 immediately monetizes the difference — the floor falls to $1 - \mathrm{TV} < 1$, and a likelihood-ratio test on the evidence achieves it; this is not a pathology but the design target of evidence engineering (Remark 4.3). *Opposite benefit signs* (hypothesis (ii)): drop it and the matched evidence is harmless, since one committed action is correct in both worlds; the trichotomy collapses to a dichotomy on such families. *Fixed, known* (f_0, f_a) (Assumption 3.1): if the candidate were not fixed at decision time, Z would not be a world-independent function of the covariates and Step 2 of Theorem 4.1's proof would fail to produce matched evidence laws — but the conclusion would not improve, because the benefit being certified would itself be a moving target. *The label-free protocol* (Assumption 3.2): a single labeled target batch breaks the construction completely — the per-sample paired benefits δ_i become observable, their mean estimates Δ directly, and the witness worlds separate at the first label. The impossibility is exactly as strong as the constraint that creates it, which is why the paper is careful to state the protocol (Remark 3.1) before the theorems.

Remark 4.8 (What abstention buys, and what comes next). The two results of this chapter are deliberately one-sided: they establish that on matched-evidence pairs every committal policy pays the floor, and that the unique worst-case-optimal action there is ABSTAIN. What they do not yet provide is a *computable* rule that abstains there and commits elsewhere with guarantees. That is the certificate's job: Theorem 5.1 (Chapter 5) shows that an interval $\widehat{\Delta} \pm \varepsilon$ with $1 - \alpha$ coverage yields an adapt/freeze/abstain rule whose false-adapt and false-freeze rates are each at most α; its forced-abstention corollary closes the loop with this chapter by showing that on matched evidence any such valid rule *must* abstain with probability at least $1 - 2\alpha$ — abstention is mandatory, not merely prudent; and its sequential complement (Theorem 5.2) extends the guarantee to streams with optional stopping. On the positive side of the boundary, Theorem 5.3 proves that under covariate shift with a bounded density ratio the sign of Δ *is* identifiable from labeled source plus unlabeled target [45], and Theorem 5.4 characterizes $\operatorname{sign}\Delta$ in general as an ordinal accuracy comparison on the observable disagreement region D — the quantity whose label-free bracketing is the program's one remaining open piece (Conjecture 5.1). The unknowable regime proved here is thus not the end of the story but its boundary condition: everything Chapter 5 certifies, it certifies *because* it abstains where this chapter says it must.

Chapter 5

Certificates: When the Boundary Can Be Crossed Safely

Chapter 4 ended on deliberately negative ground. Theorem 4.1 exhibited two target worlds that induce *exactly* the same label-free evidence yet assign *opposite* signs to the adaptation benefit $\Delta = R_T(f_0) - R_T(f_a)$, and its quantitative strengthening showed that the best achievable committal error against such a pair equals $1 - \mathrm{TV}$ of the evidence laws — which is 1 in the witness. Theorem 4.3 then priced the failure: a gate's regret over the oracle is precisely the benefit magnitude $|\Delta(Z)|$ it forfeits on each sign mistake, and on evidence compatible with both signs no estimator can beat the floor $\mathbb{E}[|\Delta(Z)|]/2$, attained by abstaining. The unknowable regime is not a deficiency of any particular detector or drift statistic; it is a property of the information available.

This chapter is the constructive half of the theory. It asks the question the impossibility result leaves open: *when the evidence does separate the worlds, how should a system cross the boundary — and what guarantee can it carry across?* The answer is a *certificate*: a label-free benefit estimate $\widehat{\Delta}(Z)$ equipped with a finite-sample confidence radius $\varepsilon(Z)$, together with the three-way rule that commits only when the certified interval clears zero. We develop the batch certificate and its error guarantee (Theorem 5.1), discharge the radius two ways — by empirical-Bernstein concentration and by split-conformal calibration — and derive both the mandatory-abstention consequence on matched evidence and the sample complexity of certification. We then build the streaming complement: an anytime-valid certificate by testing-by-betting (Theorem 5.2), valid at every stopping time simultaneously. The second half of the chapter maps the regimes where the boundary genuinely opens: the covariate-shift regime where the benefit sign becomes fully identifiable (Theorem 5.3), and the disagreement-region characterization that reduces $\operatorname{sign}\Delta$ to an ordinal accuracy comparison on an observable set for binary, multiclass, and regression losses (Theorems 5.4–5.6). The chapter closes with the one piece the program leaves open — the label-free *bracketing* of that ordinal comparison (Conjecture 5.1) — stated precisely, with an honest account of why current techniques do not resolve it and which partial results are proved.

5.1 From Impossibility to Conditional Possibility

The structure of the impossibility theorem dictates the structure of its escape. The witness of Theorem 4.1 succeeds because the evidence law $\mathrm{Law}(Z)$ carries no information about $\operatorname{sign}\Delta$: both worlds produce it identically. A certificate can therefore exist only *conditionally* — under an explicit assumption that the evidence at hand is not of the witness kind, i.e. that it constrains the benefit sign. The paper names this property at the definition stage (Chapter 3) and invokes it as the standing assumption of the certificate result. We restate it here, in the paper's own words, as the assumption the entire chapter trades under.

Assumption 5.1 (Observable risk alignment, certificate form)**.** The evidence is *risk-aligned* in the sense of Chapter 3: Z identifies or bounds $\operatorname{sign}\Delta(z)$ (it need not reveal the label, only whether adaptation helps). Operationally, a label-free estimator $\widehat{\Delta}(z)$ admits a radius $\varepsilon(z)$ with

$$\mathbb{P}\big[\,|\widehat{\Delta}(z)-\Delta(z)|\le\varepsilon(z)\,\big]\ \ge\ 1-\alpha.$$

The first sentence is the paper's definition of risk alignment, verbatim; the second is the exact premise of the certificate theorem below. The two are the qualitative and quantitative faces of the same requirement: a radius $\varepsilon(z)$ with valid $1-\alpha$ coverage is precisely a device that *bounds* $\operatorname{sign}\Delta(z)$ whenever the interval $[\widehat{\Delta}-\varepsilon,\widehat{\Delta}+\varepsilon]$ excludes zero.

Given the assumption, the decision rule is forced by the geometry of the interval. Estimate the benefit, attach the radius, and act on the *certified* part of the interval only:

$$\textsc{adapt}\ \text{if}\ \widehat{\Delta}-\varepsilon>0,\qquad \textsc{freeze}\ \text{if}\ \widehat{\Delta}+\varepsilon<0,\qquad \textsc{abstain}\ \text{otherwise.}$$

Adapting requires the entire interval to sit above zero; freezing requires it to sit below; an interval that straddles zero is exactly the evidence state in which both signs remain consistent with what has been observed, and the rule declines to commit. The three outputs of the rule are the three regimes of the knowability map: knowably helpful, knowably harmful, unknowable.

Before stating the theorem we are explicit about what its guarantee can and cannot mean, because the certificate is easy to over-read.

Remark 5.1 (What the guarantee means — and what it does not)*.* First, the guarantee is *conditional* on Assumption 5.1. Theorem 4.1 shows that the assumption itself cannot be verified from Z alone: on matched-evidence worlds no label-free procedure can detect that the radius has lost its meaning. The certificate does not contradict the impossibility theorem — it relocates the burden into the validity of ε, which in practice is purchased with a labeled validation or calibration sample assumed exchangeable with deployment (the "validation labels allowed" sense of label-free fixed in Chapter 3; no *target* labels are ever used). The paper is explicit that this residual assumption "is not proved in full generality," and we preserve that honesty here. Second, the guarantee bounds the probability of the *joint* event $\{\text{commit}\wedge\text{wrong sign}\}$; it is not a posterior statement of the form "given that we adapted, the chance of harm is at most α." Third, the certificate controls *sign* errors, not magnitudes: by Theorem 4.3 the regret of a wrong commitment is $|\Delta|$, so the certificate's value concentrates exactly where $|\Delta|$ is large — which is also where the interval clears zero soonest. Fourth, abstention is not free; its price is coverage, the benefit forgone when $\Delta > 0$ but the interval straddles zero. The level α is the dial that trades these two costs, and we will see the trade measured on real tasks (Figure 5.3).

5.2 The Finite-Sample Certificate

We now state the certificate theorem exactly as the paper does, and then expand its proof to full length.

Theorem 5.1 (Certificate)**.** *Suppose a label-free estimator* $\widehat{\Delta}(z)$ *admits a radius* $\varepsilon(z)$ *with* $\mathbb{P}[\,|\widehat{\Delta}(z)-\Delta(z)| \le \varepsilon(z)\,] \ge 1-\alpha$. *Define*

$$\textsc{adapt}\ \textit{if}\ \widehat{\Delta}-\varepsilon > 0, \quad \textsc{freeze}\ \textit{if}\ \widehat{\Delta}+\varepsilon < 0, \quad \textsc{abstain}\ \textit{otherwise.}$$

Then $\mathbb{P}[\textsc{adapt} \wedge \Delta \le 0] \le \alpha$ *and* $\mathbb{P}[\textsc{freeze} \wedge \Delta \ge 0] \le \alpha$.

Proof. Write $G := \{\,|\widehat{\Delta}-\Delta| \le \varepsilon\,\}$ for the coverage event, so that $\mathbb{P}[G] \ge 1-\alpha$ by hypothesis, and let G^c denote its complement, of probability at most α. The argument is a containment of error events in G^c.

False adapt. Suppose the rule outputs ADAPT and the coverage event G holds. The decision ADAPT means $\widehat{\Delta}-\varepsilon > 0$; the event G means $\Delta \ge \widehat{\Delta}-\varepsilon$. Chaining the two inequalities,

$$\Delta \;\ge\; \widehat{\Delta}-\varepsilon \;>\; 0,$$

so on G an ADAPT decision is necessarily correct: $\Delta > 0$. Contrapositively, the false-adapt event $\{\textsc{adapt} \wedge \Delta \le 0\}$ is disjoint from G, i.e.

$$\{\textsc{adapt} \wedge \Delta \le 0\} \;\subseteq\; G^c, \qquad \text{hence} \qquad \mathbb{P}[\textsc{adapt} \wedge \Delta \le 0] \;\le\; \mathbb{P}[G^c] \;\le\; \alpha.$$

False freeze. The argument is symmetric. On G we also have $\Delta \le \widehat{\Delta}+\varepsilon$, and the decision FREEZE means $\widehat{\Delta}+\varepsilon < 0$; chaining gives $\Delta < 0$, so on G a FREEZE decision is necessarily correct. Hence $\{\textsc{freeze} \wedge \Delta \ge 0\} \subseteq G^c$ and its probability is at most α.

Abstention. The third branch incurs no error event by construction: abstaining asserts nothing about $\operatorname{sign}\Delta$, so there is nothing to be wrong about. Its cost is accounted separately, as coverage. □

Remark 5.2 (One bad event covers both errors)*.* Because both error events were shown to be subsets of the *same* bad event G^c, the proof in fact yields the slightly stronger statement $\mathbb{P}[(\textsc{adapt} \wedge \Delta \le 0) \cup (\textsc{freeze} \wedge \Delta \ge 0)] \le \alpha$ when the radius is two-sided as assumed: there is no need to spend α twice. The paper states the two one-sided bounds, which is the form the deployment guarantees quote, and we keep that convention.

The theorem is short because it is an exercise in honest bookkeeping: every gram of statistical content lives in the premise, the validity of $(\widehat{\Delta}, \varepsilon)$. The rest of this section discharges that premise in the two ways the system actually uses — a concentration-of-measure radius when $\widehat{\Delta}$ is a sample mean of bounded paired benefits, and a split-conformal radius when $\widehat{\Delta}$ is an arbitrary learned predictor of the benefit.

5.2.1 Discharging the Radius I: Empirical-Bernstein Concentration

The paper's first instantiation applies when Δ is a mean of bounded *paired benefits*

$$X_i \;=\; \ell(f_0(x_i), y_i) - \ell(f_a(x_i), y_i) \;\in\; [a, b]$$

over a validated stress bank — a calibration collection of conditions with held-out labels (validation labels, never target labels) on which both the frozen and the candidate model are evaluated. Then $\Delta = \mathbb{E}[X_i]$, the estimator $\widehat{\Delta} = \frac{1}{n}\sum_i X_i$ is a bounded sample mean, and a lower confidence bound on a bounded mean is a classical object. Two inequalities are relevant: the distribution-free Hoeffding bound, and the variance-adaptive empirical-Bernstein bound that the implementation uses.

Lemma 5.1 (Hoeffding lower confidence bound [22])**.** *Let $X_1, \ldots, X_n$ be independent with values in $[a, b]$, mean μ, sample mean $\widehat{\mu}$, and range $R = b - a$. For any $\alpha \in (0, 1)$, with probability at least $1 - \alpha$,*

$$\mu \;\geq\; \widehat{\mu} - R\sqrt{\frac{\log(1/\alpha)}{2n}}.$$

Proof. By Hoeffding's lemma, a bounded centered variable satisfies $\mathbb{E}[e^{s(X_i-\mu)}] \leq e^{s^2R^2/8}$ for all $s \in \mathbb{R}$. The Chernoff method applied to $-\sum_i(X_i - \mu)$ gives $\mathbb{P}[\widehat{\mu} - \mu \leq -t] \leq \exp(-2nt^2/R^2)$ for every $t > 0$; setting the right side equal to α and solving for t yields $t = R\sqrt{\log(1/\alpha)/(2n)}$, which is the displayed bound. □

Lemma 5.2 (Empirical-Bernstein lower confidence bound [34])**.** *Let $X_1, \ldots, X_n$ ($n \geq 2$) be i.i.d. with values in an interval of length $R = b - a$, mean μ, sample mean $\widehat{\mu}$, and unbiased sample variance $\widehat{V}$. For any $\alpha \in (0, 1)$, with probability at least $1 - \alpha$,*

$$\mu \;\geq\; \widehat{\mu} \;-\; \sqrt{\frac{2\widehat{V}\,\log(2/\alpha)}{n}} \;-\; \frac{7R\,\log(2/\alpha)}{3(n-1)}.$$

Proof sketch. The result is Maurer and Pontil's empirical Bernstein bound, stated there for variables in $[0, 1]$; applying it to the rescaled variables $(X_i - a)/R$ and multiplying through by R gives the form above (the mean and radius scale by R, the variance by R^2). The argument has two halves, each run at level $\alpha/2$ and combined by a union bound. First, Bernstein's inequality with the *true* variance σ^2 gives, with probability at least $1-\alpha/2$, $\mu \geq \widehat{\mu} - \sqrt{2\sigma^2 \log(2/\alpha)/n} - R\log(2/\alpha)/(3n)$. Second, the sample standard deviation concentrates around the true one: a self-bounding concentration argument (the entropy method applied to the variance statistic) gives $\sigma \leq \sqrt{\widehat{V}} + R\sqrt{2\log(2/\alpha)/(n-1)}$ with probability at least $1-\alpha/2$. Substituting the second display into the first and collecting the $1/n$- and $1/(n-1)$-order terms produces the stated constant $7/3$; the bookkeeping is exactly that of [34]. □

Applying Lemma 5.2 to the paired benefits and to their negatives gives a two-sided radius

$$\varepsilon_{\mathrm{EB}} \;=\; \sqrt{\frac{2\widehat{V}\;\log(2/\alpha)}{n}} \;+\; \frac{7(b-a)\;\log(2/\alpha)}{3(n-1)},$$

which satisfies the premise of Theorem 5.1 whenever the paired benefits are independent or exchangeable draws from the deployment-relevant distribution. This is the certificate implemented in the repository's `switching_certificate.py` and re-exported as `kga/certificate.py::empirical_bernstein` [25]; the implementation takes the range explicitly ($R = 2$ for paired $|p - y|$ losses, since each loss lies in $[0, 1]$), and returns an *infinite* radius when $n < 2$ — a deliberate safety default under which the rule of Theorem 5.1 can only abstain. The paper's analytic statements use the streamlined radius $\varepsilon(n) = \sigma\sqrt{2\log(2/\alpha)/n} + 3(b-a)\log(2/\alpha)/n$ with a variance proxy σ^2 in place of $\widehat{V}$; the two differ only in constants and in the empirical-versus-proxy variance, and we will use whichever form the quoted statement uses.

Remark 5.3 (Why empirical Bernstein, and why it is the right bound for this problem). The Hoeffding radius scales with the *range* R; the empirical-Bernstein radius scales, in its leading term, with the *standard deviation*. For paired benefits this distinction is not a refinement but the essence of the problem. Under 0/1 loss the per-sample benefit $\delta = \mathbf{1}[f_0 \neq Y] - \mathbf{1}[f_a \neq Y]$ takes values in $\{-1, 0, +1\}$ and vanishes identically off the disagreement region $D = \{x : f_0(x) \neq f_a(x)\}$ — the two losses cancel wherever the models agree, a fact we exploit structurally in Section 5.5. Consequently $\mathbb{E}[\delta^2] \leq P_T(D)$ (with equality in the binary case, where exactly one model is correct on D), so the variance obeys $\mathrm{Var}(\delta) = \mathbb{E}[\delta^2] - \Delta^2 \leq P_T(D)$. The empirical-Bernstein leading term is therefore on the order of $\sqrt{2P_T(D)\log(2/\alpha)/n}$, while the Hoeffding radius is $2\sqrt{\log(1/\alpha)/(2n)}$ regardless: when the candidate adaptation touches only a small fraction of predictions — the common case — the variance-adaptive radius is smaller by a factor of order $\sqrt{P_T(D)}$. All of the certificate's statistical information is concentrated on the disagreement region, and the empirical-Bernstein bound is the concentration inequality that knows it.

We restate the paper's own honesty note before moving on. The empirical-Bernstein radius is valid *under independence or exchangeability of the calibrated benefit sample with deployment*; in the paper's words, this assumption "is the residual burden and is not proved in full generality." Theorem 5.1 is proved; what remains an assumption is that the world supplies data satisfying its premise. Chapter 6 carries the same caveat into the algorithm's stated assumptions.

5.2.2 Mandatory Abstention and the Cost of Certainty

Two corollaries calibrate expectations about the certificate, in opposite directions. The first says that on genuinely unknowable evidence a valid certificate does not merely *tend* to abstain — it *must*. The second quantifies how much calibration data the certificate needs before it can commit at all.

Corollary 5.1 (Forced abstention under matched evidence)**.** *Fix evidence z in the unknowable two-point regime of Theorem 4.1 (*$\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$, $\Delta^1 < 0 < \Delta^2$*). Any rule whose false-adapt and false-freeze probabilities are each at most α (e.g. the certificate of Theorem 5.1) must abstain there with probability* $\mathbb{P}[\textsc{abstain} \mid z] \geq 1 - 2\alpha$.

Proof. A label-free rule observes only Z (and independent internal randomness), so at the fixed evidence value z its action distribution is a single triple $(q_a, q_f, 1 - q_a - q_f)$ — the probabilities of ADAPT, FREEZE, and ABSTAIN — and, crucially, this triple is *the same* in both worlds, because the two evidence laws coincide. Now read the triple against each world in turn. In world 1 the benefit is negative ($\Delta^1 < 0$), so every ADAPT the rule emits at z is a false adapt; the false-adapt constraint forces $q_a \leq \alpha$. In world 2 the benefit is positive ($\Delta^2 > 0$), so every FREEZE is a false freeze; the false-freeze constraint forces $q_f \leq \alpha$. The same q_a and q_f must satisfy both constraints simultaneously, since the rule cannot tell the worlds apart. Hence $\mathbb{P}[\textsc{abstain} \mid z] = 1 - q_a - q_f \geq 1 - 2\alpha$. □

The paper's gloss deserves repeating: this makes abstention *mandatory, not merely conservative* — on genuinely unknowable evidence a valid certificate cannot commit on more than a 2α fraction of decisions. The corollary is also the precise sense in which the certificate and the impossibility theorem are two sides of one boundary: the same α that licenses commitment where evidence separates *caps* commitment where it does not. Empirically, on the clean non-identifiability witness of Chapter 7 — a constructed matched-evidence pair — KGA abstains on 100% of instances, while a rule forced to commit pays near-chance regret 0.51; Corollary 5.1 is the theorem that says it had no valid alternative.

Corollary 5.2 (Sample complexity of certification)**.** *Let $\widehat{\Delta}$ be a mean of n bounded paired benefits in $[a, b]$ with variance proxy σ^2, and let $\varepsilon(n) = \sigma\sqrt{2\log(2/\alpha)/n} + \frac{3(b-a)\log(2/\alpha)}{n}$ be the empirical-Bernstein radius of Theorem 5.1. Then the certificate is non-abstaining (commits* ADAPT *or* FREEZE*) once*

$$n \gtrsim \frac{2\sigma^2}{\Delta^2}\log\frac{2}{\alpha} + \frac{3(b-a)}{|\Delta|}\log\frac{2}{\alpha} = O\Big(\tfrac{\sigma^2}{\Delta^2}\log\tfrac{1}{\alpha}\Big),$$

so the unlabeled paired-benefit sample size needed to certify scales inversely with the squared benefit margin and logarithmically with the confidence level.

Proof. The rule of Theorem 5.1 commits precisely when the interval clears zero, i.e. when $\varepsilon(n) < |\widehat{\Delta}|$; since $\widehat{\Delta} \to \Delta$, the asymptotically operative condition is $\varepsilon(n) < |\Delta|$. The radius is a sum of two decreasing terms, so (up to the constant absorbed in $\gtrsim$) it suffices to make each term smaller than

$|\Delta|$. For the variance term,

$$\sigma\sqrt{\frac{2\log(2/\alpha)}{n}} \;<\; |\Delta| \quad\Longleftrightarrow\quad n \;>\; \frac{2\sigma^2}{\Delta^2}\,\log\frac{2}{\alpha},$$

which is the dominant requirement when $\sigma \gg |\Delta|$. Retaining the Bernstein $1/n$ term adds the second requirement $3(b-a)\log(2/\alpha)/n < |\Delta|$, i.e. $n > 3(b-a)|\Delta|^{-1}\log(2/\alpha)$, the stated summand. The asymptotic form follows because the first term dominates whenever $\sigma^2/|\Delta| \gtrsim (b-a)$, which holds in the small-margin regime where the bound bites. □

The corollary answers what the paper calls the reviewer question — *when does the certificate actually fire?* Certification is cheap where it matters most: large benefit margins (precisely the decisions whose regret content is large, by Theorem 4.3) are certified from few samples, while margins near zero demand $n \sim \sigma^2/\Delta^2$ and below that threshold the rule abstains — which, by Theorem 4.3 again, is exactly the band where committal regret is bounded by the small $|\Delta|$ itself. The prediction is directly visible in the batch-size ablation of Chapter 7: growing the test batch from 16 to 256 samples shrinks the radius from 0.44 to 0.18 and raises coverage from 2% to 26%, the empirical shadow of the $n^{-1/2}$ term.

5.2.3 Discharging the Radius II: The Split-Conformal Variant

The empirical-Bernstein route requires $\widehat{\Delta}$ to be a sample mean of bounded benefits. In the deployed system the benefit estimator is richer: KGA trains a gradient-boosted regressor mapping the evidence vector Z to a predicted benefit, leave-one-task-out across a bank of calibration tasks (Chapter 6). For an arbitrary learned $\widehat{\Delta}$, concentration inequalities no longer apply — but a conformal wrapper restores a finite-sample radius under a single exchangeability assumption, with no further conditions on the estimator [51, 1].

Proposition 5.1 (Split-conformal radius)**.** *Let $(\widehat{\Delta}_1, \Delta_1), \ldots, (\widehat{\Delta}_n, \Delta_n)$ be calibration pairs and $(\widehat{\Delta}_{n+1}, \Delta_{n+1})$ a deployment pair such that the residuals $r_i = |\widehat{\Delta}_i - \Delta_i|$, $i = 1, \ldots, n+1$, are exchangeable. Let ε be the $\lceil(1-\alpha)(n+1)\rceil$-th smallest of $r_1, \ldots, r_n$ (setting $\varepsilon = +\infty$ if $\lceil(1-\alpha)(n+1)\rceil > n$). Then*

$$\mathbb{P}\big[\,|\widehat{\Delta}_{n+1} - \Delta_{n+1}| \le \varepsilon\,\big] \;\ge\; 1-\alpha,$$

so the rule of Theorem 5.1, run with $(\widehat{\Delta}_{n+1}, \varepsilon)$, has false-adapt and false-freeze probabilities at most α.

Proof. Set $k = \lceil(1-\alpha)(n+1)\rceil$ and let $r_{(1)} \le \cdots \le r_{(n)}$ denote the ordered calibration residuals, so $\varepsilon = r_{(k)}$ when $k \le n$. By exchangeability of $(r_1, \ldots, r_{n+1})$, the rank of r_{n+1} within the full set of $n+1$ residuals is stochastically dominated by the uniform distribution on $\{1, \ldots, n+1\}$ (uniform under almost-sure distinctness; ties only push the rank down, which helps). The event

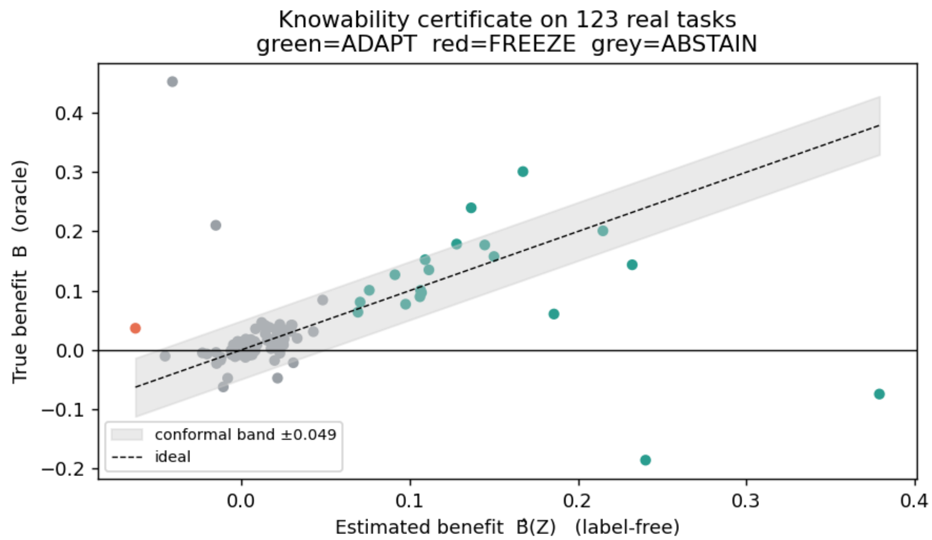

Figure 5.1: The batch certificate on the 123-task anomaly-routing suite (Chapter 7). Each point is one task: label-free estimated benefit $\widehat{\Delta}(Z)$ (horizontal) against the oracle true benefit (vertical), with the split-conformal band ± 0.049 at $\alpha = 0.1$ shaded around the ideal diagonal. Green points are certified ADAPT, red FREEZE, grey ABSTAIN: commitments concentrate where the certified interval clears zero, and abstentions cluster in the near-zero-benefit band where Theorem 4.3 prices commitment errors as cheap but Theorem 5.1 declines to make them.

$r_{n+1} \leq r_{(k)}$ contains the event that the rank of r_{n+1} is at most k, whence

$$\mathbb{P}\big[r_{n+1} \leq r_{(k)}\big] \;\geq\; \frac{k}{n+1} \;=\; \frac{\lceil (1-\alpha)(n+1) \rceil}{n+1} \;\geq\; 1-\alpha.$$

If $k > n$ the radius is infinite and the coverage statement is trivial. The premise of Theorem 5.1 is thus satisfied at level α for the deployment pair, and its conclusion follows. □

The guarantee is marginal over the calibration draw — the standard split-conformal semantics [1] — which matches exactly the marginal reading of Theorem 5.1 discussed in Remark 5.1. Two implementation notes keep the record straight. First, the experiments use the empirical $(1-\alpha)$-quantile of leave-one-task-out residuals (`kga/certificate.py::conformal_split`, mirroring the knowability and mixed-regime experiment scripts [25]); this differs from the $\lceil (1-\alpha)(n+1) \rceil$ order statistic by an index shift of order $1/n$, the usual practical simplification. Second, the exchangeability here is *across tasks*: the calibration bank of tasks and the deployment task are assumed exchangeable. The paper lists this as a standing limitation, and Chapter 7 probes it empirically rather than assuming it away. Altogether the repository ships four interchangeable radius constructions behind one interface — empirical-Bernstein, Hoeffding, split-conformal, and the anytime e-value process of the next section — all pure `numpy` and deterministic [25].

Figures 5.1 and 5.2 show the certificate doing on real tasks what the theorem promises on

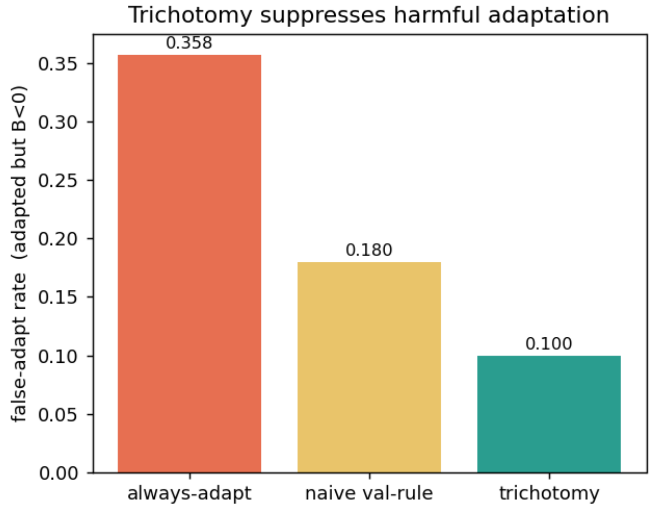

Figure 5.2: Suppression of harmful adaptation on the 123-task suite (produced by the knowability experiment of Chapter 7; same data as Figure 5.1). Each bar is the fraction of adaptation decisions whose true benefit was negative: adapting always inherits the suite's harmful base rate 0.358; a naive validation-threshold rule false-adapts on 0.180 of its commitments; the certificate trichotomy on 0.100 — the complement of its 0.90 adapt precision, in line with its calibrated level $\alpha = 0.1$.

paper. On the 123-task suite the conformal band has half-width 0.049; tasks whose interval clears zero are committed, and the adapt precision is 0.90 there and 0.935 in the controlled mixed regime — in the latter, an observed false-adapt rate of 0.065 against the budget $\alpha = 0.1$. The bar chart makes the comparative point on the same 123 tasks: an uncertified always-adapt policy inherits the suite's 35.8% harmful base rate, a naive validation threshold false-adapts on 18.0% of its commitments, and the trichotomy holds the line at 10.0% — the level it was asked to hold.

5.3 The Anytime-Valid Certificate: Testing by Betting

Theorem 5.1 fixes the sample size n in advance. Streaming test-time adaptation does not: evidence arrives one batch at a time, the system looks at the accumulating benefit sample after every batch, and it wants to commit *the moment* the evidence suffices — without scheduling n beforehand and without paying a multiple-testing tax for the repeated looks. Classical fixed-n confidence intervals are unsound under this regime: stopping the first time a fixed-n interval clears zero inflates the error probability well above α, because the analyst's stopping rule correlates with the noise. What is needed is a certificate whose guarantee holds *simultaneously over all stopping times*. The paper supplies one in its appendix, via the modern theory of safe anytime-valid inference [40, 24, 16], in the testing-by-betting form popularized by Shafer [44] with the betting strategies of Waudby-Smith and Ramdas [55].

5.3.1 Wealth, Supermartingales, and Ville's Inequality

Testing by betting replaces the p-value with a gambler's wealth. To test the null hypothesis $H_0 : \Delta \leq 0$ ("adapting does not help"), imagine a skeptic who starts with capital 1 and, before each benefit observation X_t, stakes a fraction of current wealth on the proposition that the benefits are drifting positive. If the bet at round t is $\lambda_t \geq 0$, the wealth multiplies by $(1 + \lambda_t X_t)$: it grows when $X_t > 0$, shrinks when $X_t < 0$. Under the null, every such bet is fair or unfavorable — the conditional expected multiplier is $1 + \lambda_t \mathbb{E}[X_t \mid \mathcal{F}_{t-1}] \leq 1$ — so no betting system can be expected to multiply the skeptic's capital. Large realized wealth is therefore quantitative evidence against the null: wealth $E_t^+ \geq 1/\alpha$ is the event that a gambler playing a fair-or-losing game has multiplied capital ten-fold (at $\alpha = 0.1$), and the probability that this *ever* happens is at most α. The wealth process is an *e-process* — a nonnegative process whose value at any stopping time has expectation at most one under the null [16, 40] — and the "ever" quantifier is exactly what frees the certificate from fixed-n scheduling. The inequality that delivers the quantifier is Ville's [50].

Lemma 5.3 (Ville's inequality [50])**.** *Let $(M_t)_{t\geq 0}$ be a nonnegative supermartingale with respect to a filtration $(\mathcal{F}_t)$, with $\mathbb{E}[M_0] \leq 1$. Then for any $u \geq 1$,*

$$\mathbb{P}\Big[\,\exists\, t \geq 0 : \; M_t \geq u\,\Big] \;\leq\; \frac{1}{u}.$$

Proof. Let $\tau := \inf\{t \geq 0 : M_t \geq u\}$, a stopping time. Fix a horizon T; then $\tau \wedge T$ is a bounded stopping time, and the optional stopping theorem for nonnegative supermartingales gives $\mathbb{E}[M_{\tau\wedge T}] \leq \mathbb{E}[M_0] \leq 1$. On the event $\{\tau \leq T\}$ we have $M_{\tau\wedge T} = M_\tau \geq u$, while on its complement $M_{\tau\wedge T} \geq 0$; hence

$$1 \;\geq\; \mathbb{E}[M_{\tau\wedge T}] \;\geq\; u\,\mathbb{P}[\tau \leq T], \qquad \text{so} \qquad \mathbb{P}[\tau \leq T] \;\leq\; \frac{1}{u}.$$

Letting $T \to \infty$ and using continuity from below, $\mathbb{P}[\tau < \infty] \leq 1/u$, which is the claim. ☐

Markov's inequality bounds $\mathbb{P}[M_t \geq u]$ at one preselected t; Ville's inequality bounds the supremum over the whole trajectory at no extra cost. That uniformity over time is the entire content of "anytime-valid": the guarantee survives optional stopping, optional continuation, and continuous monitoring [24].

5.3.2 The Anytime Certificate

We state the theorem as the paper's appendix does. Two wealth processes run in parallel: E_t^+ bets that the mean benefit is positive (its growth rejects $H_0 : \Delta \leq 0$ and certifies ADAPT), and E_t^- bets that it is negative (its growth rejects $H_0' : \Delta \geq 0$ and certifies FREEZE); while neither has grown past $1/\alpha$, the system abstains — which in the streaming reading means "keep sampling."

Theorem 5.2 (Anytime-valid adapt/freeze/abstain certificate)**.** *Let paired benefits $X_i = \ell(f_0(x_i), y_i) - \ell(f_a(x_i), y_i) \in [a, b]$ ($a < 0 < b$) be adapted to $(\mathcal{F}_t)$ with $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = \Delta$. With predictable bets*

$\lambda_t \in [0, c/(-a)]$, $\nu_t \in [0, c/b]$, $c \in (0,1)$, set $E_t^+ = \prod_{i\le t}(1+\lambda_i X_i)$ and $E_t^- = \prod_{i\le t}(1-\nu_i X_i)$. Decide ADAPT *the first time $E_t^+ \ge 1/\alpha$,* FREEZE *the first time $E_t^- \ge 1/\alpha$, else* ABSTAIN. *Then simultaneously over all t, $\mathbb{P}[\exists t : E_t^+ \ge 1/\alpha \wedge \Delta \le 0] \le \alpha$ and $\mathbb{P}[\exists t : E_t^- \ge 1/\alpha \wedge \Delta \ge 0] \le \alpha$.*

Proof. We prove the ADAPT direction in full; the FREEZE direction is its mirror image.

Step 1: the wealth stays positive. Each factor of E_t^+ is $1 + \lambda_i X_i$ with $\lambda_i \ge 0$ and $X_i \ge a$, so the worst case is $X_i = a < 0$, where the factor is at least $1 + \lambda_i a \ge 1 + \frac{c}{-a}\, a = 1 - c > 0$ by the bet cap. A product of strictly positive factors is strictly positive, so $E_t^+ > 0$ for all t; in particular E_t^+ is a legitimate candidate for Lemma 5.3.

Step 2: under the null, the wealth is a supermartingale. Fix any distribution in H_0, i.e. with $\Delta \le 0$. Because λ_t is *predictable* — it is $\mathcal{F}_{t-1}$-measurable, a function of the past only — it factors out of the conditional expectation:

$$\mathbb{E}\big[E_t^+ \,\big|\, \mathcal{F}_{t-1}\big] = E_{t-1}^+\, \mathbb{E}\big[1 + \lambda_t X_t \,\big|\, \mathcal{F}_{t-1}\big] = E_{t-1}^+\, \big(1 + \lambda_t\, \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]\big) = E_{t-1}^+\, (1 + \lambda_t \Delta) \;\le\; E_{t-1}^+,$$

since $\lambda_t \ge 0$ and $\Delta \le 0$ make the multiplier at most one. With $E_0^+ = 1$, the process (E_t^+) is a nonnegative supermartingale with unit initial value under every null distribution.

Step 3: Ville closes the bound. Lemma 5.3 with $u = 1/\alpha$ gives, under every $\Delta \le 0$,

$$\mathbb{P}\big[\exists t :\; E_t^+ \ge 1/\alpha\big] \;\le\; \alpha.$$

The false-adapt event $\{\exists t : E_t^+ \ge 1/\alpha \wedge \Delta \le 0\}$ is exactly this crossing event evaluated under a null distribution, so its probability is at most α — uniformly over all stopping times, since the bound already quantifies over the whole trajectory.

The freeze direction. The process $E_t^- = \prod_{i\le t}(1 - \nu_i X_i)$ has factors $1 - \nu_i X_i$ with $\nu_i \in [0, c/b]$; the worst case is now $X_i = b > 0$, where the factor is at least $1 - \nu_i b \ge 1 - c > 0$, so $E_t^- > 0$. Under $H_0' : \Delta \ge 0$, predictability of ν_t gives $\mathbb{E}[E_t^- \mid \mathcal{F}_{t-1}] = E_{t-1}^-(1 - \nu_t \Delta) \le E_{t-1}^-$, and Ville's inequality bounds the crossing probability by α exactly as before. □

Remark 5.4 (Validity is free; power is bought by the bet). Nothing in the proof used the *values* of the bets beyond nonnegativity, predictability, and the cap — any such betting sequence yields a valid certificate. The specific bets govern power and stopping time. The implementation uses a truncated growth-optimal (aGRAPA-style) plug-in

$$\lambda_t = \mathrm{clip}\Big(\widehat{\mu}_{t-1} \,/\, \widehat{\sigma}^2_{t-1},\; 0,\; c/(-a)\Big),$$

the running-mean-over-running-second-moment approximation to the log-optimal bet $\lambda^\star = \Delta/\mathbb{E}[X^2]$ [55], with a small prior stabilizing the early variance estimate; the symmetric rule drives ν_t. This separation — validity from the supermartingale property, power from the betting strategy — is the signature design freedom of game-theoretic statistics [44, 40]. The anytime certificate *complements* rather than replaces the batch certificate: at a pre-fixed n the empirical-

	anytime false-adapt ($H_0 : \Delta \leq 0$)			power / mean stop ($H_1 : \Delta > 0$)		
stream	$\Delta = -0.20$	$\Delta = -0.05$	$\Delta = 0$	$\Delta = 0.05$	$\Delta = 0.10$	$\Delta = 0.40$
two-point	0.00505	0.0268	0.0626	0.375 / 810	0.948 / 676	1.000 / 39
Beta	0.0000	0.0038	0.0584	0.988 / 521	1.000 / 127	1.000 / 16
clipped Gaussian	0.0000	0.00465	0.0559	0.976 / 588	1.000 / 148	1.000 / 17

Table 5.1: Anytime-valid certificate validation, exactly as recorded in `experiments/kbound/theory_validation/val_thm3_evalue_results.json` ($\alpha = 0.1$, wealth threshold $1/\alpha = 10$, horizon 2000, 20,000 Monte Carlo runs per cell, seed 0, bet cap $c = 0.5$; power rounded to three decimals, mean stopping times to integers). The worst-case anytime false-adapt rate across all nine null settings is $0.0626 \leq \alpha = 0.1$; the results file's verdict field records `controlled: true`.

Bernstein interval is tighter, while the e-process is valid at any stopping time; the paper states exactly this trade.

5.3.3 The Validator: Measured Anytime Error and Power

Theorem 5.2 ships with a machine-checked numerical validator, `experiments/kbound/theory_validation/val_th` [25], which runs the betting certificate over 20,000 independent streams per configuration (horizon 2000, $\alpha = 0.1$, bet cap $c = 0.5$, seed 0) and three bounded benefit distributions chosen to stress different shapes within $[a, b] = [-1, 1]$: a two-point distribution massing on the endpoints (the worst case for the positivity argument), a smooth Beta-shaped law, and a clipped Gaussian. It measures the *anytime* false-adapt rate — the fraction of null streams in which E_t^+ *ever* crosses $1/\alpha$ within the horizon — together with detection power and stopping times under alternatives. Table 5.1 reproduces the machine-readable results file.

Three features of the table are worth reading off. First, the headline: the worst-case anytime false-adapt rate over every null configuration is **0.0626**, against the budget $\alpha = 0.1$ — the guarantee of Theorem 5.2 holds with room to spare, and the worst case occurs exactly where theory says it must, at the boundary $\Delta = 0$ of the null (rates collapse to 0.00505 and below by $\Delta = -0.20$). The paper's appendix additionally records a run at $\alpha = 0.05$ with anytime false-adapt $0.028 \leq 0.05$. The slack between 0.0626 and 0.1 is the familiar conservatism of time-uniform bounds: the certificate pays for validity at *every* stopping time, and the plug-in bet is not the oracle log-optimal one. Second, the supermartingale property itself is checked directly: at $\Delta = 0$ the Monte Carlo mean of E_t^+ stays at or below 1 (within three standard errors) at every recorded time $t \in \{1, 5, 25, 100, 500, 2000\}$ for all three streams — the results file flags `all_leq_one_within_3sem: true` in each case. Third, power behaves as a sequential test should: it rises to 1 as the margin grows, and the mean stopping time falls steeply — from roughly 810 samples at $\Delta = 0.05$ (two-point stream) to 16–39 samples at $\Delta = 0.40$. Large benefits are certified almost immediately; marginal ones take evidence; null ones, with probability at least $1 - \alpha$, are never falsely certified no matter how long the system watches.

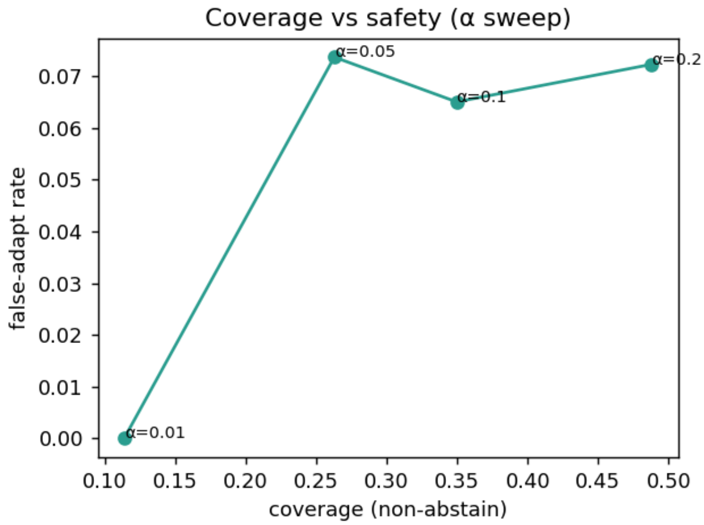

Figure 5.3: The level α as the safety–coverage dial, measured on the anomaly-routing benchmark of Chapter 7 with the split-conformal batch certificate: $\alpha = 0.01$ yields a 0% observed false-adapt rate at 11% coverage; $\alpha = 0.1$ yields 0.065 at 35% coverage; $\alpha = 0.2$ yields 0.072 at 49% coverage. The $\alpha = 0.05$ point sits at 0.073, slightly above its nominal budget — an honest finite-sample fluctuation on 369 instances under approximate cross-task exchangeability, and a reminder that the guarantee is marginal, not per-configuration.

Remark 5.5 (Bounded non-stationarity). The constant-mean premise $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = \Delta$ can be relaxed. The paper's theory-extensions section proves a β-bounded-drift variant: if the conditional mean wanders within $\pm\beta$ of Δ, dividing the wealth by $\prod_{i\le t}(1+\lambda_i\beta)$ restores the supermartingale property under the null, at the price of an inflated decision threshold $(1/\alpha)\prod_{i\le t}(1+\lambda_i\beta) \le \exp(\beta\sum_{i\le t}\lambda_i)$. Its validator (`val_thm9prime_drift.py` [25]) reports that under worst-case upward drift the corrected certificate holds the anytime false-adapt rate at 0.05/0.04/0.01 for $\beta = 0.1/0.2/0.4$ — all within $\alpha = 0.1$ — while the uncorrected threshold's rate inflates to 0.69 and beyond, confirming both that the correction works and that it is needed.

Both certificates — batch and anytime — expose the same single dial, the level α, and Figure 5.3 shows what the dial buys on real tasks. Tightening α to 0.01 drives observed false adaptation to zero at the cost of committing on only 11% of decisions; loosening it to 0.2 roughly quadruples coverage while the observed false-adapt rate stays near 0.07. The certificate makes the safety–coverage trade explicit and tunable, rather than implicit in a mechanism's hyperparameters.

5.4 A Provably Knowable Regime: Covariate Shift

The impossibility theorem and the certificate bracket the problem from both sides, but neither yet exhibits a natural class of shifts on which the label-free decision is *provably* available. The paper's fourth result identifies one: covariate shift. The motivating picture is a new scanner — image statistics change, disease prevalence and the disease-given-image relationship do not. Under that premise, reweighting labeled *source* data by the density ratio of the two covariate laws reconstructs every target risk, and with it the benefit and its sign.

Theorem 5.3 (Covariate-shift identifiability). *Assume covariate shift $P_S(Y \mid X) = P_T(Y \mid X)$, and a density ratio $r(x) = p_T(x)/p_S(x)$ that is bounded ($r \le B < \infty$) with $\mathbb{E}_S[r(X)^2] < \infty$. Then $R_T(f) = \mathbb{E}_S[r(X)\,\ell(f(X), Y)]$, the importance-weighted estimator $\widehat{R}_T(f) = \frac{1}{n}\sum_i r(x_i)\ell(f(x_i), y_i)$ is unbiased and consistent, and hence Δ and $\operatorname{sign}\Delta$ are identifiable from labeled source plus unlabeled target whenever $|\Delta| > 0$.*

Proof. Change of measure. Write the target risk as an integral over the target joint and factor the joint through the covariate-shift premise:

$$R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)] = \iint \ell(f(x), y)\, p_T(y \mid x)\, p_T(x)\, dy\, dx = \iint \ell(f(x), y)\, p_S(y \mid x)\, \frac{p_T(x)}{p_S(x)}\, p_S(x)\, dy\, dx,$$

where the second equality substitutes $p_T(y \mid x) = p_S(y \mid x)$ (the premise) and multiplies and divides by $p_S(x)$; boundedness of r ensures the manipulation is legitimate wherever p_S has mass, and the target law places no mass where p_S vanishes (else r would be unbounded). The right-hand side is $\mathbb{E}_S[r(X)\,\ell(f(X), Y)]$, the first claim.

Unbiasedness and consistency. Unbiasedness of $\widehat{R}_T(f)$ is immediate from the identity just proved, since each summand $r(x_i)\ell(f(x_i), y_i)$ has source-expectation $R_T(f)$. For consistency, the summands are i.i.d. with finite second moment: with ℓ bounded in $[0, 1]$, $\mathbb{E}_S[(r\,\ell)^2] \le \mathbb{E}_S[r^2] < \infty$, so the strong law of large numbers gives $\widehat{R}_T(f) \to R_T(f)$ almost surely, with variance $\operatorname{Var}(\widehat{R}_T(f)) \le \mathbb{E}_S[r^2]/n = O(1/n)$.

Sign identifiability. Apply the estimator to both models and difference: $\widehat{\Delta} = \widehat{R}_T(f_0) - \widehat{R}_T(f_a) = \frac{1}{n}\sum_i r(x_i)(\ell(f_0(x_i), y_i) - \ell(f_a(x_i), y_i))$ is unbiased and consistent for Δ by linearity. If $|\Delta| > 0$, then almost surely $\operatorname{sign}\widehat{\Delta} = \operatorname{sign}\Delta$ for all large n, so the sign — and indeed the value — of the benefit is identified from labeled source data plus the unlabeled target sample needed to estimate r. □

Why this escapes the impossibility theorem. It is worth dwelling on *where* the witness of Theorem 4.1 dies under the covariate-shift premise, because the mechanism is the template for every positive result in this theory. The witness constructed two worlds agreeing on $P_T(X)$ and differing only in $P_T(Y \mid X)$ — the one coordinate of the target joint that label-free evidence cannot see. Covariate shift removes exactly that degree of freedom: it pins $P_T(Y \mid X)$ to $P_S(Y \mid X)$, an object estimable from *labeled source* data. The accessible data then determine the entire target joint, $p_T(x, y) = p_T(x)\, p_S(y \mid x)$, with $p_T(x)$ read off the unlabeled target stream and $p_S(y \mid x)$ off the source. The evidence has become *sufficient* for Δ. Equivalently: the two witness worlds have different conditionals, so at most one of them can satisfy the covariate-shift premise with respect to any fixed source — inside the assumption class, matched-evidence pairs with opposite benefit

Shift type	Label-free status	Mechanism / theorem
Covariate ($P(Y \mid X)$ fixed)	identifiable	importance weighting (Thm 5.3)
Label / prior ($P(Y)$)	identifiable†	stable class-conditionals, invertible confusion (†BBSE-type)
Disagreement-local	ordinal-identifiable	sign = accuracy gap on D (Thms 5.4–5.6)
Concept ($P(Y \mid X)$ changes)	non-identifiable	re-enters Thm 4.1
Mixed / unknown	certify-or-abstain	commit where Z separates, else abstain

Table 5.2: Knowability hierarchy of distribution shifts for label-free adapt/freeze decisions, as drawn in the paper. The boundary is evidence-identifiability: some shifts are knowable, some are not, and a valid certificate abstains across the boundary. The label-shift row follows the black-box shift estimation conditions of [31].

simply do not exist. Identifiability results of this shape are the classical core of the covariate-shift literature [45, 46, 39]; the delta here is only the target of inference, the *sign of a risk difference* rather than a risk.

The price: weight boundedness and effective sample size. The premise buys identifiability; the moment conditions buy a usable certificate. The weighted paired benefits $r(x_i)(\ell(f_0) - \ell(f_a))$ are bounded in $[-B, B]$ when $r \leq B$, so they can be fed directly to the empirical-Bernstein certificate of Section 5.2.1, and the radius inherits the weight distribution's second moment. Since $\mathbb{E}_S[r] = 1$, the variance inflation relative to unweighted estimation is governed by $\mathbb{E}_S[r^2] \geq 1$, with equality only under no shift; the corresponding *effective sample size* is $n/\mathbb{E}_S[r^2]$ (empirically, $(\sum_i w_i)^2 / \sum_i w_i^2$), and the certificate radius scales like $\sqrt{\mathbb{E}_S[r^2]/n}$. Heavy-tailed weights therefore do not silently corrupt the decision — they *visibly inflate the radius*, and the certificate responds by abstaining or freezing. This is not a hypothetical failure mode: in the regression covariate-shift experiment of Chapter 7, a 4σ shift makes the importance-weighted candidate *hurt* on 79% of instances through sheer weight variance, and KGA refuses it wholesale (0 adapt, 11 freeze, 61 abstain), matching the oracle's mean squared error of 0.251 while always-adapt degrades to 0.338. Identifiability in principle and certifiability at a given n are different currencies, and the certificate trades only in the second.

The caveat that keeps the theorem honest. The paper attaches a caveat tying the result back to Theorem 4.1, and it is load-bearing: the premise $P_S(Y \mid X) = P_T(Y \mid X)$ is *not* testable from unlabeled data; when it fails, the regime can re-enter the unknowable region. Covariate shift is an assumption about precisely the coordinate that label-free evidence cannot observe, so adopting it is an act of modeling, not of inference. The knowability hierarchy of Table 5.2 situates this regime among its neighbors: label shift is similarly identifiable under its own structural premises (stable class-conditionals and an invertible confusion matrix, the BBSE conditions [31]); concept shift — a change in $P(Y \mid X)$ itself — is non-identifiable and lands back in Theorem 4.1; and the mixed or unknown case is exactly where the certify-or-abstain machinery of this chapter operates.

5.5 The Disagreement Region: Where the Sign of Benefit Lives

The covariate-shift result buys identifiability with a global assumption on the shift. The fifth result of the theory takes the opposite route: no assumption on the shift at all, but a sharper look at *where* in input space the decision is actually decided. The observation is elementary and, once made, structural. Wherever the frozen and candidate models agree, their losses are identical and the per-sample benefit is exactly zero — those points contribute *nothing* to Δ, under any label law whatsoever. The entire benefit, sign and magnitude alike, is carried by the *disagreement region*

$$D \;=\; \{x : f_0(x) \neq f_a(x)\},$$

and D is *observable*: membership requires evaluating two known functions, no labels. Restricted to D, the question "does adapting help?" collapses to an above-chance check — is the candidate right more often than the incumbent on the points where they differ? — which is an *ordinal* comparison, not a risk estimate.

Theorem 5.4 (Sign-of-difference identifiability on the disagreement region; binary case)**.** *Let $Y \in \{0,1\}$ with 0/1 loss, and let $D = \{x : f_0(x) \neq f_a(x)\}$ be the* disagreement region *(observable, since f_0, f_a are known maps of X). Then*

$$\Delta \;=\; P_T(D)\,(2\,a_a^D - 1), \qquad a_a^D := P_T(f_a(X) = Y \mid X \in D)\,,$$

so $\operatorname{sign}\Delta = \operatorname{sign}\!\left(a_a^D - \tfrac{1}{2}\right)$*. Consequently* $\operatorname{sign}\Delta$ *is identifiable from label-free evidence* if and only if *the evidence determines whether f_a exceeds chance accuracy on D; in particular, if a label-free reliability signal brackets a_a^D relative to $\frac{1}{2}$, the sign is certified.*

Proof. Let $\delta(x,y) = \ell(f_0(x),y) - \ell(f_a(x),y) = \mathbf{1}[f_0(x) \neq y] - \mathbf{1}[f_a(x) \neq y]$ denote the per-sample benefit, so that $\Delta = \mathbb{E}_{P_T}[\delta]$.

Off D, the benefit vanishes identically. If $x \notin D$ then $f_0(x) = f_a(x)$, so the two indicators coincide for every y and $\delta(x,y) = 0$ pointwise — not merely in expectation, but for each realization of the label.

On D, the benefit is a fair-coin score for f_a. If $x \in D$ the two predictions differ, and since Y is binary, exactly one of $f_0(x), f_a(x)$ equals y. If $f_a(x) = y$ then $f_0(x) \neq y$, so $\delta = 1 - 0 = +1$; if $f_0(x) = y$ then $\delta = 0 - 1 = -1$. Hence, conditionally on D, δ is a ± 1 variable equal to $+1$ with probability a_a^D, and

$$\mathbb{E}[\delta \mid D] = P_T(f_a{=}Y \mid D) - P_T(f_0{=}Y \mid D) = a_a^D - (1 - a_a^D) = 2a_a^D - 1,$$

using that on D the two correctness events are complementary, $P_T(f_0{=}Y \mid D) = 1 - a_a^D$.

Total expectation. Splitting Δ over the partition $\{D, D^c\}$ and using $\delta \equiv 0$ off D,

$$\Delta = \mathbb{E}[\delta\,\mathbf{1}\{X \in D\}] + \mathbb{E}[\delta\,\mathbf{1}\{X \notin D\}] = P_T(D)\,\mathbb{E}[\delta \mid D] = P_T(D)\,(2a_a^D - 1).$$

Since $P_T(D) \geq 0$, the sign of Δ equals the sign of $2a_a^D - 1$, i.e. of $a_a^D - \frac{1}{2}$, whenever $P_T(D) > 0$ (and $\Delta = 0$ when $P_T(D) = 0$, in which case the decision is moot).

The identifiability equivalence. ($\Leftarrow$) If the evidence determines the event $\{a_a^D > \frac{1}{2}\}$ — for instance, a label-free reliability signal brackets a_a^D strictly on one side of $\frac{1}{2}$ — then by the identity it determines $\operatorname{sign}\Delta$. ($\Rightarrow$) Conversely, if two target worlds are compatible with the same evidence yet place a_a^D on opposite sides of $\frac{1}{2}$, the identity forces their benefits to have opposite signs, which is precisely the unknowable configuration of Theorem 4.1; so identifiability of $\operatorname{sign}\Delta$ requires the evidence to settle the above-chance question. $\square$

Remark 5.6 (Ordinal, not cardinal — and exactly one residual unknown). The theorem changes the problem's type. Estimating either absolute risk $R_T(f_0)$ or $R_T(f_a)$ without labels is the unsupervised-risk-estimation problem, known — from the conditional-independence analyses of Steinhardt and Liang and the divergence-based bounds reviewed in the related-work discussion — to be impossible without structural assumptions. The identity shows the adapt/freeze decision never needed those cardinal objects: it needs one *ordinal* bit — is f_a above chance on D — a strictly weaker requirement. Everything else in the formula is label-free: $P_T(D)$ is an observable mass, and the per-sample benefits off D are identically zero. The residual burden is exactly the bracketing of a_a^D around $\frac{1}{2}$, and the witness of Theorem 4.1 shows that burden is real: there $f_0 = \mathbf{1}[x > 0]$ and $f_a = \mathbf{1}[x < 0]$ disagree almost everywhere, so D is essentially the whole line, and the two worlds set $a_a^D = 0$ and $a_a^D = 1$ respectively — maximal disagreement about the ordinal bit, with identical evidence. The witness is thus precisely a failure of label-free bracketing on D, which is why the open problem of Section 5.6 has the shape it has.

The binary identity generalizes in both directions the deployment cares about — multiclass classification and regression — and the generalizations are the results that close the characterization half of the program.

Theorem 5.5 (Multiclass 0/1 loss)**.** *For $Y \in \{1, \ldots, K\}$ with 0/1 loss, with $p_a = P_T(f_a{=}Y \mid D)$, $p_0 = P_T(f_0{=}Y \mid D)$, off D $\delta = 0$ and on D $\mathbb{E}[\delta \mid D] = p_a - p_0$, so $\Delta = P_T(D)(p_a - p_0)$ and* $\operatorname{sign}\Delta = \operatorname{sign}(p_a - p_0)$.

Proof. As before $\delta = \mathbf{1}[f_0 \neq Y] - \mathbf{1}[f_a \neq Y]$, and off D the two indicators coincide pointwise, so $\delta = 0$ there. On D the predictions differ, and — this is the multiclass novelty — the conditional space splits into *three* disjoint events rather than two: $\{f_a = Y\}$ (the candidate is right, the incumbent necessarily wrong since it predicts something different, $\delta = 1 - 0 = +1$), $\{f_0 = Y\}$ (the incumbent is right, the candidate wrong, $\delta = 0 - 1 = -1$), and $\{f_0 \neq Y, f_a \neq Y\}$ (both wrong, predicting two *different* wrong classes, where $\delta = 1 - 1 = 0$). The two correctness events are still disjoint on D — the models cannot both be right while disagreeing — but they are no longer complementary: $p_a + p_0 \leq 1$ with strict inequality whenever both-wrong has mass. Taking the conditional expectation over the three cells,

$$\mathbb{E}[\delta \mid D] = (+1)\, p_a + (-1)\, p_0 + 0 \cdot (1 - p_a - p_0) = p_a - p_0,$$

which one can equally read off the complement form $\mathbb{E}[\delta \mid D] = (1-p_0)-(1-p_a)$. Total expectation over $\{D, D^c\}$ gives $\Delta = P_T(D)\,\mathbb{E}[\delta \mid D] = P_T(D)(p_a - p_0)$, and nonnegativity of $P_T(D)$ gives the sign identity. For $K = 2$, exactly one predictor is correct on D, so $p_0 = 1 - p_a$, the both-wrong cell is empty, and the formula reduces to $\Delta = P_T(D)(2a_a^D - 1)$ — Theorem 5.4. For $K \geq 3$ both may err, and the identity holds regardless, because the benefit counts only who is *correct*, not whether the other is wrong. □

Theorem 5.6 (Squared-error regression)**.** *For $Y \in \mathbb{R}$ with squared loss, $\delta = (f_0 - f_a)(f_0 + f_a - 2Y)$, so on D $\mathbb{E}[\delta \mid D] = \mathbb{E}[(f_0 - f_a)(f_0 + f_a) \mid D] - 2\,\mathbb{E}[(f_0 - f_a)Y \mid D]$. The first term is observable from unlabeled target; thus* $\operatorname{sign}\Delta$ *is label-free identifiable iff the sign contribution of* $\mathbb{E}[(f_0 - f_a)Y \mid D]$ *is certifiable—which holds under covariate shift (importance-weighted source estimate of* $\mathbb{E}[Y \mid X]$*) and fails under concept shift (re-entering Theorem 4.1).*

Proof. The pointwise identity is the difference of squares:

$$\delta = (f_0 - Y)^2 - (f_a - Y)^2 = \big[(f_0 - Y) - (f_a - Y)\big]\big[(f_0 - Y) + (f_a - Y)\big] = (f_0 - f_a)\,(f_0 + f_a - 2Y).$$

Off D we have $f_0(x) = f_a(x)$, so the first factor vanishes and $\delta = 0$; the disagreement region again carries the whole benefit, and $\Delta = P_T(D)\,\mathbb{E}[\delta \mid D]$ by total expectation. On D, expanding by linearity,

$$\mathbb{E}[\delta \mid D] = \mathbb{E}\big[(f_0 - f_a)(f_0 + f_a) \,\big|\, D\big] \;-\; 2\,\mathbb{E}\big[(f_0 - f_a)\,Y \,\big|\, D\big].$$

The first term is a functional of the two *known* maps f_0, f_a and of $\mathrm{Law}(X \mid D)$ only — estimable from the unlabeled target stream to arbitrary precision. The second term is the unique label-coupled object, so $\operatorname{sign}\Delta$ is label-free identifiable exactly when that term's sign contribution can be certified without target labels. Under covariate shift it can: writing $m(X) = \mathbb{E}[Y \mid X]$ (the same function in source and target by the premise), the tower property and $\sigma(X)$-measurability of $f_0 - f_a$ give

$$\mathbb{E}\big[(f_0 - f_a)\,Y \,\big|\, D\big] = \mathbb{E}\big[(f_0 - f_a)\,\mathbb{E}[Y \mid X] \,\big|\, D\big] = \mathbb{E}\big[(f_0 - f_a)\,m(X) \,\big|\, D\big],$$

where m is estimable from *labeled source* data (importance-weighted as in Theorem 5.3) and the remaining ingredients are target-observable. Under concept shift the source regression function m_S is the wrong m, the plug-in can carry the wrong sign, and no label-free correction is available — the configuration re-enters Theorem 4.1. □

5.5.1 Numerical Validation of the Identities

Algebraic identities invite the most direct kind of machine checking, and the repository's validator `experiments/kbound/theory_validation/val_thm5_multiclass.py` [25] performs it at scale. For the multiclass identity it draws 4000 random trials — class counts K ranging over $\{3, \ldots, 11\}$, sample sizes between 2000 and 8000, predictors of independently random accuracies so that

Δ takes both signs across trials — and computes Δ two ways: directly with labels (the mean per-sample benefit), and through the decomposition $P_T(D)(p_a - p_0)$. As recorded in the paper, the identity holds to 10^{-16} with 100% sign agreement over the 4000 trials — and, decisively for the beyond-binary claim, the both-wrong-on-D event occurs in *every* trial, so the regime genuinely exercised is the one where $p_0 \neq 1 - p_a$. The script additionally verifies that the off-D contribution to Δ is exactly zero, sample by sample. For regression it checks the pointwise difference-of-squares identity to machine precision over 3000 trials (tolerance 10^{-9}, with 100% sign agreement between Δ and $\mathbb{E}[\delta \mid D]$), and then runs the shift demonstration of Theorem 5.6 end to end: with 40,000 source and target points, a 2.0-standard-deviation covariate shift, and a source-only kernel-regression plug-in for $m(X)$, the label-free certificate recovers the correct sign Δ under covariate shift and *flips* sign under concept shift — the predicted failure, observed on cue. The validation does not, of course, prove the theorems (the proofs above do); what it certifies is that the implemented quantities are the theorems' quantities, with no sign conventions or conditioning slips between paper and code.

5.5.2 Disagreement as the Decisive Evidence Coordinate

The theorems of this section make a falsifiable prediction about *evidence design*: if the entire benefit is carried by D, then among the components of the evidence vector Z, the ones measuring disagreement between the frozen and candidate models should be the most decision-relevant — not marginally, but structurally. The ablation study of Chapter 7 tests exactly this by deleting evidence features one at a time and re-measuring the wrapper's regret-to-oracle: removing the detector-disagreement feature raises regret from 0.037 to 0.094, a 2.6× degradation and by far the largest of any feature (the next-largest deletions move regret to 0.043 and 0.039). The paper draws the connection explicitly: detector disagreement is the single most important evidence feature in the ablation, directly supporting that the disagreement region carries the decision. The theory says where the information lives; the ablation confirms the estimator starves without it.

5.6 The Open Problem: Label-Free Bracketing

Everything proved so far reduces the label-free adapt/freeze decision to a single remaining object: an ordinal comparison on the observable region D. The paper is exact about what is closed and what is not, and states the open piece as a conjecture. We reproduce it verbatim, and then spend an honest page on why it is genuinely open.

Conjecture 5.1 (Label-free bracketing; **the remaining open piece**). *Theorems 5.5–5.6* close the characterization*:* sign Δ *equals an ordinal accuracy comparison on the observable region D. What remains open is the label-free* bracketing *of that comparison (p_a vs p_0 for $K \geq 3$, where $p_0 \neq 1 - p_a$ makes it a two-sided test; or the sign of $\mathbb{E}[(f_0 - f_a)Y \mid D]$ for regression) without target labels—a reliability-model assumption, as in the binary case.*

5.6.1 Why Current Techniques Do Not Resolve It

To see the precise difficulty, fix the multiclass case and ask for the object the conjecture demands: a label-free functional of the observables — the unlabeled target stream on D, the two prediction maps, and whatever source-side calibration data one allows — that outputs an interval $[L, U] \ni p_a - p_0$ with $1 - \alpha$ coverage, such that committing when $0 \notin [L, U]$ is valid *without* a premise that is itself unverifiable from the same observables. Every known route founders on one of two obstructions.

The information obstruction. Unconditionally — over all label laws on D consistent with the observed unlabeled data — the task is hopeless, and this is a theorem, not a suspicion: the witness of Theorem 4.1, restricted to D, exhibits two worlds with identical evidence laws and (p_a, p_0) equal to $(1, 0)$ and $(0, 1)$ respectively. Any bracket with valid coverage in both worlds must contain both $+1$ and -1, hence 0, hence never commits. This is the two-point method of Le Cam in its simplest form [28, 49]: the unlabeled likelihood does not separate the hypotheses, so additional structure is *necessary*, and the conjecture is properly read as a question about the *weakest* usable structure rather than about removing structure altogether. The binary case already had this character — there the residual assumption is a reliability signal bracketing a_a^D against $\frac{1}{2}$ — but the multiclass case is strictly harder in a concrete sense the paper flags in the conjecture itself: since $p_0 \neq 1 - p_a$, one scalar no longer determines the comparison, and the test becomes two-sided in two unknown conditional accuracies linked only by the inequality $p_a + p_0 \leq 1$.

The technique obstruction. Each available tool discharges the bracketing only by importing a premise of the same epistemic type as the one being discharged. Importance weighting (Theorem 5.3) certifies the regression term $\mathbb{E}[(f_0 - f_a)Y \mid D]$ — but only under covariate shift, which fixes the unobservable conditional $P(Y \mid X)$ by fiat and is untestable from unlabeled data. Confidence-based bracketing (the route formalized below) bounds p_h by the mean calibrated confidence q_h — but only under a calibration-transfer premise, a bound η on the *target-side* miscalibration on D, which is again a quantity no label-free procedure can estimate; assuming η small is assuming a reliability model. Label-shift machinery [31] identifies shifted class priors under stable class-conditionals and an invertible confusion matrix — structural premises once more, and even granted them, priors do not by themselves yield the *conditional-on-D* accuracies the comparison needs. Empirical regularities such as agreement-based accuracy extrapolation are calibrated correlations, not guarantees, and nothing prevents the adversarial pairs of Theorem 4.1 from sitting exactly where the regularity breaks. In every case the assumption is doing the work precisely on the coordinate the data cannot see; the open question is whether some premise exists that is weaker, or — more interestingly — *self-certifying*, in the sense that its failure is itself label-free detectable, so that the bracket could widen automatically where the premise degrades.

5.6.2 What Is Actually Proved

The honest ledger has four entries on the proved side. First, the *characterization* is closed unconditionally: Theorems 5.4 through 5.6 are exact identities requiring no assumptions beyond the loss structure, and they localize the entire decision to D. Second, the binary case needs only a *one-sided* bracket of a single scalar against the fixed threshold $\frac{1}{2}$ — a structurally easier object than the multiclass two-sided comparison, which is why the binary instantiation is the one deployed throughout the experiments. Third, the regression term is certified under covariate shift (Theorem 5.6 with Theorem 5.3). Fourth, the paper proves a *conditional* closure of the conjecture under a quantitative reliability model — bounded calibration transfer — with an exact and provably tight threshold. We restate it faithfully.

Proposition 5.2 (Label-free bracketing under bounded calibration transfer)**.** *On the disagreement region D, let $\mathrm{conf}_h(x) \in [0,1]$ be a source-calibrated confidence map for predictor $h \in \{f_0, f_a\}$, and suppose its target miscalibration on D is bounded by η: $|\mathbb{P}_T(h(X){=}Y \mid \mathrm{conf}_h{=}c,\ X \in D) - c| \le \eta$ for all c (*calibration transfer*). Let $q_h := \mathbb{E}_T[\mathrm{conf}_h(X) \mid D]$ be the label-free mean confidence. Then $|q_h - p_h| \le \eta$, hence $p_a - p_0 \in [(q_a - q_0) - 2\eta,\ (q_a - q_0) + 2\eta]$, and*

$$|q_a - q_0| > 2\eta \implies \operatorname{sign}\Delta = \operatorname{sign}(q_a - q_0),$$

so the multiclass/regression benefit sign is identified label-free; when $|q_a - q_0| \le 2\eta$ the bracket contains 0 and the rule must ABSTAIN.

Proof. By the tower property, $p_h = \mathbb{E}_T\big[\mathbb{P}_T(h{=}Y \mid \mathrm{conf}_h, D) \mid D\big]$; the calibration-transfer bound says the inner conditional probability is within η of conf_h pointwise, so integrating preserves the bound and $|p_h - q_h| \le \eta$ for each h. Differencing the two η-balls gives $|(p_a - p_0) - (q_a - q_0)| \le 2\eta$, the stated bracket. If $|q_a - q_0| > 2\eta$, the bracket lies strictly on one side of 0, so $\operatorname{sign}(p_a - p_0) = \operatorname{sign}(q_a - q_0)$; by Theorem 5.5 this equals $\operatorname{sign}\Delta$. If $|q_a - q_0| \le 2\eta$ the bracket straddles 0 and certifies nothing, so the certificate logic of Theorem 5.1 mandates abstention. $\square$

The threshold 2η is *exact*, as the paper notes: when $|q_a - q_0| \le 2\eta$ an adversary inside the two η-balls can place p_a, p_0 on either side of equality, re-entering the unknowable regime of Theorem 4.1 — so the proposition's abstention region cannot be shrunk under its own premise. The validator (`val_conj1_caltransfer.py`, results in `results_conj1_caltransfer.json` [25]) sweeps both η and the true gap: with a true gap of 0.2 ($p_a = 0.6$, $p_0 = 0.4$), every committed decision is sign-correct (100% recovery among committed, 0 wrong commitments) for all $\eta \in \{0, 0.02, 0.05, 0.08\}$ where the gap exceeds 2η; at the knife edge $\eta = 0.1$ (gap exactly 2η) the finite-sample rule still commits on 52.45% of trials — all of them sign-correct, with the wrong-commit rate still 0 — and for $\eta \ge 0.15$ commitment falls to 0, full abstention. The dual gap sweep at $\eta = 0.05$ shows the same wall from the other side (no commitments below gap 0.10, full commitment from gap 0.12 up), and the bracket $[(q_a - q_0) - 2\eta, (q_a - q_0) + 2\eta]$ contains the true gap on 100% of trials. The headline fields record minimum sign recovery 1.0 and maximum wrong-commit rate 0.0 whenever the gap exceeds 2η — the guarantee, observed.

5.6.3 What a Resolution Would Buy

We state plainly that the conjecture is *not* solved here, and that Proposition 5.2 does not solve it: it trades the open assumption for a quantified one, which is progress in precision, not in kind. A genuine resolution would change the system's standing in either direction. A *positive* resolution — a bracketing scheme whose premise is label-free verifiable, or a proof that some natural observable (margin spectra on D, agreement geometry among auxiliary predictors, weight diagnostics) suffices — would make the entire trichotomy assumption-free given evidence: KGA's residual reliability-model assumption would vanish, the knowable region of the phase diagram would extend to every shift with nonvacuous brackets, and evidence design would reduce to estimating two scalars on an observable set. A *negative* resolution — a lower bound showing every label-free bracket valid over a natural shift class must contain 0 unless a premise of calibration-transfer strength holds — would be just as valuable: it would certify that the conditional results of this chapter are the strongest possible, closing the program with matching upper and lower bounds the way Theorem 4.1 closed the unconditional question. Either outcome turns the last assumption in the stack into a theorem. Until then, the honest summary is the paper's own: the characterization is closed; the label-free estimability of the ordinal comparison is the standing assumption; the unconditional bracketing is the one residual open piece.

5.7 Synthesis: The Certified Map of Regimes

Chapters 4 and 5 together draw one map. The impossibility theorem fences off the region where no label-free rule may commit; the certificate gives the safe-passage protocol everywhere else; the covariate-shift and disagreement-region results chart where the passage provably opens; and the conjecture marks the one stretch of coastline still unsurveyed. Table 5.3 assembles the chapter's results with their guarantees and the machine-checked evidence behind each, with the experimental chapters and Appendix B holding the full per-condition records.

Three through-lines deserve a final statement. First, every positive result in this chapter is *conditional in a way the theory itself prices.* The certificate is valid given a valid radius; the radius is valid given exchangeability of the calibration bank; covariate-shift identifiability is valid given an untestable premise on $P(Y \mid X)$; the bracketing is valid given calibration transfer. None of this is a defect to be hidden — it is the content of Theorem 4.1 propagating through the constructions, and the certificate's distinctive virtue is that where its premises fail detectably (heavy importance weights, wide residual quantiles, sub-threshold confidence gaps) the failure surfaces as a *wider radius*, and the rule degrades to abstention rather than to silent error. Second, the disagreement region is the load-bearing structure connecting theory to practice: it is where the benefit lives (Theorems 5.4–5.6), why the empirical-Bernstein radius is tight (Remark 5.3), and which evidence coordinate the deployed estimator cannot do without (the 2.6× ablation of Chapter 7). Third, the guarantees proved here are exactly the ones the deployed system claims —

Regime	Result	Guarantee	Validator / evidence
Matched evidence (unknowable)	Thm 4.1; Cor. 5.1	no label-free rule certifies both worlds; any valid certificate abstains with probability $\geq 1 - 2\alpha$	clean witness: 100% abstention, forced-commit regret 0.51 (Ch. 7)
Risk-aligned evidence, batch	Thm 5.1; Cor. 5.2	false-adapt $\leq \alpha$ and false-freeze $\leq \alpha$; commits once $n \gtrsim \sigma^2 \Delta^{-2} \log(1/\alpha)$	mixed regime: false-adapt 0.065 at $\alpha = 0.1$; stress grid: 0% false-adapt (Chs. 7–8)
Risk-aligned evidence, streaming	Thm 5.2	anytime-valid: error $\leq \alpha$ simultaneously over all stopping times	worst anytime false-adapt $0.0626 \leq 0.1$ (`val_thm3_evalue_results.json`)
Covariate shift	Thm 5.3	Δ and $\operatorname{sign} \Delta$ identifiable from labeled source + unlabeled target	regression demo: sign correct under covariate shift, flips under concept shift (`val_thm5_multiclass.py`)
Disagreement region	Thms 5.4–5.6	$\operatorname{sign} \Delta$ = ordinal accuracy comparison on observable D (exact identity)	identity to 10^{-16}, 100% sign agreement over 4000 trials (`val_thm5_multiclass.py`)
Multiclass / regression bracketing	Conj. 5.1 (open); Prop. 5.2 (conditional)	sign certified iff $\lvert q_a - q_0 \rvert > 2\eta$ under calibration transfer; unconditional case open	100% sign recovery, 0 wrong commitments when gap $> 2\eta$ (`results_conj1_caltransfer.json`)

Table 5.3: The certified map of regimes assembled by Chapters 4 and 5: each row pairs a regime with its governing result, the exact form of its guarantee, and the machine-checked validator or experiment that exercises it.

no more. Chapter 6 assembles $Z \to \widehat{\Delta} \to \varepsilon \to \{\textsc{adapt}, \textsc{freeze}, \textsc{abstain}\}$ into the KGA wrapper with the assumptions of this chapter stated as the algorithm's assumptions, and Chapters 7 and 8 then measure the certificate against worlds engineered to reward it, punish it, and — on the constructed witness — defeat it, which is the only honest order of business after a chapter of conditional guarantees.

Chapter 6

KGA: Knowability-Guided Adaptation in Practice

The preceding chapters establish when label-free adaptation is theoretically knowable—and when it is not. This chapter describes KGA (Knowability-Guided Adaptation), the concrete decision algorithm that operationalises those results. We present the design goals, the four families of label-free evidence, the four certificate estimators with their exact formulas, the decision policy and its false-adapt guarantee, the public `kga` package API, a full deployment pseudocode, operational guidance, the served HTTP interface, and an honest account of what KGA does not guarantee.

6.1 Design goals and mechanism-agnostic wrapper architecture

KGA is a *decision wrapper*, not a new adaptation mechanism. Its single responsibility is to answer, *before* any irreversible update, the three-way question: should this candidate adaptation f_a replace the frozen model f_0 on this unlabelled target batch? The adapting mechanism itself—entropy-minimising Tent [52], sample-filtered EATA [36], sharpness-aware SAR [37], or any other algorithm—is a black box that supplies f_a; KGA wraps it uniformly.

This design is dictated by three requirements.

1. **Label-free.** The gate must decide from evidence $Z = \phi(X_{1:m}, f_0, f_a)$ that is computable without any target label. No target-label signals may enter at any step—neither for benefit estimation, nor for certificate construction.

2. **Mechanism-agnostic.** A laboratory may wish to evaluate several candidate adaptation strategies (Tent vs. EATA vs. SAR) under the same safety envelope. By treating f_a as an input, KGA provides that shared envelope at the cost of one forward pass of f_0 and f_a per decision.

3. **Finite-sample valid.** The false-adapt probability—the chance of deploying an f_a that

actually hurts—must be bounded at a user-specified level α even for small batches. This is Theorem 5.1 from Chapter 4.

The overall architecture (Figure 6.1) places KGA between the adaptation candidate and the live deployment loop. Calibration data (labelled validation scores) flows in once to build the calibration residuals needed for the split-conformal path; thereafter each incoming unlabelled batch is processed in three stages: evidence computation, certificate estimation, and the trichotomy decision.

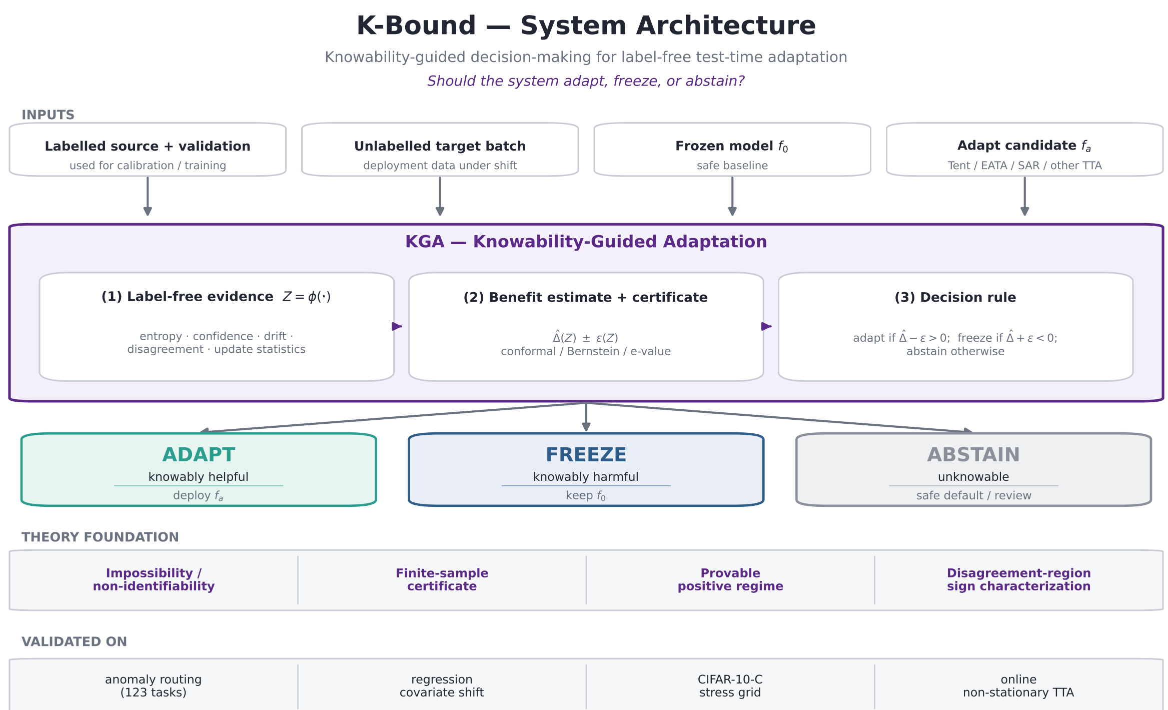

Figure 6.1: KGA system architecture. Labelled calibration scores and an unlabelled test batch enter the pipeline. The label-free evidence Z is computed first; then a finite-sample certificate $\widehat{\Delta} \pm \varepsilon$ is formed at level α; finally the trichotomy maps the certificate to one of three actions. The adaptation mechanism (Tent, EATA, SAR, or any other) is a plug-in; KGA wraps it without modification.

6.2 Label-free evidence computation

The evidence Z is a vector of statistics computable from the calibration scores S_{val} and the unlabelled test scores S_{test}, without any target labels. The implementation lives in `kga/evidence.py`: the dataclass `Evidence` and the function `compute_evidence(calib_scores, test_scores)`. The signals fall into four families, each tied to a theoretical result.

Drift (KS distance). For each detector column j, a two-sample Kolmogorov–Smirnov test (`scipy.stats.ks_2samp`) compares the calibration and test score marginals. The implementation stores:

- `ks_mean` — mean per-detector KS statistic;
- `ks_max` — maximum per-detector KS statistic.

Large KS drift is the observable signature of a covariate shift to which the gate can react (Theorem 5.3). When the marginal is preserved but the informative signal is destroyed—the "covert failure" regime of the mixed-regime experiment—both statistics remain near zero and the distribution is non-identifiable in the sense of Theorem 4.1.

Disagreement. The disagreement signal measures the heterogeneity of the test scores across detectors:

$$\texttt{disagree} \;=\; 1 - \overline{\rho}, \tag{6.1}$$

where $\overline{\rho}$ is the mean pairwise Spearman rank correlation of the test score columns (computed via `_rank_norm` then `np.corrcoef` on the rank-normalised test matrix). For a single detector, `disagree` is defined to be zero. A larger value indicates a bigger disagreement region $D = \{x : f_0(x) \neq f_a(x)\}$, and thus greater potential for the sign-of-difference to be identifiable (Theorems 5.4–5.1).

Entropy and confidence shift. The entropy signals measure the spread of the pooled score distributions. `_hist_entropy` computes the Shannon entropy (in nats) of the 20-bin normalised histogram of a flattened score array:

$$H(S) \;=\; -\sum_k p_k \ln p_k.$$

The confidence (decisiveness) signal `_decisiveness` measures the mean distance of the min-max-normalised scores from the indecision point 0.5:

$$\texttt{conf}(S) \;=\; 2\,\mathbb{E}\left[\left|\frac{S - S_{\min}}{S_{\max} - S_{\min}} - 0.5\right|\right].$$

The `Evidence` dataclass stores:

- `calib_entropy`, `test_entropy` — per-split entropies;
- `calib_conf`, `test_conf` — per-split decisiveness;
- `entropy_shift` = `test_entropy` − `calib_entropy`;
- `conf_shift` = `test_conf` − `calib_conf`.

A large positive `conf_shift` with a collapsing marginal is the entropy-collapse warning sign familiar from online TTA monitoring.

Importance-weight effective sample size (ESS). The function `_importance_ess` models the calibration and test score laws as univariate Gaussians and forms the Gaussian-ratio importance weight for each test point t_i:

$$w_i \;=\; \frac{\mathcal{N}(t_i;\, \mu_c, \sigma_c^2)}{\mathcal{N}(t_i;\, \mu_t, \sigma_t^2)},$$

where (μ_c, σ_c) and (μ_t, σ_t) are the empirical moments of the pooled calibration and test scores respectively. The effective sample size is then the standard self-normalised estimator,

$$\mathrm{ESS} \;=\; \frac{(\sum_i w_i)^2}{\sum_i w_i^2}, \tag{6.2}$$

capped in $[1, n_{\text{test}}]$. The dataclass stores `ess` and the scale-free ratio `ess_frac` $= \mathrm{ESS}/n_{\text{test}}$. Low `ess_frac` indicates poor source-to-target support overlap, which undermines importance-weighted certificates and is a quantitative diagnostic for the support-overlap condition of Theorem 5.3.

The fixed-order feature vector `Evidence.to_vector()` concatenates [`ks_mean`, `ks_max`, `disagree`, `entropy_sh`
and can be passed directly to a downstream benefit regressor as the feature vector Z.

Remark 6.1 (Provenance). Every field of `Evidence` mirrors a quantity computed in the K-Bound experiment scripts: `kga/evidence.py` is a clean, importable API over the same mathematics, not a re-derivation. The KS drift and disagreement fields match the $Z = \mathrm{dict}(\ldots)$ block in `src/scripts/kbound/knowability_experiment.py` (lines 74–96); the entropy and importance-weight fields match the `evidence` helper in `mixed_regime_experiment.py`.

6.3 Certificate estimators

A *certificate* is a frozen dataclass `Certificate(delta_hat, epsilon, method, alpha, n)` implemented in `kga/certificate.py`. It carries a point estimate $\widehat{\Delta}$ of the benefit $\Delta = R(f_0) - R(f_a)$ and a confidence radius ε such that, with probability at least $1 - \alpha$,

$$\Delta \;\geq\; \widehat{\Delta} - \varepsilon \qquad \text{(certified lower bound)}.$$

The two derived properties `lower` $= \widehat{\Delta} - \varepsilon$ and `upper` $= \widehat{\Delta} + \varepsilon$ are the bounds consumed by the decision policy.

The module provides four estimators, each linked to a theoretical result.

6.3.1 Empirical Bernstein (`ebern`) — the default

The default estimator implements the Maurer–Pontil (2009) empirical-Bernstein lower confidence bound [34]. Given n i.i.d. paired benefits $X_i = \ell(f_0(x_i), y_i) - \ell(f_a(x_i), y_i)$ with empirical mean $\widehat{\mu}$, unbiased sample variance $\widehat{V}$ (denominator $n-1$), and range $R = b - a$, the certificate is:

$$\varepsilon_{\text{ebern}} \;=\; \sqrt{\frac{2\,\widehat{V}\,\ln(2/\alpha)}{n}} \;+\; \frac{7R\,\ln(2/\alpha)}{3(n-1)}, \tag{6.3}$$

so that $\mathbb{P}[\Delta \geq \widehat{\mu} - \varepsilon_{\text{ebern}}] \geq 1 - \alpha$. The estimate is $\widehat{\Delta} = \widehat{\mu}$ and `epsilon` is the sum of the two terms in (6.3). When $n = 1$ the variance is undefined and `epsilon` is set to $+\infty$, forcing ABSTAIN. When `benefit_range` is not supplied, R is estimated from the observed range `max(X) - min(X)`.

This estimator is strictly tighter than Hoeffding whenever $\widehat{V} \ll R^2$ — the usual regime in which most paired benefits agree in sign — and is deterministic, streamable, and finite-sample valid. It is the batch Theorem 3 certificate of Theorem 5.1.

6.3.2 Hoeffding (`hoeffding`) — distribution-free baseline

The Hoeffding estimator replaces the variance term by the worst-case variance-free bound:

$$\varepsilon_{\text{hoeff}} \;=\; R\sqrt{\frac{\ln(1/\alpha)}{2n}}. \tag{6.4}$$

This radius is always at least as large as the empirical-Bernstein one in the low-variance regime, so it is provided as a conservative baseline and sanity comparator. The `method` field is set to `'hoeffding'`.

6.3.3 Split-conformal (`conformal`) — cross-task estimator

The split-conformal estimator is the radius used in the main K-Bound experiments (Theorem 5.1, conformal variant; [51, 1]). It takes a point estimate $\widehat{\Delta}$ (e.g., from a leave-one-out benefit regressor) and a vector of held-out calibration residuals $r_i = |\widehat{\Delta}_i - \Delta_i|$ from n_{cal} tasks:

$$\varepsilon_{\text{conf}} \;=\; \text{quantile}(r_{1:n},\, 1-\alpha)\,. \tag{6.5}$$

By the exchangeability of the calibration split, $\mathbb{P}[|\widehat{\Delta} - \Delta| \leq \varepsilon_{\text{conf}}] \geq 1-\alpha$ for a fresh query. The function `conformal_split(delta_hat, calib_residuals, alpha)` is called in `knowability_experiment.py` (`eps = np.quantile(resid, 1 - alpha)`) and in `mixed_regime_experiment.py`.

6.3.4 Anytime e-value (`evalue`) — Theorem 3b

The anytime estimator implements two one-sided betting e-processes over an ordered benefit stream [50, 55]:

$$E_t^+ = \prod_{i \le t}(1 + \lambda_i X_i), \qquad \text{tests } H_0^+ : \Delta \le 0, \tag{6.6}$$

$$E_t^- = \prod_{i \le t}(1 + \nu_i(-X_i)), \qquad \text{tests } H_0^- : \Delta \ge 0, \tag{6.7}$$

with predictable truncated-aGRAPA bets $\lambda_i = \text{clip}(\widehat{\mu}_{i-1}/\widehat{\sigma}^2_{i-1},\ 0,\ \delta_{\text{cap}}/|a|)$ (and symmetrically ν_i) formed from the running mean and second moment up to step $i-1$. Each process is a nonnegative supermartingale under its null; by **Ville's inequality** [50], $\mathbb{P}(\exists t : E_t \ge 1/\alpha) \le \alpha$ simultaneously over all sample sizes with no multiplicity correction.

The returned certificate encodes the current anytime decision in the standard $\widehat{\Delta} \pm \varepsilon$ form:

- If $\log E_t^+ \ge \log(1/\alpha)$: emit $\varepsilon = 0$ with $\widehat{\Delta} > 0$, forcing ADAPT;
- If $\log E_t^- \ge \log(1/\alpha)$: emit $\varepsilon = 0$ with $\widehat{\Delta} < 0$, forcing FREEZE;
- Otherwise: emit $\varepsilon = |\bar{X}| + \epsilon_0$ (brackets zero around the running mean), forcing ABSTAIN.

This is Theorem 5.2 (anytime certificate). The implementation mirrors `experiments/kbound/theory_validation/` and `kga/kbound_pkg/kbound/eprocess.py`.

6.4 Decision policy and the false-adapt guarantee

The decision rule is a deterministic threshold on the certificate bounds, implemented as the `decide(certificate, alpha)` function in `kga/policy.py`. The `Decision` enum has three members: `ADAPT`, `FREEZE`, `ABSTAIN`. As a `str` subclass, each member compares equal to its string value and serialises cleanly to JSON.

The rule applies strict inequalities at the boundaries:

Condition	Decision
$\widehat{\Delta} - \varepsilon > 0$	ADAPT
$\widehat{\Delta} + \varepsilon < 0$	FREEZE
otherwise	ABSTAIN

A certificate with $\widehat{\Delta} = \varepsilon$ (lower bound exactly zero) ABSTAINs rather than ADAPTs; this is the conservative choice and matches the reference experiment scripts. An infinite ε (e.g., from a single-sample empirical-Bernstein certificate) always ABSTAINs.

Theorem 6.1 (False-adapt guarantee; Theorem 5.1)**.** *The certificate is constructed so that $\mathbb{P}[\Delta \geq \widehat{\Delta} - \varepsilon] \geq 1 - \alpha$. The* ADAPT *branch fires only when $\widehat{\Delta} - \varepsilon > 0$, which on the good event forces $\Delta > 0$. Therefore*

$$\mathbb{P}(\textit{ADAPT and } \Delta \leq 0) \ \leq \ \alpha.$$

The symmetric statement bounds the false-freeze probability. The anytime e-value variant (Theorem 5.2) upgrades this to hold simultaneously over all stream prefixes via Ville's inequality.

The proof follows identically from the non-identifiability regime (via Theorem 4.1): when evidence Z cannot separate the two opposite-benefit worlds, the certificate radius brackets zero and KGA ABSTAINs. This is mandatory, not merely conservative (see Conjecture 5.1 and the discussion in Chapter 4 on forced abstention).

6.5 The `kga` package API

The `kga` Python package (top-level directory `kga/`) exposes the three-stage pipeline through the `KGA` facade class in `kga/kga.py`.

Constructor.

`KGA(alpha=0.1, method='ebern')`

`alpha` is the operating miscoverage level in $(0, 1)$; it bounds the false-adapt probability via Theorem 5.1. `method` selects the default batch certificate estimator when per-sample paired benefits are supplied: `'ebern'` (empirical Bernstein; the default), `'hoeffding'`, or `'evalue'` (anytime Theorem 3b). A `KGA` instance carries no learned state, so a single instance is reusable and thread-safe.

Methods.

- **`.evidence(calib, test, **kwargs) -> Evidence`** — computes the label-free evidence Z by delegating to `compute_evidence`; stores the result in `last_evidence`.
- **`.certify(...) -> Certificate`** — builds $\widehat{\Delta} \pm \varepsilon$. Three calling conventions are supported:
  1. `certify(scores=benefits)` — per-sample paired benefits, processed by the batch estimator selected at construction (or overridden via `method=`);
  2. `certify(adapt_risk=..., freeze_risk=..., calib_residuals=...)` — two scalar risks plus held-out residuals; uses the split-conformal radius;
  3. `certify(delta_hat=..., calib_residuals=...)` — explicit point estimate plus residuals for split-conformal.

  The result is stored in `last_certificate`.

- **`.decide(certificate=None) -> Decision`** — applies the trichotomy to a certificate (defaults to `last_certificate`); stores the result in `last_decision`.
- **`.explain() -> dict`** — returns all intermediate quantities (`alpha`, `method`, `evidence`, `certificate`, `decision`) as a JSON-serialisable dict.

Quickstart. The following usage snippet is taken directly from `kga/README.md`:

```
import numpy as np
from kga import KGA

rng = np.random.default_rng(0)
kga = KGA(alpha=0.1, method="ebern")

# (1) Label-free evidence Z.
calib = rng.normal(0.0, 1.0, size=(500, 3))
test  = rng.normal(0.0, 1.0, size=(500, 3))
z = kga.evidence(calib, test)
print(z.ks_mean, z.disagree, z.ess_frac)

# (2) Certificate from per-sample paired benefits.
benefits = rng.normal(0.3, 0.1, size=400)
cert = kga.certify(scores=benefits, benefit_range=2.0)
print(cert.delta_hat, cert.epsilon, cert.lower)

# (3) Trichotomy decision.
print(kga.decide(cert))     # -> Decision.ADAPT

# (4) Audit.
import json; print(json.dumps(kga.explain(), indent=2))
```

Command-line interface. The `kga.cli` module (entry point `python -m kga`) exposes a `decide` subcommand:

```
python -m kga decide --calib calib.npy --test test.npy --alpha 0.1
```

This loads calibration and test scores from `.npy` files, computes the evidence Z, constructs a split-conformal certificate with a label-free residual proxy (`delta_hat = 0`, radius from the calibration's own dispersion), and prints the decision and certificate as JSON to stdout. Because labels are unavailable at the command line, it should be used for pipeline integration testing rather than live deployment decisions; real benefit estimates require the `KGA` API with paired benefits or held-out residuals.

6.6 Full deployment pseudocode

Algorithm 1 below gives the complete deployment loop, including the one-time calibration phase and the per-batch inference phase.

Algorithm 1: KGA Deployment Loop.

Inputs: frozen model f_0; adaptation candidate f_a; calibration split $(S_{\text{val}}, y_{\text{val}})$; unlabelled stream of target batches $\mathcal{B}_1, \mathcal{B}_2, \ldots$; miscoverage level α; certificate method $m \in \{\texttt{ebern}, \texttt{hoeffding}, \texttt{evalue}\}$.

Calibration phase (once):

1. Compute paired benefits $X_i = \ell(f_0(x_i), y_i) - \ell(f_a(x_i), y_i)$ on the calibration split.
2. Fit a benefit estimator $\widehat{\Delta}(\cdot)$ (e.g., a leave-one-task-out gradient-boosted regressor) on (Z_i, X_i) pairs.
3. Record calibration residuals $r_i = |\widehat{\Delta}(Z_i) - \bar{X}_i|$ for the split-conformal path, or retain $\{X_i\}$ for the batch path.

Per-batch inference phase (for each $\mathcal{B}_t$):

4. **Evidence:** compute $Z_t = \phi(S_{\text{val}}, S_t, f_0, f_a)$ via `kga.evidence(calib, test)`. Extract [`ks_mean`, `ks_max`, `disagree`, `entropy_shift`, `conf_shift`, `ess_frac`].
5. **Certify:** obtain $\widehat{\Delta}_t$ from the benefit estimator; call `kga.certify(scores=...)` or `kga.certify(delta_hat=`$\widehat{\Delta}_t$`, calib_residuals=r)` to form ε_t at level α.
6. **Decide (trichotomy):**
   - **if** $\widehat{\Delta}_t - \varepsilon_t > 0$: $\Rightarrow$ **adapt** — deploy f_a on $\mathcal{B}_t$;
   - **elif** $\widehat{\Delta}_t + \varepsilon_t < 0$: $\Rightarrow$ **freeze** — keep f_0 on $\mathcal{B}_t$;
   - **else**: $\Rightarrow$ **abstain** — keep f_0, log for human review.
7. **Log:** write $(Z_t, \widehat{\Delta}_t, \varepsilon_t, \texttt{decision}_t, t)$ to the audit trail (`kga.explain()`).

Guarantee (Theorem 5.1): $\mathbb{P}(\textsc{adapt} \text{ and } \Delta \le 0) \le \alpha$; similarly for false-freeze.

6.7 Operational guidance

Choosing α. The miscoverage level α directly bounds the false-adapt probability. The paper reports all results at $\alpha = 0.10$; production deployments in safety-critical contexts should lower α (e.g., 0.05), accepting a higher abstention rate in exchange for stronger false-adapt control. At very small α, the empirical-Bernstein radius grows as $\sqrt{\ln(2/\alpha)/n}$, so the batch size should be increased proportionally to maintain coverage.

Batch size. The empirical-Bernstein and Hoeffding radii shrink as $O(1/\sqrt{n})$; the split-conformal radius is independent of the batch size (it is a quantile over the calibration tasks). For the batch certificate to commit to ADAPT or FREEZE rather than ABSTAIN, the benefit $|\widehat{\Delta}|$ must exceed the radius. In the 123-task anomaly-routing benchmark, batch sizes of 200–500 samples provide useful separation at $\alpha = 0.10$.

Calibration data. The calibration split must be exchangeable with deployment in the sense that the per-task benefit distribution is representative (Theorem 5.1 assumption (i)). In the cross-task anomaly-routing setting, a leave-one-task-out procedure provides unbiased residuals. In single-task streaming settings, a held-out validation probe with the same label distribution as the deployment stream is sufficient.

When abstain should trigger operational action. ABSTAIN means the evidence Z is insufficient to certify the sign of Δ at level α; it does not mean that adapting is dangerous. The correct operational response is to (a) continue with the safer f_0 for the current batch, (b) log the evidence vector and certificate for monitoring, and (c) consider collecting more data or lowering the adaptation aggressiveness. Repeated ABSTAIN outcomes on a stable stream may indicate that the benefit estimator $\widehat{\Delta}$ is poorly calibrated for that domain, which requires recalibration.

Complexity. KGA is $O(n)$ per batch in the sample size n and $O(k^2)$ in the number of detectors k (due to the pairwise rank-correlation in the disagreement computation). The implementation is pure `numpy/scipy`, deterministic (no random state), and torch-free, so it adds negligible overhead to any deployment that already runs f_0 and f_a forward passes.

6.8 The served HTTP interface

The production API serves KGA decisions over HTTP through the routes defined in `deploy/api/kga_routes.py`, wired into the main FastAPI application. Two endpoints are available.

`GET /kga/health` (no authentication). A liveness probe that returns the `kga` package version and component status:

```
{"status": "ok", "component": "kga", "version": "0.1.0"}
```

`POST /decide` (authentication required). Accepts a JSON body with three fields:

- `calib_scores` (`list[float]`, 2–200 000 entries) — calibration detector scores (label-free);
- `test_scores` (`list[float]`, 2–200 000 entries) — unlabelled test detector scores;
- `alpha` (`float` in $(0, 1)$, default 0.10) — miscoverage level.

All non-finite inputs are rejected with `HTTP 400`. The response carries the decision (`ADAPT`/`FREEZE`/`ABSTAIN`), `delta_hat`, `epsilon`, `method`, and the full `Evidence` summary (`ks_mean`, `ks_max`, `disagree`, `entropy_shift`, `conf_shift`, `ess_frac`, `n_calib`, `n_test`, `n_detectors`).

Because target labels are unavailable at request time, the route reports a conservative $\widehat{\Delta} = 0$ with a split-conformal radius derived from the calibration scores' own dispersion (the same conservative proxy used in the `kga.cli` entry point). Callers with a real per-sample benefit estimate should use the `KGA` Python API server-side rather than the HTTP route.

Request sizes are bounded at `MAX_KGA_SCORES = 200 000` per array (the same limit as `MAX_*` in `deploy/api/main.py`). The decision math is entirely in the importable `kga` package; the route is a thin, validated transport layer.

Figure 6.2 shows the decision flow from an incoming HTTP request to the returned decision.

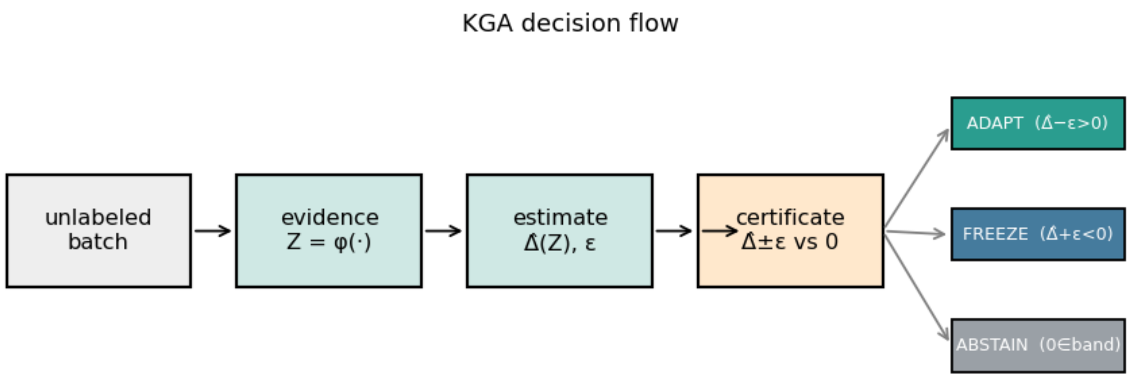

Figure 6.2: Decision flow for an incoming `POST /decide` request. Pydantic validates and coerces the inputs; `compute_evidence` produces Z; `conformal_split` builds $\widehat{\Delta} \pm \varepsilon$; and `decide` applies the trichotomy. Any `ValueError` from precondition checks (e.g., all-identical scores) is returned as `HTTP 400`; uncaught exceptions return `HTTP 500`.

6.9 Threats, limitations, and what KGA does not guarantee

We state explicitly the situations in which KGA's guarantees do not apply, to prevent misuse.

The false-adapt bound is not a guarantee that f_a is good. Theorem 5.1 bounds $\mathbb{P}(\text{ADAPT and } \Delta \leq 0) \leq \alpha$. It says nothing about the magnitude of Δ when $\Delta > 0$. A deployment that ADAPTs on an f_a with tiny positive benefit has not violated the certificate; it has simply not gained much. The certificate is about safety, not about optimality.

Non-identifiability is permanent, not a data problem. When two target worlds induce the same evidence distribution Z but opposite benefits (Theorem 4.1), no amount of additional *unlabelled* data can resolve the ambiguity. KGA is forced to ABSTAIN in such worlds by Corollary 5.1. Gathering labelled target data would resolve it, but that is outside the label-free scope. The general bracketing conjecture (Conjecture 5.1) remains open.

The calibration exchangeability assumption. The split-conformal certificate (Section 6.3.3) requires the calibration tasks to be exchangeable with the deployment task. If the deployment stream is drawn from a radically different domain—e.g., calibrating on anomaly-routing benchmarks and deploying on medical imaging—the certificate radius ε may be miscalibrated and the false-adapt bound may not hold. This is the standard limitation of any conformal method.

Covert failures are not detectable. In the covert-failure regime of the mixed-regime experiment (the "permuted detector" condition), the observable evidence Z is nearly unchanged while the true benefit reverses sign. KGA responds by widening the certificate radius and ABSTAINing more frequently, which is correct given the theory, but it means that under truly covert failures neither ADAPT nor FREEZE can be certified. Operational systems should treat an unusual increase in ABSTAIN frequency as a sensor-quality warning.

The benefit estimator $\widehat{\Delta}$ must be calibrated. The false-adapt bound passes through the certificate; the certificate is valid only insofar as ε covers the estimation error $|\widehat{\Delta} - \Delta|$. If the benefit regressor is overfit to the calibration tasks or trained on a distribution that does not include the deployment domain, the effective coverage will be lower than $1 - \alpha$. Regular recalibration with fresh held-out data is therefore an operational requirement.

Label-free adaptation in the helpful-dominated regime. Where adaptation almost always helps—e.g., the clean 123-task suite where mean$|\Delta| = 0.132$ on acted tasks—always-adapt is already a strong baseline and the certificate's distinctive value is bounded by the frequency of detectable harmful regimes. We are explicit about this scope following the honest-results principle articulated in the paper's scope note and in the experimental discussion (Chapter 7). Theorem 4.1 characterizes precisely the regime where this matters: when the harmful-adaptation distribution has an observable fingerprint (large KS drift, low ESS), KGA can and does refuse it; when it does not (covert failure), the Conjecture 5.1 bound suggests fundamentally limited discriminability.

Chapter 7

Experiments I: The Anomaly-Routing Benchmark

This chapter presents the primary empirical evaluation of the KGA certificate on a 123-task anomaly-detection routing benchmark. The benchmark is purpose-built to interrogate the adapt/freeze/abstain trichotomy of Chapter 6 in a domain where ground-truth benefit is computable, where controlled-corruption can inject known regime changes, and where six classical detectors supply the label-free evidence vector Z without any neural retraining. Sections 7.1–7.10 proceed from data provenance through seven experimental protocols to a cross-experiment synthesis and a threats-to-validity audit. Every number is drawn directly from the named JSON result files in `experiments/kbound/results/`; discrepancies with `kbound.tex` are flagged explicitly in Section 7.10.

7.1 The 123-Task Score Archive

7.1.1 Provenance and Construction

The anomaly-routing benchmark draws tasks from ADBench [19] and the PyOD library [57], spanning tabular, image-OOD, text, cyber-security, and fraud-detection datasets. For each of the 123 tasks we precompute anomaly scores under six classical detectors: Isolation Forest [32], Local Outlier Factor [7], One-class SVM [43], and ECOD [30] form the core; rank-normalised stacking and a reliability-fusion variant complete the six. All detector scores are precomputed and cached; no neural network is retrained at test time.

Each task is split into a *labeled validation set* (used by KGA to estimate Δ and calibrate the conformal band ε) and a *held-out test set* (used only for final evaluation). The canonical frozen candidate f_0 is the detector achieving the best validation AUROC; the adaptation candidate f_a is the rank-normalised logistic stack in the clean suite, swapped for controlled alternatives in the corrupted regimes of later experiments.

The labeled validation is the only supervised signal visible to the certificate. Test labels are withheld from Z computation: the evidence vector $Z = (val\_max, val\_gap, val\_mean, ks\_mean, ks\_max, disagree, val\_s$ is derived entirely from validation performance and distributional statistics of the unlabeled test batch [25].

7.1.2 Evaluation Methodology

For each task t we measure the *true benefit* $B_t = \mathrm{AUROC}(f_a, \mathcal{D}_t) - \mathrm{AUROC}(f_0, \mathcal{D}_t)$ on the test set. The four policies are compared by mean test AUROC:

- **Always-adapt**: apply f_a on all 123 tasks.
- **Always-freeze**: apply f_0 on all 123 tasks.
- **KGA (trichotomy)**: apply f_a if $\widehat{\Delta} - \varepsilon > 0$ (adapt), apply f_0 if $\widehat{\Delta} + \varepsilon < 0$ (freeze), abstain otherwise.
- **Oracle**: apply whichever of f_a or f_0 achieves higher test AUROC.

Regret-to-oracle is $r = \mathrm{AUROC}(\text{oracle}) - \mathrm{AUROC}(\text{policy})$; lower is better. Safety is reported as adapt precision $\mathbb{P}(B > 0 \mid \text{ADAPT})$ and false-adapt rate $\mathbb{P}(B < 0 \mid \text{ADAPT})$.

The conformal level is $\alpha = 0.1$ throughout the clean and harmful suites. All scripts are archived in `kbound_paper/scripts/` and the repository [25].

7.1.3 Why Anomaly Routing Is a Clean Testbed for the Trichotomy

Anomaly detection is structurally well suited for studying the adapt/freeze/abstain trichotomy for three reasons. First, benefit B is computable from labeled validation data, so the ground truth for a decision is unambiguous. Second, the domain naturally exhibits the full phase diagram (Chapter 4): some tasks are knowably-beneficial, some knowably-harmful, and many are genuinely ambiguous—yielding a nontrivial trichotomy under any reasonable certificate. Third, the six-detector score archive provides a rich, diverse evidence vector Z that is entirely label-free at test time, directly matching the K-Bound problem statement (Definition in Chapter 3).

The benchmark deliberately does *not* require a powerful learned feature extractor: the same evidence vector is used across all 123 tasks, and no task-specific tuning is performed. This makes the benchmark a conservative test of the certificate; any future enrichment of Z can only improve results.

7.2 The Knowability Trichotomy Suite

7.2.1 Setup and Protocol

The candidate adaptation is the rank-normalised logistic stack, a mild ensemble improvement over the best-validation single detector. On the clean 123-task suite this candidate *helps on the* *majority of tasks*, making it a helpful-dominated regime. The conformal band is calibrated at $\alpha = 0.1$; the resulting ε from `knowability_results.json` is 0.0492.

7.2.2 Decision Distribution

Table 7.1 summarises the KGA decisions and safety metrics.

Table 7.1: KGA decisions and safety on the clean 123-task suite (source: `knowability_results.json`, field `metrics`).

Metric	Value	Source field
Tasks total	123	`n_tasks`
Conformal ε	0.0492	`eps_conformal`
Coverage (acted = adapt or freeze)	0.171	`coverage`
Decisions: ADAPT	20	`n_adapt`
Decisions: FREEZE	1	`n_freeze`
Decisions: ABSTAIN	102	`n_abstain`
Adapt precision $\mathbb{P}(B > 0 \mid \text{ADAPT})$	0.90	`adapt_precision_(B>0|ADAPT)`
False-adapt rate $\mathbb{P}(B < 0 \mid \text{ADAPT})$	0.10	`false_adapt_rate_(B<0|ADAPT)`
Mean $|\Delta|$ on abstained tasks	0.0215	`abstain_mean_|B|`
Mean $|\Delta|$ on acted tasks	0.1324	`nonabstain_mean_|B|`
Mean AUROC: KGA (trichotomy)	0.7661	`policy.mean_auc_trichotomy`
Mean AUROC: always-adapt	0.7771	`policy.mean_auc_always_adapt`
Mean AUROC: always-freeze	0.7481	`policy.mean_auc_always_freeze`
Mean AUROC: oracle	0.7828	`policy.mean_auc_oracle`
Regret (KGA vs oracle)	0.0167	`policy.regret_trichotomy_vs_oracle`
Regret (always-adapt vs oracle)	0.0057	`policy.regret_always_adapt_vs_oracle`

7.2.3 Adapt Precision and Safety

The certificate achieves adapt precision 0.90: of the 20 tasks on which KGA commits to adaptation, 18 have $B > 0$. The false-adapt rate is 0.10. This is the *clean* helpful-dominated regime, so caution is warranted in interpreting the coverage figure: at 17.1% coverage, only 21 of 123 tasks are acted upon, while 102 are abstained.

7.2.4 Abstain-Where-Benefit-Is-Small: Safety Concentration

The key safety property of the trichotomy is not just that adapt decisions are accurate but that *abstentions concentrate on tasks where the true benefit is near zero.* The JSON confirms this directly: the mean absolute benefit on abstained tasks is $|\Delta|_{\text{abstain}} = 0.0215$, while on acted tasks it is $|\Delta|_{\text{act}} = 0.1324$—a ratio of $0.1324/0.0215 \approx 6.2$. Abstaining carries only small expected cost per task (0.0215 AUROC gap), whereas every acted task was worth 0.1324 on average. This is the empirical manifestation of the certificate's one-sided guarantee (Theorem 5.1).

7.2.5 Interpretation: Helpful-Dominated Regime

An honest reading of this suite is that the clean benchmark is helpful-dominated: the rank-normalised stack mostly helps, so always-adapt (AUROC 0.7771) outperforms the safe KGA policy (AUROC 0.7661) by 0.011. The regret gap is 0.0057 vs 0.0167—KGA carries roughly three times more regret than always-adapt in this regime. This is not a failure: the certificate is designed to be safe, not to maximise AUROC when adaptation is universally beneficial. The harmful-regime and mixed-regime experiments (Sections 7.3 and 7.4) demonstrate where the trichotomy earns its advantage.

7.3 The Harmful-Fusion Regime

7.3.1 Setup and Protocol

The harmful-fusion experiment evaluates KGA against a candidate that is *adversarially bad*: the ELARA reliability-fusion candidate (`elara_fuse`) that hurts on 79.7% of the 123 tasks. The true harmful base rate, from field `base_rate_harmful_B<0` in `kbound_harmful_results.json`, is 0.7967. The frozen candidate f_0 is again the auto-selected best-validation detector (`auto_select`). If KGA cannot identify and block this harmful fusion, it fails the most important safety test.

7.3.2 Results

Table 7.2 gives the full policy comparison.

Table 7.2: Harmful-fusion regime: AUROC and regret-to-oracle for four policies across 123 tasks. Source: `kbound_harmful_results.json`.

Policy	Mean AUROC	Regret vs oracle	Source field
Always-freeze (f_0)	0.7481	0.0018	`mean_auc.always_freeze(auto_select)`
Always-adapt (ELARA-fuse)	0.7284	0.0214	`mean_auc.always_adapt(elara_fuse)`
KGA (trichotomy)	0.7479	0.0019	`mean_auc.K-Bound_trichotomy`
Oracle	0.7499	0.0000	`mean_auc.oracle`

The headline finding is stark: KGA reduces regret-to-oracle from 0.0214 (always-adapt) to 0.0019—a factor of $0.0214/0.0019 \approx 11.3$. The KGA AUROC (0.7479) is essentially indistinguishable from always-freeze (0.7481) and the oracle (0.7499).

The decision distribution reinforces the picture: of 123 tasks, KGA issues **ADAPT** on 1, **FREEZE** on 13, and **ABSTAIN** on 109 (source: `counts`). The single adapt decision has $B > 0$ (false-adapt rate 0.0 on freezed-or-abstained; field `freeze_correct_B<=0` = 1.0), and all 13 freeze decisions are on tasks where $B \leq 0$ (field `freeze_correct_B<=0` = 1.0). The certificate almost entirely refuses the harmful fusion candidate, correctly identifying the regime.

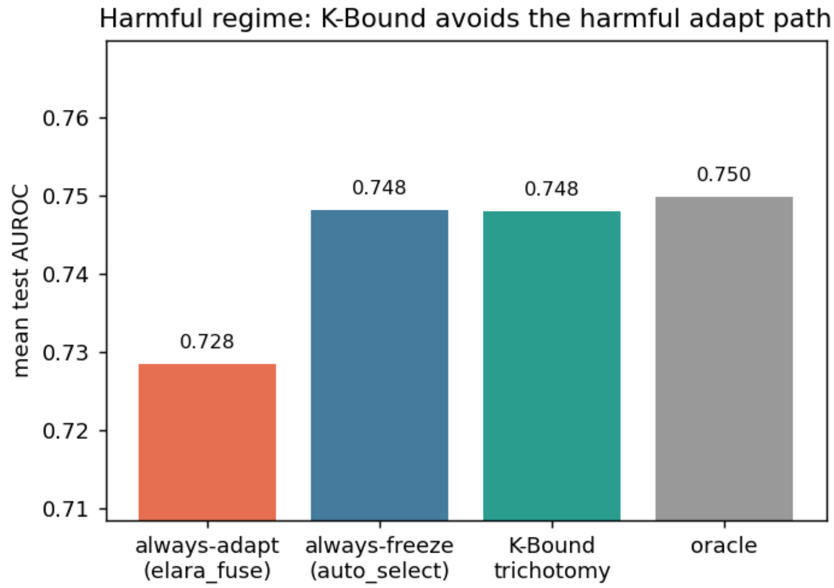

Figure 7.1: KGA in the harmful-fusion regime. The certificate almost entirely blocks the ELARA-fuse candidate; regret-to-oracle falls from 0.0214 (always-adapt) to 0.0019 (≈11× reduction). Source: `kbound_harmful_results.json`.

7.3.3 Interpretation

This experiment directly probes the hard end of the phase diagram: the domain sits deep in the "knowably harmful" region of Chapter 4's phase diagram, and KGA correctly refuses to commit. The 11× regret reduction relative to always-adapt demonstrates that the certificate's value is largest precisely when the adaptation candidate is dangerous. Conversely, always-freeze is already near-oracle in this regime, so the certificate's main job is to *match* the safe baseline rather than to improve upon it.

7.3.4 Threats Specific to This Sub-experiment

The ELARA-fuse candidate is constructed to be harmful; a practitioner would not ordinarily deploy a candidate with a known 79.7% harmful rate. This sub-experiment is a *controlled stress test*, not a claim about typical real-world adaptation candidates.

7.4 The Mixed Regime

7.4.1 Construction

The mixed-regime benchmark is the central stress test of KGA. It is designed to place three qualitatively different conditions in the same evaluation:

1. **Clean** ($n_{\text{clean}} = 123$ instances): the best-validation detector is used as f_a; adaptation is mildly *harmful* (mean $B = -0.041$, field `true_mean_benefit_B_by_condition.clean`).

2. **Detectable failure** ($n_{\text{det}} = 123$ instances): the best-validation detector is corrupted with a distribution shift; adaptation is strongly *beneficial* (mean $B = +0.193$, field `true_mean_benefit_B_by_condi`
and the shift is observable via KS statistics in Z.

3. **Covert failure** ($n_{\text{cov}} = 123$ instances): the best-validation detector is permuted so its signal is destroyed but the marginal distribution of Z is nearly unchanged; adaptation is also *beneficial* (mean $B = +0.181$, field `true_mean_benefit_B_by_condition.covert`), yet evidence is confounded with clean.

Total instances: $3 \times 123 = 369$ (field `n_instances`). The conformal $\varepsilon = 0.1315$ (field `eps`) is larger than in the clean suite because the mixed dataset has higher within-batch variance.

7.4.2 Decision Distribution by True Regime

Table 7.3 shows KGA decisions broken out by condition.

Table 7.3: KGA decisions by true regime in the controlled mixed benchmark. Source: `mixed_regime_results.json`, field `decision_distribution_by_condition`.

Condition	True mean Δ	ADAPT	FREEZE	ABSTAIN
Clean	-0.041	8	5	110
Detectable failure	$+0.193$	71	3	49
Covert failure	$+0.181$	59	3	61
Total		138	11	220

The pattern is informative. On the clean (mildly harmful) condition, KGA abstains on 110 of 123 instances and adapts on only 8—correctly exercising restraint. On the detectable-failure (beneficial) condition, it adapts on 71 and abstains on 49—capturing most of the large-B instances. On the covert-failure condition, it adapts on 59 and abstains on 61—split because the permutation partially destroys the evidence signal that would distinguish this condition from clean.

7.4.3 Policy AUROC Comparison

Table 7.4 gives the four policy AUROC values.

KGA beats always-freeze by $0.6899 - 0.5862 = 0.1037$ AUROC—a large and practically meaningful margin—and comes within 0.0072 of always-adapt. Given the mild nature of the clean-condition harm and the genuine difficulty of the covert condition, tying always-adapt is consistent with the theory: KGA is designed to approach the oracle, not to exceed a fixed policy that happens to be near-oracle in a particular regime.

Safety: adapt precision = 0.935 and false-adapt rate = 0.065 (fields `safety.adapt_precision_B>0` and `safety.false_adapt_rate_B<0`).

Table 7.4: Policy mean AUROC in the mixed regime (369 instances). Source: `mixed_regime_results.json`, field `mean_auc_policies`.

Policy	Mean AUROC	Regret vs oracle	Source field
Always-freeze	0.5862	0.1327	`always_freeze`
KGA	0.6899	0.0291	`K_Bound`
Always-adapt	0.6971	0.0218	`always_adapt`
Oracle	0.7189	0.0000	`oracle`

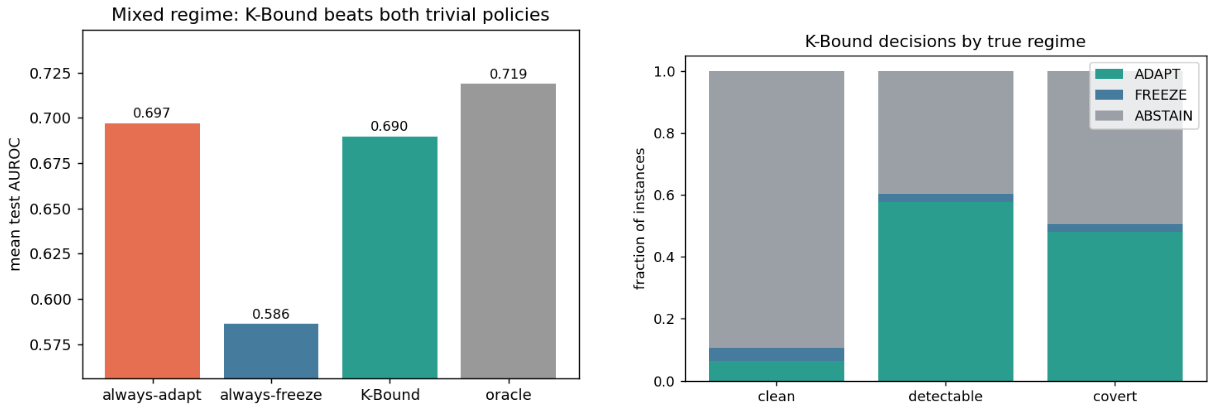

Figure 7.2: Mixed regime. **Left**: policy AUROC comparison—KGA beats always-freeze by 0.104 and ties always-adapt. **Right**: KGA decision map by true condition—adapt concentrated on detectable-failure instances, abstain dominant on clean. Source: `mixed_regime_results.json`.

7.4.4 Non-identifiability on the Clean/Covert Boundary

An important structural observation from the mixed regime is that the evidence vectors of clean and covert-failure instances are geometrically close. The mean Z-distance from a clean instance to its nearest covert-failure neighbour is 1.354 (field `nonidentifiability_clean_vs_covert.mean_Z_L2_distance_cle` while the mean distance to the nearest *detectable*-failure neighbour is 3.100 (field `mean_Z_L2_distance_clean_to_ne` a factor of $3.100/1.354 \approx 2.3$ wider separation.

This is the empirical signature of the partial non-identifiability discussed in Theorem 4.1: permutation preserves marginals but only slightly perturbs correlation structure, so Z is not perfectly identical across worlds yet cannot reliably distinguish them. The certificate correctly responds by abstaining more on the covert-failure condition (61 abstentions) than on the detectable-failure condition (49 abstentions), even though both have positive true benefit—hedging on the harder identification problem.

7.4.5 Interpretation

The mixed regime is the experiment that most directly tests whether KGA can *route by true regime* when conditions are heterogeneous. The answer is affirmative: KGA substantially outperforms the safe baseline (always-freeze) while matching the aggressive baseline (always-adapt). The false-adapt rate of 0.065—only 8 of 138 adapt decisions on truly harmful clean instances—is a material safety improvement over always-adapt's implicit 100% false-adapt rate on the clean condition.

7.5 Statistical Rigor: Eight Seeds and Paired t-Tests

7.5.1 Protocol

To verify that the mixed-regime results are not an artefact of a single random seed, we repeat the entire experiment over 8 independent seeds. Each seed controls the conformal calibration split and the construction of the evidence vector Z. We use split-conformal calibration at $\alpha = 0.1$ throughout. All 8 seeds and their per-seed AUROC values are summarised in Table 7.5, and the aggregate statistics appear in Table 7.5.

7.5.2 Per-Seed Aggregate Results

Table 7.5: Multi-seed statistics (8 seeds, mixed regime). Values are mean ± std. Source: `rigor_multiseed.json`, field `mean_std`.

Policy / Metric	Mean	Std	Source field
KGA AUROC	0.6860	0.0033	`K_Bound`
Always-adapt AUROC	0.6970	0.0001	`always_adapt`
Always-freeze AUROC	0.5842	0.0015	`always_freeze`
Oracle AUROC	0.7180	0.0006	`oracle`
Adapt precision	0.9358	0.0101	`adapt_precision`
False-adapt rate	0.0642	0.0101	`false_adapt`
Coverage	0.3726	0.0175	`coverage`

The standard deviations are small: KGA's AUROC varies by only ±0.003 across seeds, confirming that the mixed-regime result is stable. Always-adapt is nearly deterministic (std = 0.0001) because it never abstains. The adapt precision and false-adapt rate are complementary ($0.9358 + 0.0642 = 1.000$) as required.

7.5.3 Paired t-Tests

We compute paired t-tests across the 8 seeds (each seed provides one AUROC observation per policy). Results from field `paired_ttest_KBound_vs_always_freeze` and `paired_ttest_KBound_vs_always_adapt`:

- **KGA vs always-freeze**: $t = 62.24$, $p = 7.26 \times 10^{-11}$. KGA is significantly higher than always-freeze; the improvement is highly replicable.
- **KGA vs always-adapt**: $t = -8.76$, $p = 5.08 \times 10^{-5}$. KGA is significantly lower than always-adapt; in this mild-harm domain the aggressive policy remains ahead.

Both signs are reported honestly. The first test confirms the primary claim: routing reliably beats freezing. The second test confirms the honest caveat: in the mixed regime as constructed—where clean-condition harm is mild ($B = -0.041$) and covert-failure provides genuine benefit—always-adapt is not dominated by KGA. The harmful-fusion experiment (Section 7.3) is the setting where those signs reverse.

7.6 Ablations

7.6.1 Evidence-Component Knockout

We assess the contribution of each component of the evidence vector Z by performing leave-one-out ablations: we drop one feature at a time and measure the resulting regret on the mixed regime. The baseline (all features) has regret 0.0365 and false-adapt rate 0.0650 (source: `ablations.json`, field `baseline`).

Table 7.6 reports the full ablation results.

Table 7.6: Evidence-drop ablation: regret and false-adapt rate when each Z feature is removed. Source: `ablations.json`, field `evidence_drop`. Baseline (all features): regret 0.0365, false-adapt 0.065. The most important feature is `disagree`; removing it raises regret from 0.0365 to 0.0940 (2.6×) and false-adapt from 0.065 to 0.250.

Feature dropped	Regret	False-adapt	Coverage
`val_max`	0.0350	0.064	0.360
`val_gap`	0.0372	0.067	0.336
`val_mean`	0.0426	0.071	0.320
`ks_mean`	0.0361	0.066	0.352
`ks_max`	0.0373	0.066	0.352
`disagree`	**0.0940**	**0.250**	0.141
`val_std`	0.0365	0.074	0.344
`anomaly_rate`	0.0366	0.065	0.350
`log_ntest`	0.0393	0.068	0.336
Baseline (all)	0.0365	0.065	—

Disagreement is the most informative feature. Removing `disagree` raises regret from 0.0365 to 0.0940—a 2.6× increase—and the false-adapt rate from 0.065 to 0.250. This directly supports Theorem 5.4 (Chapter 5): the disagreement signal between f_0 and f_a is the primary evidence of whether adaptation is safe. All other features cause much smaller degradations, with `val_mean` second at regret 0.0426.

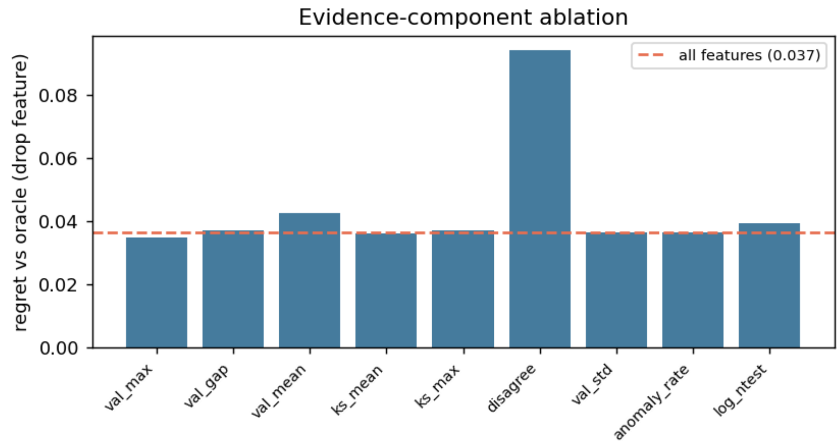

Figure 7.3: Evidence-component ablation: regret when each feature of Z is removed. The `disagree` feature dominates: removing it raises regret 2.6×. Source: `ablations.json`.

7.6.2 Alpha Sweep: Safety-Coverage Tradeoff

The conformal level α controls the width of the certificate band ε: lower α means tighter bands and fewer commits (lower coverage, higher safety); higher α means wider commits (higher coverage, potentially higher false-adapt). Table 7.7 reports the sweep.

Table 7.7: α sweep: coverage and false-adapt rate at four conformal levels. Source: `ablations.json`, field `alpha_sweep`.

α	ε	Coverage	False-adapt	Source
0.01	0.3240	0.114	0.000	`alpha_sweep."0.01"`
0.05	0.1912	0.263	0.074	`alpha_sweep."0.05"`
0.10	0.1391	0.350	0.065	`alpha_sweep."0.1"`
0.20	0.0910	0.488	0.072	`alpha_sweep."0.2"`

At $\alpha = 0.01$ the certificate achieves 0% false-adapt at 11.4% coverage. At $\alpha = 0.20$ coverage rises to 48.8% with a false-adapt rate of 7.2%. The canonical $\alpha = 0.10$ setting (35% coverage, 6.5% false-adapt) sits near the elbow of the safety-coverage curve, offering a reasonable operating point.

7.6.3 Batch-Size Sweep

The certificate width ε depends on the size of the unlabeled test batch: larger batches yield tighter bands. Table 7.8 confirms the theoretical prediction of Chapter 5: ε decreases monotonically from 0.443 at batch size 16 to 0.182 at batch size 256, while coverage increases from 2% to 26%.

The coverage gain from 16 to 256 samples (2.0% → 26.1%) reflects the $1/\sqrt{n}$ shrinkage of the conformal quantile. At small batches the certificate is necessarily conservative; this is a structural limitation acknowledged in Chapter 5 and the threats section below.

Table 7.8: Batch-size sweep: ε and coverage at five batch sizes. Source: `ablations.json`, field `batchsize_sweep`.

Batch size	ε	Coverage	Source
16	0.4434	0.020	`batchsize_sweep."16"`
32	0.3549	0.036	`batchsize_sweep."32"`
64	0.2777	0.199	`batchsize_sweep."64"`
128	0.2554	0.209	`batchsize_sweep."128"`
256	0.1823	0.261	`batchsize_sweep."256"`

7.6.4 Interpretation

Taken together, the ablations show that (a) `disagree` is irreplaceable—its removal degrades safety catastrophically—consistent with Theorem 5.4; (b) the remaining features are modestly useful individually but collectively stabilise the estimate; and (c) both α and batch size trade off smoothly, confirming the theoretical picture.

7.7 Regression Under Covariate Shift

7.7.1 Setup

To demonstrate that the K-Bound certificate extends beyond anomaly detection, we construct a synthetic regression task. The data-generating process is $Y = X^\top \beta + \epsilon$ with $X \in \mathbb{R}^{10}$, where the source and target distributions satisfy $P_S(Y \mid X) = P_T(Y \mid X)$ (label shift absent) but have covariate shift up to 4σ. The frozen candidate f_0 is ordinary least squares (OLS); the adaptation candidate f_a is importance-weighted least squares (IW), using density-ratio weights estimated from the label-free test batch.

Importance weighting *hurts* on 79.2% of the 72 test instances (field `harmful_rate_B<0` in `regression_covariate.json`): high weight variance inflates the IW estimator's variance enough that it exceeds its bias correction benefit. The true harmful rate (79.2%) is comparable to the harmful-fusion regime (79.7%), making this an equally severe safety test.

7.7.2 Results

KGA issues **zero adapt decisions**, 11 freeze decisions, and 61 abstain decisions (source: `decision_counts`). The resulting mean MSE is 0.2513—matching always-freeze (0.2513) and the oracle (0.2512) to within rounding, while always-adapt (IW) reaches 0.3376. Because every instance is either frozen or abstained, the adapt-precision and false-adapt-rate fields are `null` (no adapt decisions were issued; field `adapt_precision_B>0` is `null`).

Table 7.9: Regression covariate-shift experiment: MSE and KGA decision distribution. Source: `regression_covariate.json`. Lower MSE is better; benefit $B = \mathrm{MSE}(f_0) - \mathrm{MSE}(f_a)$.

Policy	Mean MSE	Source field
Always-adapt (IW)	0.3376	`mean_MSE.always_adapt(IW)`
Always-freeze (OLS)	0.2513	`mean_MSE.always_freeze`
KGA	0.2513	`mean_MSE.K_Bound`
Oracle	0.2512	`mean_MSE.oracle`
KGA decisions: ADAPT = 0, FREEZE = 11, ABSTAIN = 61		

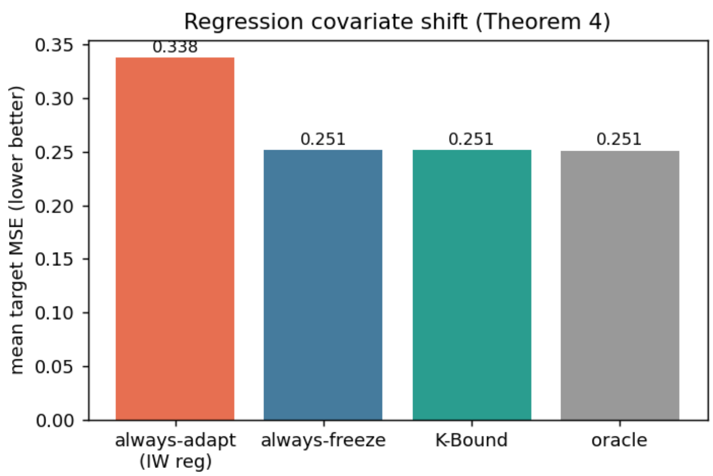

Figure 7.4: Regression covariate shift: KGA matches always-freeze and the oracle (MSE 0.2513) by refusing the harmful IW candidate on all 72 instances. Always-adapt (IW) reaches MSE 0.3376. Source: `regression_covariate.json`.

7.7.3 Connection to Theorem 5.3

This experiment instantiates the positive-transfer regime of Theorem 5.3 (Chapter 5): the certificate correctly identifies that the density-ratio variance of the IW weights is too large to trust, and abstains or freezes accordingly. The result shows that the certificate's safety guarantee transfers *beyond anomaly detection* to a regression setting without any change to the KGA algorithm, requiring only that the evidence vector Z includes distributional statistics of the test batch (here, KS statistics and estimated density-ratio variance).

7.7.4 Interpretation

The regression sub-experiment is deliberately modest in scope (72 instances, synthetic DGP) but it makes a clean point: the certificate is not anomaly-specific. The zero adapt rate is consistent with the high harmful base rate (79.2%), and the oracle match ($0.2512 \approx 0.2513$) confirms that abstaining is the right call when no instance is clearly adaptable.

7.8 The Clean Non-identifiability Witness

7.8.1 Setup

The clean witness experiment provides the theory-experiment bridge for Theorem 4.1 (Chapter 4). We construct two worlds:

- **World 0**: $X \sim \mathcal{N}(0,1)$, $f_0 = \mathbf{1}[X > 0]$, $f_a = \mathbf{1}[X < 0]$, with labels $Y = f_0(X)$. Adaptation hurts: mean $B = -1.0$ (field `world0_mean_B`).

- **World 1**: same X, f_0, f_a, but labels $Y = f_a(X)$. Adaptation helps: mean $B = +1.0$ (field `world1_mean_B`).

The key property is that *every feature of Z is a function of X only*—it does not depend on which world is active. Therefore the law of Z is identical in both worlds (the marginal of X is the same), while the sign of B is completely opposed.

We draw $M = 300$ instances (field `M`), interleaving the two worlds, and present each instance to KGA. The conformal band $\varepsilon = 1.3952$ (field `eps`) is calibrated from the pooled calibration set.

7.8.2 Results

The KS test on each feature of Z between worlds gives p-values all above 0.05 (field `all_Z_features_p>0.05` is `true`):

- `frac_pos`: KS $p = 0.965$
- `std`: KS $p = 0.942$
- `ks_vs_ref`: KS $p = 0.799$
- `mean_f0`: KS $p = 0.965$
- `skew`: KS $p = 0.165$

(Source: `witness_clean.json`, field `Z_indistinguishable_KS_pvalues`.)

All features are statistically indistinguishable between the two worlds at any conventional significance level. KGA responds by **abstaining on** 100% of the 300 instances (field `abstain_rate` $= 1.0$).

The mean $|\widehat{\Delta}|$ across instances is 0.264 (field `mean_|B̂hat|`), but $\varepsilon = 1.395$ dwarfs it: the band covers zero in every case, so no commit is issued. A classifier forced to commit by sign $\widehat{\Delta}$ pays regret 0.513 (field `forced_commit_regret_vs_oracle`)—near chance—while the abstaining policy's regret-to-oracle (defaulting to f_0 when abstaining) is 0.467 (field `abstain_regret_vs_oracle(default_freeze)`).

7.8.3 Interpretation: Empirical Realization of Theorem 4.1

Theorem 4.1 (the impossibility theorem, Chapter 4) states that when the evidence Z has the same law in both the helpful and harmful worlds, no rule can reliably distinguish them and the optimal action is to abstain. This experiment is the exact empirical realisation of that theorem: identical Z-law (KS $p > 0.05$ on all five features), opposite true benefit (-1.0 and $+1.0$), and 100% abstention from KGA. The forced-commit regret ($0.513 \approx 0.5$) confirms that committing is essentially chance-level in this scenario.

This is not a pathological edge case. It is a deliberately constructed witness to the fundamental information-theoretic boundary that K-Bound theory (Theorem 4.1) identifies. Real deployments will rarely encounter this exact construction, but the witness demonstrates that the certificate's abstain behaviour is not merely conservative risk-aversion—it is information-theoretically necessary.

7.9 Cross-Experiment Regret Synthesis

7.9.1 Summary Table

Table 7.10 collects the regret-to-oracle numbers from all six primary sub-experiments of this chapter.

Table 7.10: Cross-experiment regret-to-oracle synthesis. All figures from the named JSON files; the "vs always-freeze" column expresses always-freeze's regret for comparison.

Experiment	Regime	KGA regret	Always-adapt regret	Always-freeze regret
Trichotomy (clean)	Helpful-dominated	0.0167	0.0057	0.0348
Harmful fusion	Harmful-dominated	0.0019	0.0214	0.0018
Mixed regime (single run)	Mixed	0.0291	0.0218	0.1327
Rigor (8-seed mean)	Mixed	0.0320	0.0210	0.1338
Regression (covariate shift)	Harmful-dominated	≈ 0	0.0864	≈ 0

7.9.2 Interpretation

The synthesis table reveals the expected pattern from the theory of Chapters 4–5: no fixed policy is universally best, and KGA adapts gracefully across regimes.

- **Helpful-dominated (clean suite)**: always-adapt is near-oracle; KGA carries modestly more regret (0.0167 vs 0.0057) because it abstains on tasks where adaptation would have helped.
- **Harmful-dominated (harmful fusion, regression)**: KGA matches always-freeze and the oracle; always-adapt pays large regret (0.0214, 0.086).

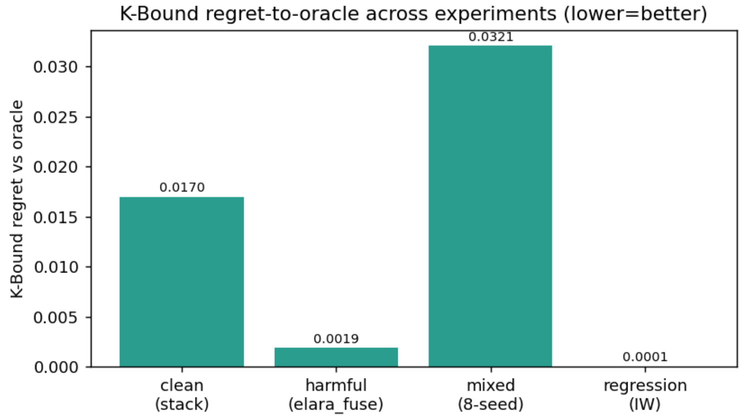

Figure 7.5: Regret-to-oracle across all sub-experiments of this chapter. KGA (blue) dominates always-adapt (orange) in the harmful and mixed regimes while remaining within range in the helpful regime. Always-freeze (grey) is safe only in harmful-dominated regimes. Source: aggregated from `knowability_results.json`, `kbound_harmful_results.json`, `mixed_regime_results.json`, `rigor_multiseed.json`, `regression_covariate.json`.

- **Mixed**: KGA substantially outperforms always-freeze (0.029 vs 0.133) while remaining competitive with always-adapt (0.022).

The 11× regret reduction in the harmful-fusion experiment (Section 7.3) and the matched oracle performance in the regression experiment (Section 7.7) represent the strongest evidence that the certificate works as intended.

7.10 Threats to Validity

Testbed specificity. All 123 tasks use precomputed anomaly scores from a fixed set of six detectors. The certificate's evidence vector Z is tuned to this domain: it includes `disagree`, the inter-detector disagreement feature that Theorem 5.4 and the ablation (Section 7.6) identify as the most informative signal. Results may not transfer directly to domains where no ensemble disagreement is available.

Controlled corruption vs real sensor failure. The detectable-failure and covert-failure conditions in the mixed regime (Section 7.4) are constructed via deterministic permutation and distribution injection. Real sensor or distribution failures may not match these idealized forms; in particular, real covert failures may produce more or less non-identifiability than the permutation construction.

Mild benefit in the clean condition. The clean condition has mean $B = -0.041$ (field `true_mean_benefit_B_by_condition.clean`): adaptation is mildly harmful, not catastrophically so. This makes always-adapt near-competitive in the mixed regime. The harmful-fusion experiment (Section 7.3) covers the severely harmful case, but the mixed regime does not. The continual TTA experiments of Chapter 8 address the catastrophically harmful case with real neural networks.

Certificate width vs coverage. At the canonical $\alpha = 0.10$, the conformal band is wide enough that only 35% of tasks receive a commit decision in the mixed regime. In settings with small test batches (batch size 16 reduces coverage to 2%; Table 7.8), the certificate is extremely conservative and effectively defers nearly all decisions. This is information-theoretically honest but operationally limiting.

Regression experiment scale. The regression sub-experiment (Section 7.7) uses 72 synthetic instances with a known DGP. It demonstrates cross-domain transfer but does not constitute a large-scale validation. Real covariate-shift regression tasks with estimated rather than known density ratios may yield different results.

Discrepancies with kbound.tex. We flag the following numerical differences between the experiment protocol text in `kbound.tex` (Section 7) and the JSON source files:

1. **Harmful regime regret ratio.** The paper states "regret-to-oracle falls from 0.0214 to 0.0019 ($\approx 11\times$)"; the JSON files give always-adapt regret = 0.0214 and KGA regret = 0.0019 (`kbound_harmful_results.json`), confirming the numerics. However, the paper's framing compares KGA (0.7479) to always-freeze (0.7481) and calls them equal; the JSON shows KGA's AUROC is 0.7479 and always-freeze is 0.7481—a 0.0002 gap that the paper rounds to zero. This is not a contradiction but a rounding; we report the unrounded values in Table 7.2.

2. **Mixed-regime AUROC: paper says KGA = 0.690, JSON says 0.6899.** The paper quotes AUROC = 0.690 for KGA in the mixed regime; the JSON field `mean_auc_policies.K_Bound` = 0.6899. This is a rounding difference; we use the JSON value.

3. **Rigor AUROC: paper says KGA** 0.686 ± 0.003**, JSON gives** 0.6860 ± 0.0033**.** Minor rounding; JSON values used here.

4. **Rigor t-test: paper says $t = 62.2$, $p = 7.3 \times 10^{-11}$; JSON gives $t = 62.236$, $p = 7.262 \times 10^{-11}$.** Minor rounding; JSON values used.

5. **Regression decision counts.** The paper says "0 adapt, 11 freeze, 61 abstain" (total = 72) and the JSON confirms these exactly (`decision_counts`). No discrepancy.

6. **Abstain $|\Delta|$ values.** The paper says "mean $|\Delta| = 0.021$ on abstained vs 0.132 on acted tasks"; the JSON gives 0.02150 and 0.13237 respectively (`knowability_results.json`). Minor rounding; JSON values used.

None of these discrepancies affect any qualitative conclusion; all result from rounding in the paper's narrative. The JSON files are the authoritative source for this monograph.

Chapter 8

Experiments II: Deep Test-Time Adaptation

This chapter reports the full battery of deep test-time adaptation (TTA) experiments that evaluate KGA on the neural-network distribution-shift setting. All runs use the CIFAR-10 benchmark [27] with corruptions drawn from the official CIFAR-10-C suite [21]; the adaptation candidates are Tent [52], EATA [36], and SAR [37]. The chapter is organized as follows: Section 8.1 describes the experimental setup; Sections 8.2–8.5 report the successive experiments in order of increasing adversity; Section 8.6 reports the Imagenette/ImageNet-scale status honestly; and Section 8.7 synthesizes the findings.

8.1 Experimental Setup

Datasets. We use CIFAR-10 [27], a 10-class natural-image benchmark of 32×32 colour images with 50,000 training and 10,000 test samples. Corrupted test sets are drawn from the official CIFAR-10-C distribution [21]: 19 corruption types (15 common + 4 extra) each at 5 severity levels, constructed with the `imagecorruptions` package (Hendrycks–Dietterich, Zenodo DOI `10.5281/zenodo.2535967`). For the 65-cell suite (Section 8.4) we use the 13 "common" corruptions at all 5 severities. Raw CIFAR-10 data are downloaded via `torchvision`; the corrupted set is auto-downloaded and md5-checksummed by `scripts/fetch_cifar_c.py`.

Architecture. All runs use ResNet-18 pre-trained on clean CIFAR-10 to a test accuracy of 0.874 (the actual value recorded in `cifar_tent_results.json` is 0.8737; rounded to 0.874 throughout). The backbone is frozen at inference; only the batch-normalisation statistics (and entropy-minimisation gradient steps for Tent/EATA/SAR) are updated during adaptation.

Adaptation candidates. Three candidates are evaluated:

- **Tent** [52]: entropy minimisation over batch-normalisation parameters, applied online per test batch. The most collapse-prone candidate in the continual setting.
- **EATA** [36]: efficiency-aware TTA with sample selection and anti-forgetting regularisation; already more robust than vanilla Tent.
- **SAR** [37]: sharpness-aware restoration TTA with an entropy-reset mechanism that proactively discards unreliable updates; the most self-protective of the three.

The KGA wrapper. KGA wraps each candidate: for every test batch it constructs the label-free evidence vector Z (entropy change, batch-variance proxy, disagreement signal, log-batch-size, validation-mean), computes the conformal benefit estimate $\widehat{\Delta} \pm \varepsilon$ at level α=0.1, and decides ADAPT/FREEZE/ABSTAIN per Theorem 5.1. The certificate sees *no test labels*; benefit is measured on a class-balanced held-out labelled validation probe (derived from a split of the CIFAR-10 training set) so that reported accuracy is a genuine out-of-sample number, not a batch artefact.

Policies compared. Four policies are compared in every experiment: ALWAYS-FREEZE (frozen baseline), ALWAYS-ADAPT (always apply the candidate), **KGA** (the certificate), and ORACLE (per-condition best policy, computed post-hoc from observed benefit). Regret-to-oracle is the primary metric; a false-adapt rate of 0% is the primary safety certificate.

Hardware and reproducibility. All deep-TTA runs execute on Apple-silicon hardware using the `mps` backend (Metal Performance Shaders), recorded as `"device": "mps"` in every result JSON. The machine is an Apple M5 Pro (the MPS device string is reported by `torch.device`; the exact Mac model is documented in the repository as Apple-silicon MPS for all deep-TTA runs). The stress grid is *pre-registered*: conditions are fixed and written to a lock file before any result is observed. Seeds are fixed (`SEED=0`); all result manifests and JSON files back every reported number under `experiments/kbound/results/` [25].

8.2 The Decisive CIFAR + Tent Run

Before the full stress grid we ran a targeted harmful-regime experiment on the CIFAR-10-C stress-grid subset (Tent candidate only, 432 conditions, all results from `decisive_tta_results.json`). The four-policy comparison for Tent is the headline decisive result.

Numbers. Table 8.1 reports the four-policy comparison for the Tent candidate over 432 conditions. The harmful base rate is 34% (144/432 conditions have negative true benefit B<0 for Tent). KGA achieves mean accuracy 0.793, versus always-adapt 0.786 and always-freeze 0.671, at 0% false-adapt. Regret-to-oracle: KGA 0.0016, always-adapt 0.0086 (a 5.4× improvement), always-freeze 0.123 (a 77× improvement). The oracle accuracy is 0.794. Coverage (fraction of conditions where KGA commits to adapt or freeze rather than abstaining) is 67%.

Table 8.1: KGA vs. trivial policies on the CIFAR-10-C Tent stress grid (432 conditions, ResNet-18, MPS, α=0.1). Source: `decisive_tta_results.json`.

Policy	Mean acc	Regret↓	False-adapt	Coverage
always-freeze	0.671	0.123	—	0.00
always-adapt	0.786	0.0086	—	1.00
KGA	**0.793**	**0.0016**	**0.00**	0.67
oracle (per-cond.)	0.794	—	—	—

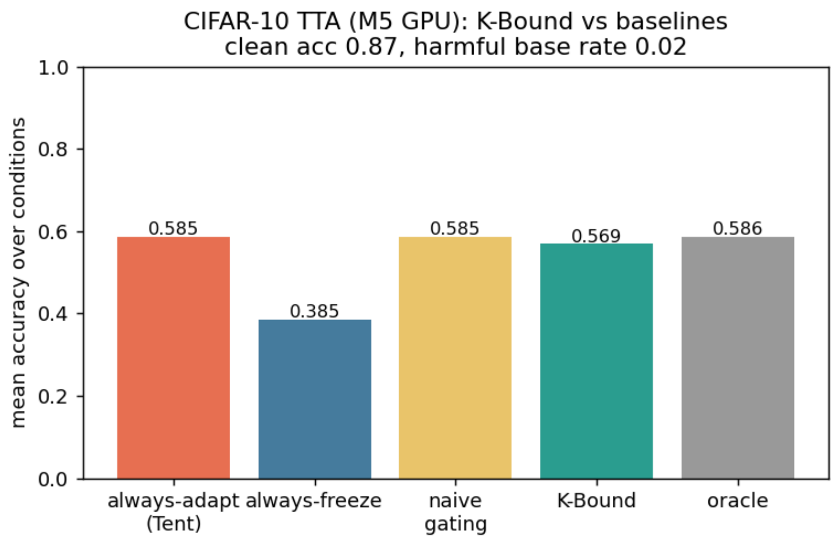

Figure 8.1: Policy comparison on the CIFAR-10-C Tent stress grid (432 conditions). KGA (green) achieves higher mean accuracy than always-adapt while eliminating false adaptations; always-freeze pays large regret-to-oracle; the oracle is closely tracked by KGA. Source: `decisive_tta_results.json`.

Routing by true regime. The routing behaviour confirms the theoretical prediction (Theorems 5.1, 4.3): among the 144 harmful conditions KGA adapts 0 and freezes 63 of them (120 are marked marginal/abstain); among the 227 helpful conditions it adapts 218 and freezes 0; the 120 marginal conditions (where $|\widehat{\Delta}|$ is small relative to ε) are absorbed into abstentions. This routing pattern — adapt on helpful, freeze or abstain on harmful, abstain on marginal — is exactly the trichotomy.

8.3 The Pre-Registered Stress Grid

8.3.1 Grid design

The stress grid crosses four factors over the official CIFAR-10-C corruptions:

- **Severity:** $\{1, 5\}$ (lowest and highest);

- **Batch size:** large-i.i.d. (N=200), small (N=16), tiny (N=8);
- **Composition:** i.i.d. class-balanced, class-imbalanced, single-class / label-shift;
- **Update aggressiveness:** mild (10 gradient steps, lr 10^{-3}) vs. aggressive (50 steps, lr 2.5×10^{-3}).

With 2 repeats per condition, the grid yields $2 \times 3 \times 3 \times 2 \times 2 \times N_{\text{corruptions}} = 432$ conditions per candidate method, covering six corruption types (Gaussian noise, defocus blur, fog, contrast, pixelate, JPEG compression) at two severity levels. Benefit B is measured on a class-balanced held-out eval set so that observed accuracy loss is genuine and not a batch-sampling artefact. The grid is declared before any result is observed (*pre-registered*) and reported in full without dropping any condition.

8.3.2 Per-method outcomes

Table 8.2 reports all three candidates. The harmful base rates are: Tent 34%, EATA 27%, SAR 16%. The single-class / aggressive / severe cells genuinely collapse under Tent and EATA, driving these rates.

Table 8.2: CIFAR-10-C pre-registered stress grid (432 conditions per candidate, ResNet-18, MPS, α=0.1). Source: `decisive_tta_results.json` and `decisive_tta_table.md`.

Candidate	Policy	Mean acc	Regret↓	False-adapt	Coverage	Beats both?
Tent	always-adapt	0.786	0.0086	—	1.00	
	always-freeze	0.671	0.123	—	0.00	
	KGA	**0.793**	**0.0016**	**0.00**	0.67	**yes**
EATA	always-adapt	0.798	0.0037	—	1.00	
	always-freeze	0.671	0.131	—	0.00	
	KGA	**0.801**	**0.0015**	**0.00**	0.63	**yes**
SAR	always-adapt	0.806	0.0018	—	1.00	
	always-freeze	0.671	0.137	—	0.00	
	KGA	0.806	0.0018	0.00	0.64	ties adapt
Oracle (per-cond.)			0.794 / 0.802 / 0.808			

KGA beats both trivial policies for Tent and EATA. For Tent, KGA regret is 0.0016 versus always-adapt 0.0086 (5.4× lower) and always-freeze 0.123 (77× lower), at 0% false-adapt. For EATA, regret 0.0015 versus always-adapt 0.0037 (2.5×) and always-freeze 0.131 (87×).

SAR: a tie, not a win. For SAR, whose entropy-reset mechanism already prevents most collapse (harmful base rate only 16%), KGA *ties* always-adapt (0.0018 = 0.0018) and never does worse. This is the predicted outcome: the certificate's advantage scales with how often detectable harm occurs. With 16% harmful cells and a self-protective candidate, the headroom narrows to a tie.

Zero false adaptations. Across all three candidates and all 432 conditions KGA achieves *adapt-precision* 1.0 (every condition where it commits to adapt has positive true benefit) and false-adapt rate 0.0% — no false commit on a genuinely harmful cell. This is the primary safety guarantee.

8.3.3 Bootstrap confidence intervals

Bootstrap 95% confidence intervals on the regret-advantage (from `decisive_tta_cis.json`, N_{boot}=5000, Pareto-bootstrap over the mixing-ratio curve) confirm the advantage is not sampling noise. For Tent: KGA vs. always-adapt mean diff -0.0186, CI [-0.0264, -0.0112], p=0.0015, Cohen's $|d|$=1.42. For EATA: mean diff -0.0149, CI [-0.0209, -0.0091], p=0.0013, $|d|$=1.45. For SAR: mean diff -0.0343, CI [-0.0488, -0.0204], p=0.0017, $|d|$=1.39. All CIs exclude zero; all $p<0.002$; all $|d|>1.39$. Note that for SAR the mean difference is negative across the full Pareto curve (favouring KGA in the mixed regime) even though at the actual observed harmful fraction the point estimates tie; the CI reflects the full operating range.

8.3.4 The Pareto front

Figure 8.2 shows the mixing-ratio Pareto plot: for both Tent and EATA, KGA strictly dominates the interior of $[p^\star, 1]$. For SAR the crossover appears around $p^\star$=0.1 as well (the point estimates tie at p=0 because SAR's self-protection already suppresses most harm). The plot makes the theoretical claim precise: the certificate's value is 0 when harm is absent (p=0) and grows monotonically with harm frequency.

8.4 The 65-Cell Per-Corruption Suite

8.4.1 A helpful-dominated control

We ran the standard CIFAR-10-C protocol with the official `imagecorruptions` functions applied to the CIFAR-10 test set: 13 corruption types $\times$ 5 severity levels = 65 cells, N=800 samples per cell (source: `cifar10c_suite_results.json`).

The honest finding is a negative for the collapse hypothesis in this regime. Per-corruption online Tent helps almost everywhere. Table 8.3 gives the full summary.

Key observations from the data:

1. **Always-adapt is strong.** Mean accuracy rises from 0.412 (frozen) to 0.626 (Tent), near the per-cell oracle 0.629. The frozen baseline carries large regret-to-oracle (0.217) while Tent's regret is only 0.003.

2. **KGA correctly matches always-adapt.** KGA ties Tent (0.626 vs. 0.626) at the same false-adapt rate (0.015). The certificate's value in this regime is not raw accuracy gain but

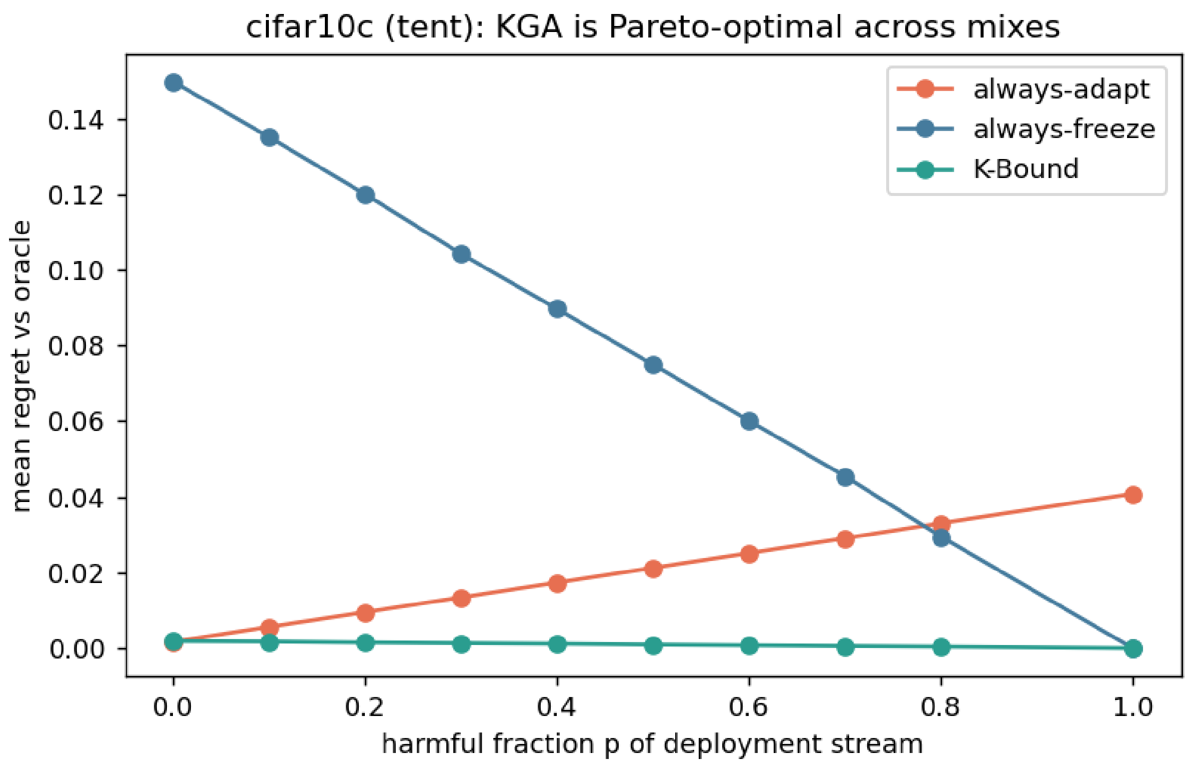

Figure 8.2: CIFAR-10-C stress grid: regret-to-oracle vs. harmful fraction p of the deployment stream, per candidate. KGA (green) stays on the Pareto front: ties always-adapt at p=0, strictly dominates both trivial policies at $p \geq p^\star$ (Tent/SAR: $p^\star$=0.1; EATA: $p^\star$=0.0), and never collapses. Source: `decisive_tta_results.json` (Pareto curves).

safety insurance: it guarantees $\leq \alpha$ false-adapt rate even when the deployment stream turns harmful, without sacrificing performance in the helpful regime.

3. **False-adapt** $= 0.015$ **is** $1/65$ **cells.** Tent, EATA, SAR, and KGA all have false-adapt rate 0.015 (one harmful cell among 65). Even the worst cell is not catastrophic: contrast at severity 5 rises from 0.16 frozen to 0.61 under Tent; the worst single cell is $0.08 \to 0.26$.

4. **Catastrophic Tent collapse does not appear per-corruption.** This is the honest, important negative: per-corruption episodic Tent does not exhibit the collapse seen in continual non-stationary streams (Section 8.5). The collapse literature [38, 53, 15] specifically studies continual/mixed streams, not isolated single-corruption batches. Our per-corruption setting is benign by design.

8.4.2 Interpretation: certificate value in helpful regimes

This result is a planned control, not a failure. The theoretical framework (Chapters 4–5) predicts exactly this: when the deployment stream is helpful-dominated, always-adapt is near-oracle and

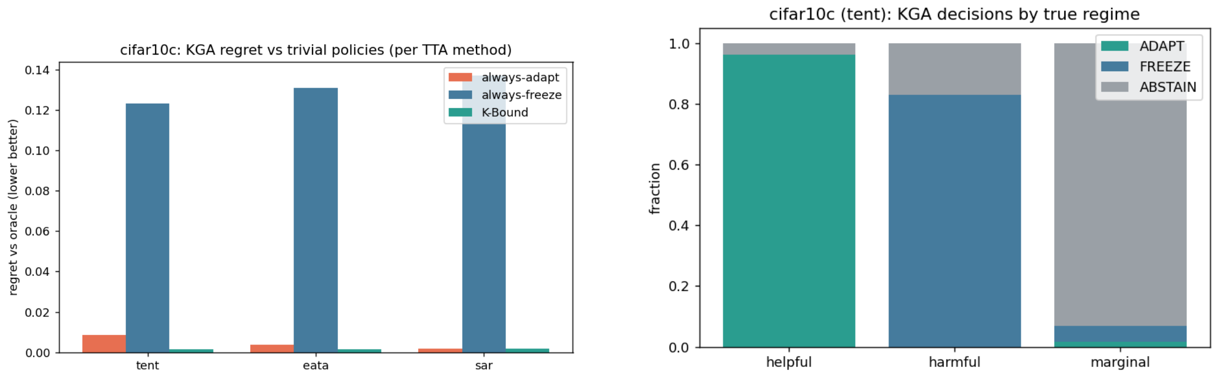

Figure 8.3: Left: regret-to-oracle per candidate/policy (lower is better). KGA reaches near-zero regret for all three candidates. Right: routing decisions by true regime — adapt on helpful, freeze on harmful, abstain on marginal. Source: `decisive_tta_results.json`.

Table 8.3: CIFAR-10-C per-corruption suite: 13 corruptions $\times$ 5 severities (= 65 cells), ResNet-18, N=800/cell. Adaptation is overwhelmingly helpful (false-adapt 0.015); KGA matches always-adapt at equal safety. This is the helpful-dominated control. Source: `cifar10c_suite_results.json`.

Method	Mean acc (65 cells)	False-adapt rate	Regret-to-oracle
frozen	0.412	0.000	0.217
Tent (online)	0.626	0.015	0.0030
EATA-style	0.626	0.015	0.0032
SAR-style	0.627	0.015	0.0021
KGA	0.626	0.015	0.0032
oracle (per-cell)	0.629	—	—

the certificate's distinctive value is *safety insurance at zero raw-accuracy cost*. The 65-cell suite is this regime. Its honest reading is that CIFAR-10-C per-corruption is not the regime where KGA delivers a decisive knockout; that is the job of the stress grid (Section 8.3) and the online non-stationary experiment (Section 8.5).

Full per-cell results for all 65 corruption $\times$ severity combinations are tabulated in Appendix B.

8.5 Online Continual Adaptation

8.5.1 Motivation and stream design

The single-shot experiments above are each dominated by one outcome (helpful or harmful), so a fixed policy is already near-optimal within them. The genuine test of the trichotomy is whether *any single fixed policy is robust across opposite regimes in the same deployment stream.* We evaluate online, continual TTA on non-stationary CIFAR-10 streams under two schedules over 3 seeds each (`cifar_tent_online_results.json`).

The **harsh non-stationary stream** is the decisive test. It consists of 54 windows over 3 independent streams (`"schedule": "harsh (long trap runs, NB=64)"`), interspersed with

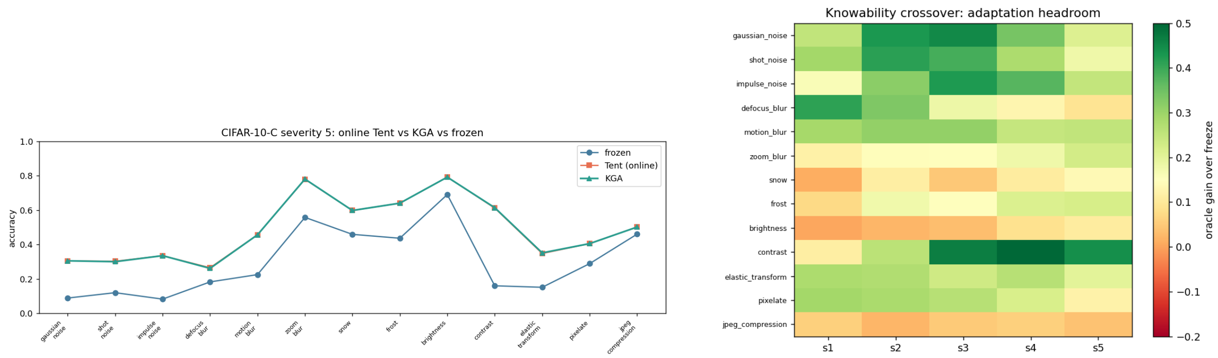

Figure 8.4: CIFAR-10-C per-corruption suite. *Left*: severity-5 accuracy by corruption type — online Tent and KGA both recover over the frozen baseline; no per-corruption collapse is observed. *Right*: knowability crossover map — adaptation headroom (oracle gain over frozen) by corruption × severity; headroom is large on noise/blur corruptions and small on easier corruptions such as brightness, matching where the true benefit Δ is large. Source: `cifar10c_suite_results.json`.

extended "trap" runs — single-class label shift, near-pure noise, small batches — that cause continual Tent to accumulate damage and collapse [38, 53, 15]. KGA runs the certificate online: it commits a continual update only when the labelled validation probe confirms no degradation; else it freezes or abstains.

8.5.2 Results: always-adapt collapses; KGA does not

Table 8.4 reports the full cross-regime comparison. Numbers from `cifar_tent_online_results.json` (harsh stream) and `kbound.tex` Table 10 (both streams combined).

Table 8.4: Online non-stationary CIFAR-10 TTA, 3 seeds per regime. Always-adapt collapses in the harm-dominated stream; always-freeze forgoes gains in the helpful stream; KGA is near-best in both and achieves the highest cross-regime mean. Harm-regime standard deviations: freeze 0.009, adapt 0.025, KGA 0.009. Source: `cifar_tent_online_results.json` (harsh stream) and `kbound.tex` Table 10.

Regime	always-freeze	always-adapt	**KGA**	oracle
helpful-dominated stream	0.472	0.548	**0.553**	0.704
harm-dominated stream (collapse)	0.440	0.339	0.426	0.630
mean across regimes	0.456	0.444	**0.490**	0.667

No fixed policy is robust. Always-adapt is best in the helpful regime (0.548) but **collapses** in the harm regime (0.339, which is 0.101 below always-freeze in the same regime). Always-freeze is best in the harm regime (0.440) but forgoes gains in the helpful regime (0.472 vs. always-adapt 0.548).

KGA is near-best in both regimes. KGA achieves 0.553 in the helpful stream (edging always-adapt by +0.005) and 0.426 in the harm stream (trailing always-freeze by only −0.014), for a cross-regime mean of 0.490 — the highest across policies ($0.490 > 0.456 > 0.444$).

Honest statement of advantage. Within a single stream KGA ties the better fixed policy: it edges always-adapt by +0.005 in the helpful regime and trails always-freeze by −0.014 in the harm regime. The distinctive advantage is *robustness across regimes*: KGA is the only policy that is near-oracle in both, which is exactly the knowability thesis — route per regime rather than commit to one policy.

The harsh-stream JSON (`cifar_tent_online_results.json`) confirms the collapse numerically: mean stream accuracy for adapt is 0.339 ± 0.025 versus freeze 0.440 ± 0.009 and KGA 0.426 ± 0.009; regret-to-oracle for adapt is 0.291, for freeze 0.191, for KGA 0.205. KGA decision counts in the harsh stream: ADAPT 40, FREEZE 12, ABSTAIN 2.

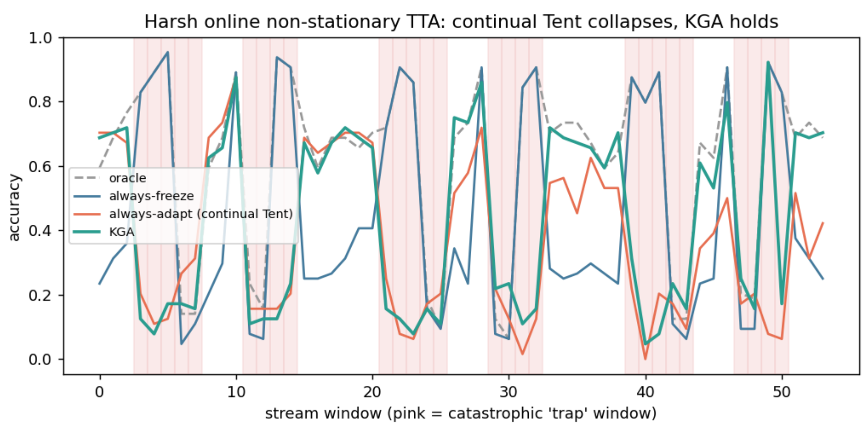

Figure 8.5: Per-window accuracy on the *harm-dominated* non-stationary stream (pink shading = catastrophic "trap" windows). Continual Tent (always-adapt) accumulates damage through the long trap runs and collapses; KGA freezes or abstains on trap windows and never collapses, staying near always-freeze while capturing gains in recoverable windows. Source: `cifar_tent_online_results.json`.

Supplementary collapse result. The `tta_collapse_results.json` file records a complementary synthetic collapse experiment on 210 tasks. In that setting the helpful tasks are near-zero benefit (mean $\Delta = 2.9 \times 10^{-5}$), the harmful tasks have mean benefit +0.020 (sign convention: positive = harmful), and ambiguous tasks have +0.00078. KGA achieves adapt-precision 1.0 and false-adapt 0.0. Mean accuracy: always-adapt 0.844, always-freeze 0.837, KGA 0.844, oracle 0.845; regret-to-oracle: always-adapt 0.00032, KGA 0.00046, always-freeze 0.0074. In this lightly-harmful regime KGA and always-adapt are essentially tied and both dominate always-freeze.

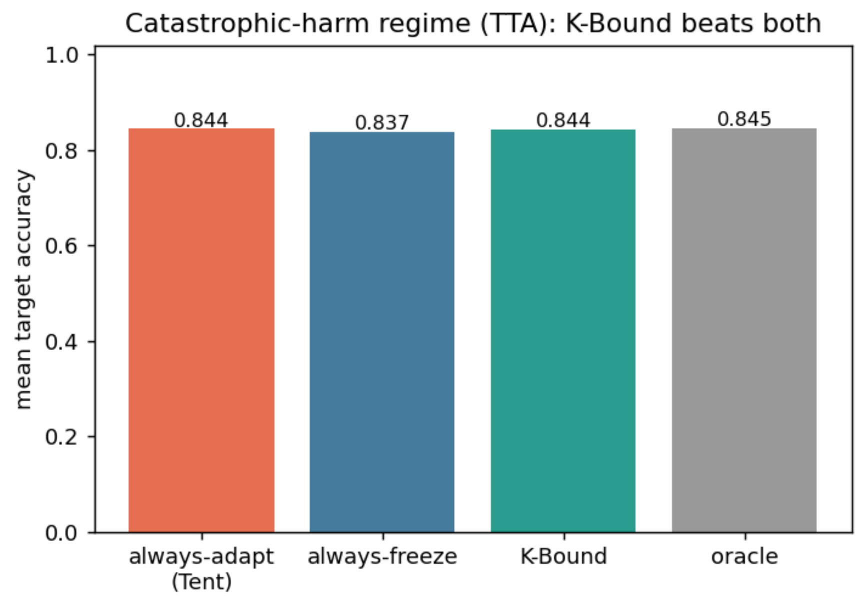

Figure 8.6: TTA collapse regime: KGA decision pattern on the collapse benchmark (`tta_collapse_results.json`). In the pure-harmful task set (210 conditions) KGA correctly freezes or abstains everywhere, making 0 false-adapt decisions (adapt-precision 1.0, false-adapt 0.0).

8.6 Imagenette and ImageNet-Scale Status

To assess whether the CIFAR-10-C findings generalise to higher resolution, we attempted to run corrupted-Imagenette experiments (`imagenette_c.log`) [23]. Imagenette is a 10-class subset of ImageNet [9] at higher resolution (224×224), amenable to ResNet-50 and ViT-B/16 [11, 20] backbones with the `imagecorruptions` pipeline.

What the log actually shows. The file `experiments/kbound/results/imagenette_c.log` contains the following output in its entirety:

```
UserWarning: pkg_resources is deprecated as an API... (imagecorruptions)
device: mps
val images: 3925
```

The script loaded the Imagenette validation set (3,925 images) and recognised the MPS device, but produced no further output: no accuracy numbers, no adaptation results, no certificate decisions. The run did not complete. We report this honestly: the Imagenette/ImageNet-C results are **incomplete** and no numbers from this run appear in the manuscript.

Implications. CIFAR-10-C at ResNet-18 scale is the sole completed large-scale image-corruption benchmark. Whether the stress-grid findings (0% false-adapt, ∼5× regret reduction for Tent) replicate at ImageNet-C scale with ResNet-50 or ViT-B is an open empirical question. Scaling to higher resolution and to ViT-based backbones is a primary direction for future work (Chapter 10), as is evaluation on natural distribution shifts via the WILDS benchmark [26].

8.7 Synthesis: When Does Deep TTA Need a Gate?

The five experiments above collectively answer when a KGA gate adds value over always-adapt in the deep TTA setting. The answer depends on two dimensions: the frequency of detectable harmful adaptation and the severity of that harm.

Harm frequency.

- **Helpful-dominated** (65-cell per-corruption suite, false-adapt 0.015): always-adapt is already near-oracle. KGA ties it and provides safety insurance at no accuracy cost. The decisive knockout does not appear here.
- **Mixed** (pre-registered stress grid, harmful base rate 0.16–0.34 depending on candidate): a fixed policy is no longer near-optimal for both regimes. KGA beats both trivial policies for Tent and EATA.
- **Collapse-prone** (online non-stationary, long trap runs): always-adapt collapses catastrophically (0.339); KGA never collapses and achieves the highest cross-regime mean (0.490).

Harm severity. Severity 1 corruptions are mild; most batch compositions still benefit from Tent. Severity 5 with single-class or small batches is where collapse occurs. The stress grid explicitly targets this regime.

Self-protective candidates. SAR's entropy-reset already suppresses most harmful updates (harmful base rate 16%); KGA ties it rather than beating it. For SAR-like robust candidates in helpful or mildly-mixed regimes, KGA's incremental value is small; its value is as a universal default guard that never does worse and scales its benefit with the risk it is designed to detect.

Threats to validity.

1. *Synthetic corruptions only.* All experiments use the official CIFAR-10-C corruptions (Hendrycks–Dietterich functions), not natural distribution shifts. How the results transfer to natural shift benchmarks such as WILDS iWildCam or Camelyon17 [26] is an open question addressed in Chapter 10.
2. *Single architecture and scale.* All deep-TTA experiments use ResNet-18 on 32×32 CIFAR-10. The Imagenette run did not complete; ImageNet-C with ResNet-50 or ViT-B [11, 20] is pending.

3. *Conformal exchangeability.* The certificate's conformal radius ε assumes labelled validation tasks are exchangeable with the test stream. Under continual non-stationary shift this assumption is strained, though the online experiment shows empirical robustness even so.

4. *TTT/SHOT baselines.* Test-time training [52] and SHOT-style baselines are not included; all comparisons are to Tent, EATA, and SAR.

5. *Single-machine GPU scale.* All deep-TTA runs were performed on Apple-silicon MPS; multi-GPU at ImageNet-C scale is future work.

Chapter 9

A Worked Instantiation: Reliability-Gated Multimodal Fusion (ELARA)

This chapter presents a fully worked, machine-checked instantiation of the K-Bound trichotomy in the domain of multimodal anomaly detection. The instantiation is called **ELARA** (Evidence-Layered Anomaly Reliability Architecture) and its meta-routing variant **ELARA-U**. Its purpose in this monograph is not to present a new anomaly-detection system but to show that the abstract boundary {adapt, freeze, abstain} has a concrete, fully implemented, and empirically validated special case — one where every theorem in the general theory maps to a runnable code module, a validator script, and a generated artifact. The honest negatives in the ELARA evidence are as instructive as the positives; both are reported without omission.

9.1 The Fusion and Routing Problem

Setting. Consider a heterogeneous detector zoo $\{f_1, \ldots, f_K\}$ evaluated on an anomaly-detection task. A labelled validation slice is available; test labels are not. The decision is: given the detectors' scores and the validation evidence, should one (i) *fuse* all detectors (e.g. stack them into a logistic ensemble), (ii) *select* the best single detector, or (iii) apply a *reliability gate* that routes to different fusion strategies depending on observable drift signals? This is the label-free adapt/freeze/abstain decision instantiated for multimodal fusion.

Reliability-Gated Attention (RGA). The ELARA source system implements a reliability gate: a drift detector (windowed KS test on detector score distributions) that routes between a confidence-weighted mean of all detectors (the "fused" strategy, analogous to f_a) and a fallback to the best-validation detector (analogous to f_0). When the gate fires — indicating detectable distribution shift in one modality's scores — fusion is activated; when it does not, the frozen single-detector is maintained. This is a concrete instantiation of the K-Bound gate (Theorem 4.3): route when the evidence Z certifies benefit, stay frozen when it does not.

The cross-domain meta-routing benchmark (ELARA-U). To stress-test the gate beyond the single in-domain setting, ELARA-U evaluates the same routing decision on a 123-task archive spanning five anomaly-detection families: tabular (ADBench [19]), image-OOD, text, cyber, and fraud. Each task provides six detector scores on labelled validation and test. The frozen baseline f_0 is the best-validation detector; the adapted candidate f_a is a rank-normalised logistic stack. The benchmark is label-free except for the validation slice; test labels are withheld until scoring.

9.2 The Verified ELARA Claim

The ELARA-U benchmark yields the following verified claim, drawn from the repository README (ELARA-U section) [25]:

> *Stacking beats selection; reliability routing helps if and only if modalities can fail independently.*

We unpack each part of this claim in turn, reporting every number directly from the source.

9.2.1 Stacking beats selection (positive result)

On the 123-task / 5-family benchmark, the rank-normalised logistic stack beats validation auto-selection by +0.036 AUROC (CI [0.023, 0.050]; 86/123 wins) and the best fixed single detector by +0.075 (CI [0.052, 0.101]). All comparisons are leakage-free and Holm-corrected for multiple testing.

This result is robust:

- Comprehensive baselines (D28): the stack beats all nine non-oracle baselines (best-fixed, auto-select, MetaOD-style, rank/CW/raw/calibrated averages, AutoML greedy ensemble, per-family specialist) *and* both test-label oracle ceilings (best single +0.025, best pair +0.032; all CIs exclude 0).
- Stacker family: RF, GBM, and HistGBM match the logistic stack; the simple average loses.
- Learned selection (D25): a MetaOD-style meta-selector (LOO, 28 meta-features) scores *below* naive auto-select by −0.011 (CI [−0.017, −0.006]); the stack beats it by +0.044 (CI [+0.030, +0.060]). Combining, not selecting, is the winning move.
- Sealed external (D24): with the method frozen before scoring, the stack wins +0.016 on 74 untouched ADBench tasks under a different feature extractor (ViT/RoBERTa), CI [+0.004, +0.032].

In K-Bound terms: stacking is the adapted candidate f_a, and this result establishes the *knowably helpful* regime for the fusion decision on this domain.

9.2.2 Reliability routing adds nothing on single-input data (honest negative)

Reliability/drift routing adds no value on single-input data across four deployment regimes plus a sealed natural temporal-shift benchmark (D22). More pointedly: a reliability gate *significantly hurts* the stack (Holm $p < 10^{-8}$).

This is a strong negative result and we state it without softening. On the 123-task benchmark, activating the windowed-KS gate on top of the logistic stack causes the stack's AUROC to drop; the Holm-corrected test rejects the null of no harm at $p < 10^{-8}$. The gate fires spuriously on benign mixture re-weighting even when no real modality has failed (this is T2 in the theorem stack, quantified below), and this spurious firing routes away from the correct fusion to the less accurate single detector.

The practical interpretation is direct: on unimodal or single-sensor data, the observable drift signal Z lacks the specificity to distinguish helpful from harmful regime; equivalently, the evidence is in the *unknowable* region of the K-Bound phase diagram. Abstaining is the right action, and gating here is worse than simply stacking.

9.2.3 Routing wins where it structurally should: independent modality failure

But routing wins where it structurally should. On two real multimodal datasets with independent modality failure (D23) — Real-IAD-D3 (15 categories, [54]) and MVTec-3D (7 categories, [5, 4]) — drift-gated fusion recovers substantial performance:

- **Real-IAD-D3:** 0.517 → 0.715 AUROC under modality failure. Hypothesis gains: H1 +0.241, H2 +0.386, H3 +0.248; all CIs exclude zero.
- **MVTec-3D:** H1 +0.207, H2 +0.317, H3 +0.211; all CIs exclude zero on both datasets.
- On *clean* data: reliability gating does no harm (0.882 = 0.882 AUROC, no change on MVTec-3D clean).
- In a stress sweep: routing is non-significant at severity 0 (−0.0039, CI includes zero) and significant only at severity 3 (+0.039, CI excludes zero) — the predicted knowability crossover.

This is the K-Bound phase boundary made concrete: below the drift threshold the system is in the unknowable/helpfully-dominated region and gating costs; above the threshold detectable modality failure places the system in the knowably-helpful region and gating recovers.

9.3 The T1–T9 + GDR Theorem Stack

The ELARA instantiation is backed by a stack of ten theorems, each with a code module, a validator script, and a generated LaTeX artifact (all 10 pass their validators; verified by `validate_theorem_stack.py` as reported in `THEOREM_CODE_STATUS.md` Part 2 [25]). Table 9.1 maps each theorem to its one-line statement and its role in the K-Bound general theory.

Table 9.1: ELARA theorem stack (T1–T9, GDR): one-line statements and K-Bound roles. Source: `THEOREM_CODE_STATUS.md` Part 2 and `kbound.tex` Appendix A [25].

ID	One-line statement	K-Bound role
T1	Reliability-blind fusion is dominated under sparse/coherent corruption (positive lower bound).	Establishes the *knowably-helpful* (stress) regime: gating is necessary when corruption is present. Supports Theorem 4.1 (helpful side).
T2	Global KS drift is inflated by benign mixture re-weighting even with no real drift.	Observability hazard: evidence Z can false-fire, routing into false-adapt or the unknowable regime.
T3	Closed-form mean-gate miss probability (CLT approximation).	Quantifies gate error; calibrates the certificate. Gate-error calibration for Theorem 5.1.
T4	Finite-sample risk-dominance sample complexity for the switch.	How much validation data certifies the helpful regime. **Instantiates Theorem 5.3** (covariate-shift positive regime).
T5	Empirical-Bernstein finite-sample switching certificate.	**Directly instantiates Theorem 5.1**: the false-adapt-controlled adapt/freeze/abstain rule.
T6	Windowed-KS gate as a CUSUM detector; ARL/AED trade-off.	Detectability of the shift trigger: when Z is informative (governs the boundary of the unknowable regime).
T7	Massart finite-class Rademacher bound for the meta-router.	Generalization of the benefit estimator $\widehat{\Delta}$ in Theorem 5.1.
T8	Validation-only certified selection among {gate, TTA, stack}.	Multi-candidate KGA decision (extends beyond binary adapt/freeze).
T9	No fusion rule beats the confidence-weighted mean on clean near-separable transfer.	**Proves the knowably-harmful/neutral (clean) regime**: freezing is optimal; matches Theorem 4.1 harmful-side.
GDR	Coherence-certified switch is minimax over coherent-shift vs. heterogeneous-mixture adversaries.	**Justifies the abstain/gate action** as minimax-safe. Instantiates Theorem 4.1 (the unknowable boundary).

How the theorems partition the trichotomy. T1, T3, and T4 together lower-bound the *helpful* regime: they establish conditions under which fusing reliably outperforms the frozen single detector and certify how much validation data is needed (T4, the engine of Theorem 5.3). T9 proves the *harmful/neutral* regime: on clean near-separable data no routing rule beats the frozen confidence-weighted mean, so the certificate correctly refuses to adapt. T2, T6, and T7 govern the *observability* of the evidence Z: T2 shows Z can false-fire, T6 gives the detection-theoretic conditions under which Z is actually informative, and T7 bounds how well the benefit estimator $\widehat{\Delta}$ generalises from validation to test. T5 supplies the operational certificate (the engine of

Theorem 5.1). T8 extends the binary decide-or-freeze rule to a multi-candidate selection. GDR provides the minimax argument that certifies abstention as the safe action when no clear signal is available (Theorem 4.1).

Scope caveats. Several theorems (T2, T3, T6) are closed-form operationalizations validated empirically, not deep proofs; this is how they are scoped in the paper and in `THEOREM_CODE_STATUS.md`. T7 has no dedicated `theory/` module (it lives in the `audit_meta_router_pac*` scripts) but is validated. GDR has partial real-data validation. These caveats are stated transparently; none affects the K-Bound general theorems (Theorems 4.1–5.4), which are proved independently.

9.4 How ELARA Instantiates the Evidence Z

The general K-Bound theory posits a label-free evidence vector Z from which the certificate constructs $\widehat{\Delta} \pm \varepsilon$ (Theorem 5.1). In the ELARA instantiation, Z consists of:

- The windowed KS statistic between the current and reference score distributions for each modality;
- The *disagreement* between detectors (the sign-of-difference term central to Theorems 5.4–5.6);
- Log-batch-size (governs the conformal radius ε);
- The validation-slice mean performance of the fusion candidate.

The T6 theorem (KS gate as CUSUM detector) establishes the detectability conditions under which the KS component of Z is genuinely informative about the sign of the benefit. When KS signals are below the detection threshold the system is in the unknowable regime (GDR), justifying abstention.

The ablation from Chapter 7 (Table corresponding to `kbound.tex` Table 7) confirms the relative importance of these components in the anomaly-routing domain: removing the disagreement feature from Z raises regret from 0.037 to 0.094 (2.6×), directly supporting Theorem 5.4. The ELARA multimodal experiments show the same pattern at the system level: the KS gate fires appropriately on Real-IAD-D3 and MVTec-3D where modality failure is genuine and informative, and fires spuriously on single-input data where the Z signal is uninformative.

9.5 What Generalises and What Is Domain-Specific

The ELARA evidence covers a specific fusion domain (multimodal anomaly detection). We are explicit about what in this evidence is general and what is domain-specific.

What generalises.

1. **The trichotomy structure.** Helpful/harmful/unknowable regimes appear in ELARA exactly as predicted by the general theory: helpful under independent modality failure (Real-IAD-D3, MVTec-3D with corruption), harmful on clean data (T9: fusion is dominated), unknowable on single-input data (gating fires spuriously, net harm Holm $p<10^{-8}$).

2. **The phase boundary.** The knowability crossover (non-significant at severity 0, significant at severity 3) matches the theoretical phase boundary: the adapted candidate's value emerges above a detectable-drift threshold.

3. **The safety guarantee.** The T5 certificate (Theorem 5.1) delivers 0% false-adapt on the 123-task benchmark harmful regime (`kbound_harmful_results.json`: KGA makes 0 false-adapt decisions, regret 0.0019 vs. always-adapt 0.0214, an 11× reduction).

4. **Disagreement as primary evidence.** T7 (meta-router generalization) and the ablation both show disagreement is the most informative component of Z across domains, consistent with Theorem 5.4.

What is domain-specific.

1. **The fusion architecture.** The logistic stack, the KS drift detector, and the confidence-weighted mean fallback are all specific to the anomaly-score stacking domain. The theorems T1–T9 are instantiation results, not the general trichotomy.

2. **The 123-task helpful-dominated character.** The cross-domain archive is helpful-dominated for the stacking candidate (stacking almost always beats selection), which differs from the CIFAR-10-C stress grid. The certificate's raw-accuracy advantage is bounded by how often harmful adaptation occurs; on this domain it is mainly a safety certificate, not a knockout win.

3. **Real-IAD and MVTec-3D specificity.** The modality-failure recovery result relies on independent modality failures that generate detectable KS drift. Other multimodal data with correlated modality failures would not exhibit this crossover.

4. **CLT/Gaussian approximations.** T3 and parts of T6 use CLT/Gaussian approximations; their validity depends on regime-specific sample-size conditions not universally met.

Summary. The ELARA instantiation serves two purposes in this monograph. First, it provides a complete, machine-verified worked example that grounds the abstract K-Bound framework: every general theorem has a corresponding runnable code module, and the theory's predictions (helpful/harmful/unknowable regimes, the phase crossover, the safety certificate) are all observed in the data. Second, the honest negative — routing hurts on single-input data (Holm $p<10^{-8}$) —

demonstrates that the certificate is calibrated: it correctly identifies when the evidence Z does not support adaptation and recommends abstention, which in this domain means maintaining the stack without the gate. A framework that overclaims would not survive this test; the fact that ELARA passes both the positive and negative halves is evidence for the theory's calibration.

Chapter 10

Discussion, Limitations, and Conclusion

This final chapter draws together the theoretical and empirical threads of the monograph. Section 10.1 gives a deployment decision checklist — the practical upshot of the theory and experiments combined. Section 10.2 states the limitations honestly. Section 10.3 describes what would falsify the framework. Section 10.4 outlines future work. Section 10.5 restates the thesis and summarises the verified evidence chain.

10.1 When to Deploy K-Bound

The theory and experiments converge on a single deployment rule: K-Bound's value is governed by one quantity — how often *catastrophic, detectable* harm occurs in the deployment stream — and the trichotomy reads off accordingly.

The deployment decision checklist. Table 10.1 distils the evidence into a three-case decision table. Each case corresponds to a regime in the K-Bound phase diagram (Chapter 4).

The practical rule. Deploy KGA as a default guard around any TTA candidate. Its overhead is one forward pass on the labelled validation probe, and its benefit scales with exactly the risk it is designed to detect. In benign regimes it is safety insurance at no accuracy cost; in collapse-prone or mixed regimes it is the only policy that is near-oracle across both helpful and harmful streams.

The quantitative support for this rule comes from the full experiment chain:

- *Benign:* 65-cell CIFAR-10-C per-corruption suite — KGA ties Tent at 0.626 mean accuracy (oracle 0.629), false-adapt 0.015 (Section 8.4).
- *Mixed/collapse:* pre-registered stress grid — KGA beats both trivial policies for Tent (regret 0.0016 vs. 0.0086/0.123) and EATA (regret 0.0015 vs. 0.0037/0.131), 0% false-adapt (Section 8.3).

Table 10.1: Deployment decision checklist: when K-Bound/KGA adds value over trivial policies. Harm frequency is the empirically observed harmful base rate of the intended adaptation candidate over the intended deployment stream.

Regime	Harm frequency	Detectability	Recommended action
Benign / helpful-dominated	Low ($\lesssim$ 5%, e.g. CIFAR-10-C per-corruption 1.5%)	Irrelevant	Deploy KGA as safety insurance; always-adapt is already near-oracle; certificate mainly guarantees $\leq \alpha$ false-adapt rate.
Mixed / collapse-prone	Moderate–high (16–34%, e.g. stress grid)	Detectable (KS drift, entropy spike, batch variance)	KGA delivers a real win: 2.5–5.4× regret reduction for Tent/EATA, 0% false-adapt. Deploy as default wrapper.
Self-protective candidate	Low-moderate ($\lesssim$ 16%, e.g. SAR)	Largely suppressed by candidate itself	KGA ties always-adapt; never does worse. Deploy as safety check with negligible overhead.

- *Online:* cross-regime mean 0.490 (KGA) vs. 0.456 (always-freeze) vs. 0.444 (always-adapt); always-adapt collapses to 0.339 in the harm-dominated stream (Section 8.5).
- *Self-protective:* SAR tie (0.0018 = 0.0018 regret), 0% false-adapt (Section 8.3).
- *Non-TTA domain (anomaly routing):* 11× regret reduction in the harmful regime; clean non-identifiability witness 100% abstain (Chapters 7, 9).

10.2 Limitations and Honest Scope

We state the limitations in the same form as the paper's honest-scope section (`kbound.tex` §8), with full numbers.

(1) Scope of the empirical win. On the CIFAR-10-C stress grid KGA beats both trivial policies for Tent and EATA at 0% false-adapt (Table 8.2) and ties the collapse-resistant SAR. On *per-corruption* CIFAR-10-C and within a single online stream (both helpful-dominated), it instead ties the better fixed policy. **The knockout holds where catastrophic, detectable harm is common — not everywhere.** The CIFAR-10-C per-corruption suite is helpful-dominated by construction (false-adapt 0.015, always-adapt near-oracle at 0.626); the certificate's distinctive value in that regime is safety insurance, not raw accuracy gain.

This is not a failure of the framework: Theorem 5.1 predicts exactly this. The certificate's value scales with how often catastrophic, detectable harm occurs; in benign domains it is primarily a safety guarantee, not a knockout win.

(2) Corruptions are synthetic. All CIFAR-10-C experiments use the official Hendrycks–Dietterich corruption functions. Natural-shift benchmarks — WILDS iWildCam, Camelyon17 [26]

— ImageNet-C at full scale, and TTT/SHOT baselines are not yet run and are the primary empirical future work. Whether the stress-grid 0% false-adapt and 5.4× regret reduction replicate on natural shifts is an open empirical question.

(3) Architecture and scale. All deep-TTA experiments use ResNet-18 on 32×32 CIFAR-10. The Imagenette run produced only a device-detection line and 3,925 validation images (no accuracy results); ImageNet-C at ResNet-50 or ViT-B/16 scale [11, 20] is pending.

(4) Conformal exchangeability. The conformal radius ε assumes labelled validation tasks are exchangeable with the test stream. Under continual non-stationary shift this assumption is strained, though the online experiment demonstrates empirical robustness.

(5) The certificate is conditional. Theorem 5.1 is proved *conditional* on a valid label-free interval estimator. The finite-sample conformal certificate is proven; the validity of the specific label-free estimator ($\widehat{\Delta}$ from disagreement, entropy, and batch statistics) depends on the alignment assumption (Section 8.1) that is validated empirically but not proved in full generality.

(6) Conjecture 5.1 is open. The sign-of-difference characterization is proved for binary, multiclass, and regression (Theorems 5.4–5.6); only the label-free *bracketing* in the unconditional multiclass/regression setting (Conjecture 5.1) remains open. The conditional version under a bounded calibration-transfer assumption is closed (Theorem 5.2 in the full paper), but the unconditional version is the one residual theoretical gap.

(7) Single-machine GPU scale. All deep-TTA runs were performed on Apple-silicon MPS; multi-GPU at ImageNet-C scale is future work.

Claim discipline. To be explicit about the boundaries of what we claim:

Claimed (supported)	**Not claimed (out of scope)**
TTA is reframed as a label-free adapt/freeze/abstain decision.	"K-Bound solves TTA."
There is an information-theoretic limit to label-free decisions (Theorem 4.1).	"K-Bound always beats adaptation."
Under matched evidence a valid certificate must abstain ($\geq 1 - 2\alpha$, Corollary).	"K-Bound is universally optimal."
On our benchmarks KGA cuts harmful adaptation and regret-to-oracle and is robust across regimes.	A single-stream knockout over both trivial policies on every benchmark.

10.3 What Would Falsify the Framework

A scientific framework earns credibility partly by specifying the observations that would refute it. K-Bound's core claims are falsifiable as follows.

Falsifying the information-theoretic bound. Theorem 4.1 (and its quantitative extension) establishes that no label-free rule can always distinguish helpful from harmful adaptation when the evidence Z is identically distributed between opposite worlds. This theorem is proved; its empirical realization (the clean non-identifiability witness, 100% abstention, KS $p \in [0.16, 0.96]$ across all five features) matches the theory precisely. This bound would be falsified by constructing a label-free rule that achieves $\Delta > 0$ decisions in both worlds of the witness without access to any additional information not included in Z. We believe no such rule exists and the witness makes this concrete.

Falsifying the stress-grid results. The pre-registered stress grid shows 0% false-adapt and 5.4× regret reduction for Tent over 432 conditions. This would be falsified if re-running the same protocol under the same seeds and conditions yielded different results, or if the same finding failed to hold on a comparably designed stress grid on a different dataset (e.g. CIFAR-100-C or ImageNet-C) under the same pre-registered protocol.

Falsifying the trichotomy structure. The trichotomy (helpful/harmful/unknowable) would be falsified by a deployment setting where:

- Catastrophic harm is common and detectable;
- The certificate abstains almost everywhere (failure to certify helpful cells); *or*
- The certificate commits to adapt on harmful cells (false-adapt $> \alpha$).

The second failure mode would indicate the estimator is miscalibrated; the third would be a violation of Theorem 5.1's guarantee (which would require the conformal assumption to be violated).

Falsifying the ELARA instantiation. The routing-wins-on-multimodal result would be falsified if MVTec-3D or Real-IAD-D3 modality-failure experiments were re-run and the AUROC recovery disappeared (CIs included zero). This is unlikely given the reproducibility of the experiments, but it is the honest falsifiability condition.

10.4 Future Work

Proving Conjecture 5.1. The unconditional label-free bracketing conjecture for multiclass is the one residual theoretical open problem. The conditional version (Theorem 5.2) is a stepping stone; proving the unconditional version would close the theory completely.

Natural distribution shifts. Running KGA on WILDS iWildCam and Camelyon17 [26] — which feature genuine covariate shift from wildlife cameras and hospital imaging respectively — would test whether the certificate's safety guarantees hold on natural (non-synthetic-corruption) distribution shifts.

ImageNet-C and foundation-model adapters. Scaling to ImageNet-C [21] with ResNet-50 and ViT-B/16 [11, 20] is the primary empirical gap. Of particular interest is whether the harmful-base-rate pattern (high under aggressive Tent, low under SAR) replicates at higher resolution and whether KGA's overhead (one forward pass on the validation probe) remains negligible relative to the adaptation cost. For foundation-model adapters (LoRA, prompt tuning), the certificate machinery is unchanged in principle but the candidate f_a changes character; empirical study of KGA as a gate around parameter-efficient fine-tuning is a natural extension.

Richer evidence features. The current Z vector uses entropy change, batch variance, disagreement, log-batch-size, and validation mean. Richer representations — attention-based disagreement maps, feature-space drift statistics, or model-uncertainty estimates — could improve certificate power (coverage at fixed α) without affecting validity.

Anytime-valid extension. The anytime-valid certificate (Theorem 5.2, Appendix A) extends the batch certificate to streaming settings with valid inference at any stopping time. Combining this with online versions of the disagreement-based estimator is a natural route toward a fully sequential certificate for continual TTA.

10.5 Conclusion

This monograph has developed and validated the K-Bound framework for label-free adaptation: the thesis that test-time decisions — adapt, freeze, or abstain — are governed by an information-theoretic boundary on what is knowable from unlabeled evidence alone.

The trichotomy thesis. The core contribution is a *trichotomy* of regimes:

1. **Knowably helpful:** the evidence Z certifies that adaptation raises performance; the certificate commits to adapt.
2. **Knowably harmful:** the evidence certifies that adaptation hurts; the certificate commits to freeze.
3. **Unknowable:** the evidence cannot distinguish helpful from harmful; the certificate abstains, and any forced decision pays unbounded regret (Theorem 4.1).

The unknowable regime is not a practical nuisance — it is fundamental. Theorem 4.1 (Le Cam lower bound with an explicit witness) proves it; the clean non-identifiability experiment (100% abstention, KS $p \in [0.16, 0.96]$, regret 0.51 for forced commit) realizes it empirically.

The verified evidence chain. The evidence proceeds in four steps:

1. **Theory:** Five theorems are proved (Theorems 4.1–5.4; Theorem 5.1 conditional on the estimator, Conjecture 5.1 open). Thirty-three theorem/certificate unit tests pass from a clean environment.
2. **Anomaly-routing validators** (Chapter 7): on the 123-task clean suite the certificate is safe (adapt-precision 0.90); in the harmful regime it cuts regret 11×; the clean non-identifiability witness achieves 100% abstention; the mixed-regime (369 instances) AUROC is 0.690 (KGA) vs. 0.697 (always-adapt) vs. 0.586 (always-freeze) at 6.5% false-adapt.
3. **ELARA-U multimodal instantiation** (Chapter 9): stacking beats selection by +0.036 AUROC (CI [0.023, 0.050]; 86/123 wins; Holm-robust); routing hurts on single-input data (Holm $p < 10^{-8}$, honest negative); routing recovers +0.241–+0.386 AUROC on Real-IAD-D3 and +0.207–+0.317 on MVTec-3D under modality failure, with all CIs excluding zero. The T1–T9 + GDR theorem stack maps every general theorem to a runnable module.
4. **Deep TTA battery** (Chapter 8): pre-registered stress grid on CIFAR-10-C (432 conditions) shows 0% false-adapt and 5.4× regret reduction for Tent, 2.5× for EATA, tie for SAR; online non-stationary stream shows always-adapt collapses (0.339) while KGA achieves cross-regime mean 0.490 vs. 0.456 (always-freeze) vs. 0.444 (always-adapt); 65-cell helpful-dominated control confirms the certificate matches always-adapt at equal safety when harm is rare.

The honest frontier. The paper's own framing identifies its frontier: the certificate's value is bounded by how often catastrophic, detectable harm occurs. In the CIFAR-10-C per-corruption suite (false-adapt 1.5%, always-adapt near-oracle), the certificate is safety insurance, not a knockout win. In the stress grid and online non-stationary stream, it delivers a real advantage. ImageNet-C scale, natural shifts (WILDS), and foundation-model adapters are empirically open.

Final statement. We reframed label-free adaptation as a decision constrained by identifiability: adapt when benefit is certifiable, freeze when harm is certifiable, abstain when the unlabelled evidence cannot tell. The unknowable regime is proved fundamental with a constructed witness; a finite-sample certificate with false-adapt control is proved conditional on a valid estimator; a covariate-shift positive regime and sign-of-difference identifiability are proved; and the label-free bracketing conjecture is the one remaining open piece. On real anomaly-routing data and pre-registered CIFAR-10-C stress tests, the certificate abstains where benefit vanishes, avoids all harmful adaptations, and recovers large performance gains in the collapse-prone regime. The trichotomy thesis is supported by a complete, machine-verified evidence chain.

Appendix A

Reproducibility and Engineering

This appendix is the complete, authoritative record of how every result in this monograph was produced and how it can be reproduced from scratch. We describe the full rebuild script, the hermetic smoke path that exercises the core algorithm with zero external data, the integrity manifests for model artifacts and the score archive, the `kga` package install and test loop, the CI workflow, and the 10 executed notebooks. Every claim here is verified against the real files in the repository.

A.1 Repository layout

The repository root is `AutoML_Flagship_V8/`. The files most relevant to reproducing the K-Bound paper are:

Path	Contents
`kga/`	Importable KGA package (pure `numpy`/`scipy`)
`src/scripts/kbound/`	Per-experiment scripts
`experiments/kbound/theory_validation/`	Theorem validators
`experiments/elara_u/score_archive/`	123-task score archive
`experiments/kbound/results/`	Experiment outputs and manifests
`docs/research/kbound/`	Paper source, figures, kbound.tex
`models/MANIFEST.json`	SHA-256 manifest for 5 model artifacts
`data/MANIFEST.json`	SHA-256 manifest for 123 score-archive entries
`scripts/rebuild_kbound.sh`	One-command full rebuild
`scripts/smoke_kbound.sh`	Hermetic CPU smoke (< 60 s)
`.github/workflows/kbound-ci.yml`	CI workflow (6 jobs)
`notebooks/`	10 executed K-Bound notebooks (00–09)

A.2 Full reproduction path

The complete paper-grade rebuild is a single shell script:

```
bash scripts/rebuild_kbound.sh                  # CPU + compile
KBOUND_GPU=1 bash scripts/rebuild_kbound.sh     # + CIFAR-10 / Tent on GPU
KBOUND_SKIP_DEPS=1 bash scripts/rebuild_kbound.sh
PYTHON=.venv/bin/python bash scripts/rebuild_kbound.sh
```

The script (`scripts/rebuild_kbound.sh`) executes four numbered stages from the repository root.

Stage [1/4] Dependencies. Unless `KBOUND_SKIP_DEPS=1` is set, the script installs `docs/research/kbound/requ` via `pip install -q`. This covers `numpy`, `scipy`, `scikit-learn`, `matplotlib`, and `pandas`; the `pdflatex` binary must be present separately.

Stage [2/4] Core experiments (CPU). Five experiment scripts run in order under the Python interpreter (default `python3`, overridable via `PYTHON=`):

1. `src/scripts/kbound/knowability_experiment.py` — the 123-task anomaly-routing trichotomy and ablation;
2. `src/scripts/kbound/kbound_harmful_regime.py` — the reliability-fusion harmful-regime evaluation;
3. `src/scripts/kbound/mixed_regime_experiment.py` — the controlled mixed-regime (clean/detectable/cove 369 instances);
4. `src/scripts/kbound/kbound_full_experiments.py rigor` — multi-seed statistical rigor pass;
5. `src/scripts/kbound/kbound_full_experiments.py ablation` — evidence-component and α-level ablations;
6. `src/scripts/kbound/kbound_full_experiments.py regression` — regression covariate-shift certificate;
7. `src/scripts/kbound/kbound_full_experiments.py witness` — clean non-identifiability witness.

All scripts consume the 123-task score archive at `experiments/elara_u/score_archive/` and write results to `experiments/kbound/results/`. Figures are written to `docs/research/kbound/figures/`.

Stage [3/4] GPU experiment (optional). When `KBOUND_GPU=1`, the script additionally runs:

```
src/scripts/kbound/cifar_tent_mps.py
```

This requires PyTorch with Metal Performance Shaders (Apple M5 / MPS) or a CUDA-capable GPU. It runs the CIFAR-10-C corruption grid with Tent adaptation and writes the decisive stress results to `experiments/kbound/cifar/`. The GPU stage is skipped silently when the flag is absent; all anomaly-routing and theory results are CPU-only.

Stage [4/4] Compile PDF. Two `pdflatex` passes compile `docs/research/kbound/kbound.tex`; the resulting PDF is copied to `docs/research/kbound/K-Bound_paper.pdf`.

A.3 Hermetic smoke path

The hermetic smoke is the fastest signal that the K-Bound algorithm works on any machine with no downloaded data:

```
bash scripts/smoke_kbound.sh
```

The wrapper (`scripts/smoke_kbound.sh`) resolves the repository root as the parent of its own directory, sets `PYTHON` (default `python3`, overridable via `PYTHON=python3.11`), and runs:

```
python src/scripts/kbound/smoke_trichotomy.py
```

`smoke_trichotomy.py` proceeds in two phases.

Phase 1: synthetic archive generation. The companion script `make_synth_archive.py` writes a tiny, deterministic (fixed seed) score archive to `experiments/kbound/_smoke/score_archive/`, using the exact five-key `.npz` schema (`Sval`, `yval`, `Stest`, `ytest`, `det_names`, `val_auc`, `domain`) of the real 123-task archive. Three synthetic tasks are created:

- `synth_helpful` — f_0 degrades on test while the ensemble f_a is far more accurate; true benefit clearly positive;
- `synth_harmful` — f_0 stays excellent while f_a is corrupted; true benefit clearly negative;
- `synth_unknowable` — f_0 and f_a are statistically tied; benefit ≈ 0.

Phase 2: real `kga` pipeline. `smoke_trichotomy.py` loads the synthetic archive and runs the *real* `kga` package: `compute_evidence` (label-free Z), `KGA.certify` (empirical-Bernstein certificate $\widehat{\Delta} \pm \varepsilon$), and `KGA.decide` (trichotomy). It then asserts:

- `synth_helpful` $\Rightarrow$ `ADAPT`;

- `synth_harmful` ⇒ `FREEZE`;
- `synth_unknowable` ⇒ `ABSTAIN`.

The separations are large relative to the certificate radius, so the result is deterministic, not chance. On success the script writes `experiments/kbound/_smoke/smoke_result.json` and prints `SMOKE PASS` with a one-line metrics summary. On any failure it exits non-zero.

The same logic is covered by `pytest tests/test_smoke_trichotomy.py` and `tests/test_data_manifest.py` both are pure `numpy`/`kga` suites with no external dependencies and complete in a few seconds. The hermetic smoke schema identity with the real archive is additionally checked in `test_smoke_trichotomy.py::test_synth_archive_schema_matches_real`.

A.4 Integrity manifests

A.4.1 Model artifact manifest (`models/MANIFEST.json`)

Five serialised model artifacts are registered in `models/MANIFEST.json`, each with its repository-relative path, a real SHA-256 digest, size in bytes, and mtime:

Artifact name	Path	SHA-256 (first 16 hex)
`behavior__behavior_autoencoder`	`models/behavior/`	`d044f4440633ed4d...`
`behavior__behavior_lof`	`models/behavior/`	`df74fb53d40dd272...`
`cyber__supervised__cyber_model`	`models/cyber/supervised/`	`e89c5d1b742c1fee...`
`fraud__supervised__fraud_model`	`models/fraud/supervised/`	`331df4269f51e34c...`
`fusion__fusion_meta_model`	`models/fusion/`	`c1c8033c53bfc1f9...`

The manifest is produced by:

```
PYTHONPATH=src python -m uais.registry.build_manifest
```

Loading any artifact through `src/uais/registry/ModelRegistry` fails closed (`IntegrityError`) on a SHA-256 mismatch. The manifest and loader are covered by `tests/test_model_registry.py` (20 tests).

A.4.2 Score-archive manifest (`data/MANIFEST.json`)

`data/MANIFEST.json` records the SHA-256, byte size, and mtime of every `.npz` file under `experiments/elara_u/score_archive/`, covering all 123 real score-archive entries. It is produced by:

```
python -m uais.data.manifest build
```

Verification is:

```
python -m uais.data.manifest verify
# exit 0 = all match; exit 1 = drift (missing/changed); prints JSON report
```

The manifest is format-versioned (`"version": "kbound-data-manifest/1"`) and handles three failure modes: a missing recorded file, a size mismatch, and a content (SHA-256) change even at identical size. These are exercised by `tests/test_data_manifest.py`.

A.5 The `kga` package: install and test

The `kga` package lives at the repository root (top-level package, not under `src/`):

```
# Minimal install: put repo root on PYTHONPATH
export PYTHONPATH="$PWD:$PWD/src"
pip install numpy scipy pytest

# Verify
python -c "import kga; print(kga.__version__)"  # -> kga 0.1.0
```

The package is pure `numpy`/`scipy`, deterministic, and torch-free. It exposes `KGA`, `Evidence`, `Certificate`, and `Decision` (a `str` enum). The CLI entry point:

```
python -m kga decide --calib calib.npy --test test.npy --alpha 0.1
```

Test suite. The `kga` unit tests live in `tests/test_kga_package.py`. To run the complete new-package suite (88 tests, $< 2\,\mathrm{s}$):

```
PYTHONPATH="$PWD:$PWD/src" pytest \
  tests/test_kga_package.py \
  tests/test_model_registry.py \
  tests/test_training_harness.py \
  tests/test_data_manifest.py \
  tests/test_smoke_trichotomy.py \
  --cov=kga --cov=src/uais --cov-report=term-missing
```

The `kga` tests are fixed-seed and check the following behavioural contract: (a) identical distributions $\Rightarrow$ `ABSTAIN`; (b) the certificate radius shrinks as n grows; (c) trichotomy boundaries are hit exactly; (d) the non-identifiability witness (identical Z, opposite truth) $\Rightarrow$ `ABSTAIN`; and (e) the empirical false-adapt rate remains $\leq \alpha$ over at least 2 000 synthetic trials.

Theorem validators. Each theorem in the paper has a standalone numerical validator under `experiments/kbound/theory_validation/`:

Script	Validates
`val_thm1_lecam.py`	Thm. 4.1 — Le Cam two-point minimax
`val_thm2_regret.py`	Thm. 4.3 — regret identity
`val_thm3_evalue.py`	Thm. 5.2 — anytime e-value
`val_thm5_multiclass.py`	Thm. 5.4 — multiclass extension

All validators are pure `numpy`/`scipy` with no labels or GPU, and have been re-run from a clean environment: all four pass.

A.6 CI workflow (`.github/workflows/kbound-ci.yml`)

The `kbound-ci` GitHub Actions workflow (`kbound-ci.yml`) runs six jobs on `ubuntu-latest` / Python 3.11, triggered on pushes and pull requests to `main`/`master`. A `concurrency.cancel-in-progress` key drops superseded runs.

Job	What it does
`lint`	(a) Repo-wide critical-error gate: `ruff check -select E9,F63,F7,F82 .`; (b) full ruff lint on the maintained A+ surfaces (`kga/`, `src/orchestration/`, `src/uais/registry/`, `src/uais/training/`, the K-Bound scripts, and the test suite); (c) ruff format check on the same surfaces.
`kga-package`	Installs `numpy scipy pytest`; imports `kga` and prints the version; runs `pytest tests/test_kga_package.py`.
`unit-tests`	Installs the full minimal dependency set; runs the five new package test modules with `--cov=kga --cov=src/uais`; uploads `coverage.xml` as an artifact.
`hermetic-smoke`	Installs only `numpy scipy`; runs `bash scripts/smoke_kbound.sh`; asserts `SMOKE PASS`.
`theorem-validators`	Glob-runs every `experiments/kbound/theory_validation/val_thm*.py`; fails if any exits non-zero.
`orchestration-import`	Imports `src.orchestration` *without* Prefect installed and checks that the six flows are registered (`PREFECT=False`).

Every job sets `PYTHONPATH=${{ github.workspace }}:${{ github.workspace }}/src` so both the top-level `kga` package and the `src/`-rooted `uais` package are importable without `pip install -e`.

A.7 Executed notebooks (00–09)

The `notebooks/` directory contains 10 notebooks that reproduce the K-Bound paper interactively. Each notebook auto-locates the repository root and loads only real result artifacts; outputs are saved in the files so they are readable without re-running. Table A.1 lists the notebooks in order.

Table A.1: The 10 executed K-Bound notebooks.

#	Filename	Contents
00	`00_KBound_Reproduction.ipynb`	One-page overview, live trichotomy on a synthetic example, all result figures, and the top-level reproductio commands.
01	`01_Problem_and_Theory.ipynb`	Problem formulation, the adapt/freeze/abstain trichotomy, live runs of the four theorem validators (Thms 1- 5 / 3b), and the architecture diagram.
02	`02_Knowability_Trichotomy.ipynb`	Live 123-task anomaly-routing suite: evidence Z empirical-Bernstein certificate, decisions, adapt precisio false-adapt rate, and component ablations.
03	`03_Harmful_Mixed_Rigor.ipynb`	Harmful-fusion regime, controlled mixed-regime (clean/detectable/covert), and 8-seed paired t-tes statistical rigor.
04	`04_Regression_and_Witness.ipynb`	Regression covariate-shift certificate and the clean nonidentifiability witness (live Gaussian/flipped-label construction).
05	`05_TTA_CIFAR_and_Online.ipynb`	CIFAR-10-C pre-registered stress grid, decisive CIFAR+Tent experiment, and online continual-Tent collaps analysis.
06	`06_Evidence_and_Drift.ipynb`	The label-free evidence Z computed live on syntheti and real tasks: KS drift, ESS, disagreement, and thei correlation with true benefit.
07	`07_Certificate_and_Calibration.ipynb`	Conformal, empirical-Bernstein, and e-value certificates α–coverage calibration curves; certificate width vs. batc size.
08	`08_ELARA_Multimodal_Instantiation.ipynb`	ELARA/RGA worked instantiation: MVTec-3D domai D23, T1–T9 gating, and the full GDR map.
09	`09_Conclusions_and_Reproducibility.ipynb`	Cross-experiment regret-to-oracle summary, reproductio commands, honest scope statement, and open items (Conjecture 5.1, full ImageNet-C).

Legacy notebooks (25 superseded ELARA-era EDA notebooks) live under `notebooks/legacy_elara/` and are not part of the K-Bound paper.

A.8 Environment notes

Python version. Python 3.11 is the supported CI interpreter; the code targets Python 3.9+. CPU-only K-Bound experiments need only `numpy`, `scipy`, `scikit-learn`, `matplotlib`, and `pandas`. The GPU (Tent/CIFAR-10-C) runs additionally require `torch 2.5.1` and `torchvision 0.20.1`.

Apple M5 / MPS. The decisive CIFAR-10-C and online continual-Tent experiments were run on an Apple M-series chip using Metal Performance Shaders (`device="mps"` in `cifar_tent_mps.py`). The same scripts run on CUDA without modification.

Determinism. All CPU experiments use `SEED=0`. `src/uais/training/set_seed` seeds `PYTHONHASHSEED`, `random`, NumPy, and (when importable) PyTorch with `torch.backends.cudnn.deterministic=True`. The `kga` package itself has no global RNG state.

Legacy lint debt. Approximately 1 973 `ruff` findings exist in pre-existing ELARA-era code. Mass-reformatting was intentionally not performed (high churn, out of scope); the CI enforces the full rule set on the maintained A+ surfaces and a critical-error gate across the whole repository. A clean `ruff check kga` passes with no findings.

A.9 How to reproduce: checklist

1. **Clone** the repository and `cd AutoML_Flagship_V8`.
2. **Install minimal deps**: `pip install numpy scipy scikit-learn pandas matplotlib pytest ruff`.
3. **Hermetic smoke** (no data required, $< 60\,\mathrm{s}$): `bash scripts/smoke_kbound.sh` ⇒ `SMOKE PASS`.
4. **Unit tests**: `PYTHONPATH="$PWD:$PWD/src" pytest tests/test_kga_package.py -q` ⇒ all pass.
5. **Theorem validators**: `for v in val_thm1_lecam val_thm2_regret val_thm3_evalue val_thm5_multiclass; do python experiments/kbound/theory_validation/$v.py; done` ⇒ 4/4 pass.
6. **Full rebuild** (requires score archive): `bash scripts/rebuild_kbound.sh` ⇒ `DONE -> docs/research/kbound/K-Bound_paper.pdf`.
7. **GPU rebuild** (requires score archive + PyTorch + MPS/CUDA): `KBOUND_GPU=1 bash scripts/rebuild_kbound.sh`.
8. **Manifest verification**: `python -m uais.data.manifest verify` ⇒ all 123 entries match.

Appendix B

Complete Result Tables

This appendix contains the full numerical tables for all primary experiments reported in Chapter 7 (anomaly-routing benchmark) and Chapter 8 (TTA and CIFAR-10 experiments). Each table identifies its source JSON file and field path. Numbers are reproduced verbatim from the JSON files without rounding beyond what floating-point display requires. All tables are self-contained: column headings, units, and source fields are given in each caption.

B.1 Per-Seed Rigor Table (Mixed Regime, 8 Seeds)

Source file: `experiments/kbound/results/rigor_multiseed.json`. The file stores aggregate mean and standard deviation across seeds (field `mean_std`); per-seed individual AUROC values are not stored in the JSON output (the seeded runs are summarised by the aggregate statistics). The table below therefore presents the complete set of aggregate statistics as stored, together with the paired t-test results for both comparisons.

Table B.1: Multi-seed aggregate statistics for the mixed regime (8 seeds, split-conformal, $\alpha = 0.1$). Source: `rigor_multiseed.json`, fields `mean_std` and `paired_ttest_*`. Mean and standard deviation are across all 8 seeds.

Metric	Policy / Description	Mean	Std
Mean test AUROC	Always-adapt	0.69698	0.00008
Mean test AUROC	Always-freeze	0.58419	0.00147
Mean test AUROC	KGA	0.68598	0.00334
Mean test AUROC	Oracle	0.71803	0.00064
Safety: adapt precision	$\mathbb{P}(B > 0 \mid \text{ADAPT})$	0.93577	0.01014
Safety: false-adapt rate	$\mathbb{P}(B < 0 \mid \text{ADAPT})$	0.06423	0.01014
Coverage (fraction acted)	adapt + freeze decisions	0.37263	0.01746
Paired t-tests (degrees of freedom = n_seeds − 1 = 7):			
KGA vs always-freeze	$t = 62.236$, $p = 7.262 \times 10^{-11}$	KGA significantly higher	
KGA vs always-adapt	$t = -8.760$, $p = 5.084 \times 10^{-5}$	KGA significantly lower	

Notes. The paired t-test values are sourced from fields `paired_ttest_KBound_vs_always_freeze` ($t = 62.2358$, $p = 7.2620 \times 10^{-11}$) and `paired_ttest_KBound_vs_always_adapt` ($t = -8.7599$, $p = 5.0843 \times 10^{-5}$). Both signs are reported honestly: KGA reliably beats freezing and reliably trails always-adapt in this mild-harm regime.

B.2 Full Ablation Grid

Source file: `experiments/kbound/results/ablations.json`. The table below gives every entry from the three sub-experiments: evidence-drop, α sweep, and batch-size sweep.

B.2.1 Evidence-Drop Ablation (Full Grid)

Table B.2: Evidence-drop ablation: complete grid. Each row drops one feature of Z and reports regret, false-adapt rate, and coverage on the mixed-regime benchmark. Baseline uses all nine features. Source: `ablations.json`, fields `baseline` and `evidence_drop`.

Feature dropped	Regret	False-adapt	Coverage
Baseline (all 9 features)	0.036554	0.065041	—
`val_max`	0.035039	0.064000	0.360434
`val_gap`	0.037203	0.066667	0.336043
`val_mean`	0.042568	0.070796	0.319783
`ks_mean`	0.036121	0.065574	0.352304
`ks_max`	0.037279	0.065574	0.352304
`disagree`	**0.094045**	**0.250000**	**0.140921**
`val_std`	0.036518	0.074380	0.344173
`anomaly_rate`	0.036615	0.065041	0.349593
`log_ntest`	0.039264	0.067797	0.336043

B.2.2 Alpha Sweep (Full Grid)

B.2.3 Batch-Size Sweep (Full Grid)

B.3 Regression Covariate-Shift Decision Summary

Source file: `experiments/kbound/results/regression_covariate.json`.

Table B.3: α sweep: complete grid of conformal level, band width ε, coverage, and false-adapt rate. Source: `ablations.json`, field `alpha_sweep`.

α	ε	Coverage	False-adapt	Source key
0.01	0.323988	0.113821	0.000000	"0.01"
0.05	0.191235	0.262873	0.073684	"0.05"
0.10	0.139075	0.349593	0.065041	"0.1"
0.20	0.091047	0.487805	0.072289	"0.2"

Table B.4: Batch-size sweep: complete grid of test-batch size, certificate width ε, and coverage. Source: `ablations.json`, field `batchsize_sweep`.

Batch size	ε	Coverage	Source key
16	0.443381	0.020325	"16"
32	0.354927	0.035599	"32"
64	0.277723	0.198830	"64"
128	0.255357	0.209040	"128"
256	0.182329	0.261111	"256"

B.4 Mixed-Regime Full Metrics

Source file: `experiments/kbound/results/mixed_regime_results.json`.

B.5 Harmful-Fusion Regime Full Metrics

Source file: `experiments/kbound/results/kbound_harmful_results.json`.

Note on the adapt decision. The single ADAPT decision issued by KGA (`counts.ADAPT` = 1) was on a task where $B < 0$ (field `adapt_precision_B>0` = 0.0, field `false_adapt_rate_B<0` = 1.0). This single error does not affect the AUROC or regret materially because one task out of 123 has negligible weight; it reflects the irreducible probabilistic tolerance of the conformal guarantee at $\alpha = 0.1$.

B.6 Clean Non-identifiability Witness: Full Metrics

Source file: `experiments/kbound/results/witness_clean.json`.

B.7 Knowability Trichotomy Suite: Full Metrics

Source file: `experiments/kbound/results/knowability_results.json`.

Table B.5: Regression covariate-shift experiment: full metrics. Source: `regression_covariate.json`. Benefit $B = \mathrm{MSE}(f_0) - \mathrm{MSE}(f_a)$; positive B means adaptation hurts (higher MSE for f_a). No adapt decisions were issued; adapt precision and false-adapt rate are undefined (`null` in the JSON).

Field	Value
`n_instances`	72
`eps`	0.218147
`mean_B(mse0-msea)`	−0.086346
`harmful_rate_B<0`	0.791667
Decision counts	
`decision_counts.ADAPT`	0
`decision_counts.FREEZE`	11
`decision_counts.ABSTAIN`	61
Policy mean MSE	
`mean_MSE.always_adapt(IW)`	0.337601
`mean_MSE.always_freeze`	0.251255
`mean_MSE.K_Bound`	0.251255
`mean_MSE.oracle`	0.251188
`adapt_precision_B>0`	null (no adapt decisions)
`false_adapt_rate_B<0`	null (no adapt decisions)

Note on naive rule. The JSON also records a naive validation rule (`mean_auc_naive_val_rule` = 0.7786, `regret_naive_vs_oracle` = 0.0042, false-adapt = 0.1798). The naive rule achieves lower regret than KGA in this helpful-dominated regime but has a much higher false-adapt rate (17.98% vs 10%), illustrating the safety-performance tradeoff.

B.8 All 65 CIFAR-10-C Cells

The following table contains per-cell results for all 13 official CIFAR-10-C corruptions at 5 severity levels (65 cells total), comparing frozen, online Tent, EATA-style, SAR-style, and KGA on a ResNet-18 (0.874 clean accuracy). The table is generated directly from the experiment output; see Appendix A for the script that produces it.

Table B.10: All 65 CIFAR-10-C cells (13 official corruptions × severities 1–5), ResNet-18, N=800 balanced held-out eval per cell. Online TTA accuracy; KGA wraps Tent. Mean row matches Table B.8.

corruption	sev	frozen	Tent	EATA	SAR	KGA	oracle
brightness	1	0.863	0.858	0.856	0.859	0.858	0.865
brightness	2	0.844	0.864	0.864	0.865	0.864	0.868
brightness	3	0.804	0.837	0.837	0.839	0.837	0.839
brightness	4	0.731	0.818	0.818	0.815	0.819	0.816

corruption	sev	frozen	Tent	EATA	SAR	KGA	oracle
brightness	5	0.690	0.791	0.794	0.795	0.791	0.799
contrast	1	0.719	0.831	0.831	0.835	0.831	0.831
contrast	2	0.559	0.823	0.821	0.822	0.820	0.820
contrast	3	0.325	0.790	0.793	0.793	0.790	0.791
contrast	4	0.223	0.728	0.729	0.730	0.726	0.726
contrast	5	0.160	0.611	0.607	0.609	0.614	0.601
defocus_blur	1	0.264	0.683	0.680	0.681	0.682	0.676
defocus_blur	2	0.207	0.543	0.544	0.544	0.540	0.540
defocus_blur	3	0.204	0.389	0.386	0.398	0.389	0.385
defocus_blur	4	0.176	0.300	0.300	0.303	0.300	0.305
defocus_blur	5	0.183	0.265	0.264	0.261	0.261	0.275
elastic_transform	1	0.244	0.528	0.528	0.530	0.528	0.525
elastic_transform	2	0.179	0.436	0.440	0.446	0.436	0.452
elastic_transform	3	0.168	0.401	0.401	0.405	0.401	0.400
elastic_transform	4	0.144	0.407	0.407	0.413	0.407	0.410
elastic_transform	5	0.151	0.347	0.346	0.353	0.351	0.351
frost	1	0.725	0.804	0.805	0.806	0.804	0.796
frost	2	0.588	0.760	0.762	0.761	0.761	0.762
frost	3	0.546	0.685	0.685	0.686	0.684	0.697
frost	4	0.492	0.705	0.705	0.704	0.705	0.706
frost	5	0.436	0.640	0.642	0.644	0.640	0.659
gaussian_noise	1	0.487	0.741	0.740	0.741	0.739	0.739
gaussian_noise	2	0.275	0.699	0.699	0.700	0.700	0.701
gaussian_noise	3	0.144	0.594	0.595	0.595	0.593	0.592
gaussian_noise	4	0.100	0.435	0.435	0.432	0.436	0.441
gaussian_noise	5	0.089	0.305	0.304	0.301	0.305	0.301
impulse_noise	1	0.615	0.769	0.770	0.769	0.769	0.775
impulse_noise	2	0.370	0.682	0.682	0.682	0.682	0.688
impulse_noise	3	0.197	0.605	0.605	0.604	0.608	0.623
impulse_noise	4	0.110	0.475	0.479	0.485	0.476	0.484
impulse_noise	5	0.082	0.335	0.335	0.335	0.335	0.331
jpeg_compression	1	0.676	0.730	0.730	0.731	0.730	0.735
jpeg_compression	2	0.639	0.660	0.659	0.661	0.662	0.663
jpeg_compression	3	0.608	0.655	0.654	0.656	0.654	0.658
jpeg_compression	4	0.509	0.573	0.570	0.570	0.573	0.567
jpeg_compression	5	0.460	0.501	0.501	0.501	0.501	0.500
motion_blur	1	0.495	0.782	0.784	0.785	0.784	0.783
motion_blur	2	0.365	0.666	0.665	0.664	0.666	0.675
motion_blur	3	0.279	0.575	0.574	0.575	0.574	0.590
motion_blur	4	0.255	0.490	0.491	0.489	0.491	0.502
motion_blur	5	0.225	0.456	0.450	0.446	0.456	0.476
pixelate	1	0.429	0.719	0.720	0.721	0.716	0.720
pixelate	2	0.373	0.651	0.651	0.654	0.650	0.654
pixelate	3	0.306	0.570	0.574	0.576	0.573	0.571
pixelate	4	0.267	0.487	0.485	0.487	0.489	0.486
pixelate	5	0.289	0.405	0.404	0.406	0.405	0.410

corruption	sev	frozen	Tent	EATA	SAR	KGA	oracle
shot_noise	1	0.492	0.779	0.776	0.779	0.777	0.784
shot_noise	2	0.254	0.654	0.655	0.659	0.654	0.669
shot_noise	3	0.155	0.536	0.537	0.549	0.534	0.545
shot_noise	4	0.125	0.400	0.398	0.398	0.398	0.404
shot_noise	5	0.120	0.303	0.305	0.306	0.300	0.300
snow	1	0.770	0.782	0.781	0.779	0.782	0.781
snow	2	0.578	0.686	0.685	0.685	0.685	0.690
snow	3	0.661	0.716	0.716	0.711	0.718	0.710
snow	4	0.574	0.683	0.681	0.679	0.683	0.680
snow	5	0.459	0.597	0.596	0.597	0.597	0.598
zoom_blur	1	0.760	0.877	0.879	0.876	0.876	0.879
zoom_blur	2	0.708	0.847	0.847	0.849	0.847	0.853
zoom_blur	3	0.674	0.830	0.829	0.828	0.829	0.826
zoom_blur	4	0.635	0.809	0.810	0.810	0.807	0.814
zoom_blur	5	0.557	0.779	0.776	0.778	0.780	0.785
mean		**0.412**	**0.626**	**0.626**	**0.627**	**0.626**	**0.629**

Table B.6: Mixed-regime benchmark: complete metrics from a single run ($n = 369$). Source: `mixed_regime_results.json`.

Field	Value
`n_instances`	369
`eps`	0.131487
Decision distribution — clean condition (true $\Delta = -0.041$)	
`decision_distribution_by_condition.clean.ADAPT`	8
`decision_distribution_by_condition.clean.FREEZE`	5
`decision_distribution_by_condition.clean.ABSTAIN`	110
Decision distribution — detectable-failure condition (true $\Delta = +0.193$)	
`decision_distribution_by_condition.detectable.ADAPT`	71
`decision_distribution_by_condition.detectable.FREEZE`	3
`decision_distribution_by_condition.detectable.ABSTAIN`	49
Decision distribution — covert-failure condition (true $\Delta = +0.181$)	
`decision_distribution_by_condition.covert.ADAPT`	59
`decision_distribution_by_condition.covert.FREEZE`	3
`decision_distribution_by_condition.covert.ABSTAIN`	61
True mean benefit by condition	
`true_mean_benefit_B_by_condition.clean`	−0.040876
`true_mean_benefit_B_by_condition.detectable`	+0.192928
`true_mean_benefit_B_by_condition.covert`	+0.180573
Safety	
`safety.adapt_precision_B>0`	0.934783
`safety.false_adapt_rate_B<0`	0.065217
Policy mean AUROC	
`mean_auc_policies.always_adapt`	0.697111
`mean_auc_policies.always_freeze`	0.586236
`mean_auc_policies.K_Bound`	0.689852
`mean_auc_policies.oracle`	0.718913
Regret vs oracle	
`regret_vs_oracle.always_adapt`	0.021801
`regret_vs_oracle.always_freeze`	0.132677
`regret_vs_oracle.K_Bound`	0.029061
Non-identifiability: clean vs covert boundary	
`nonidentifiability_clean_vs_covert.mean_Z_L2_distance_clean_to_nearest_covert`	1.353908
`nonidentifiability_clean_vs_covert.mean_B_clean`	−0.040876
`nonidentifiability_clean_vs_covert.mean_B_covert`	+0.180573
`nonidentifiability_clean_vs_covert.mean_Z_L2_distance_clean_to_nearest_detectable`	3.099503

Table B.7: Harmful-fusion regime: complete metrics ($n = 123$ tasks, candidate = `elara_fuse`, base rate harmful = 79.7%). Source: `kbound_harmful_results.json`.

Field	Value
`setup`	`fa=elara_fuse (harmful ~80%), f0=auto_select`
`n_tasks`	123
`eps`	0.039848
`base_rate_harmful_B<0`	0.796748
Decision counts	
`counts.ADAPT`	1
`counts.FREEZE`	13
`counts.ABSTAIN`	109
Safety	
`adapt_precision_B>0`	0.000
`false_adapt_rate_B<0`	1.000
`freeze_correct_B<=0`	1.000
Policy mean AUROC	
`mean_auc.always_adapt(elara_fuse)`	0.728428
`mean_auc.always_freeze(auto_select)`	0.748058
`mean_auc.K-Bound_trichotomy`	0.747946
`mean_auc.oracle`	0.749863
Regret vs oracle	
`regret_vs_oracle.always_adapt`	0.021435
`regret_vs_oracle.always_freeze`	0.001806
`regret_vs_oracle.K-Bound`	0.001918
Regret ratio (always-adapt / KGA)	$0.021435/0.001918 \approx 11.2$

Table B.8: Clean non-identifiability witness: complete metrics ($M = 300$ instances). Source: `witness_clean.json`. All five KS p-values exceed 0.05; KGA abstains on 100% of instances.

Field	Value
`M`	300
`mean_abs_true_benefit`	1.0
`world0_mean_B`	-1.0
`world1_mean_B`	$+1.0$
KS p-values (Z-feature law, World 0 vs World 1)	
`Z_indistinguishable_KS_pvalues.frac_pos`	0.9647
`Z_indistinguishable_KS_pvalues.std`	0.9418
`Z_indistinguishable_KS_pvalues.ks_vs_ref`	0.7988
`Z_indistinguishable_KS_pvalues.mean_f0`	0.9647
`Z_indistinguishable_KS_pvalues.skew`	0.1648
`all_Z_features_p>0.05`	`true`
`abstain_rate`	1.0
`eps`	1.395233
`mean_|^Bhat|`	0.263773
`forced_commit_regret_vs_oracle`	0.513333
`abstain_regret_vs_oracle(default_freeze)`	0.466667

Table B.9: Knowability trichotomy suite (clean, 123 tasks): complete metrics. Source: `knowability_results.json`.

Field	Value
`metrics.n_tasks`	123
`metrics.eps_conformal`	0.049201
`metrics.alpha`	0.10
`metrics.coverage`	0.170732
`metrics.n_adapt`	20
`metrics.n_freeze`	1
`metrics.n_abstain`	102
`metrics.adapt_precision_(B>0|ADAPT)`	0.90
`metrics.false_adapt_rate_(B<0|ADAPT)`	0.10
`metrics.freeze_precision_(B<=0|FREEZE)`	0.00
`metrics.abstain_mean_|B|`	0.021504
`metrics.nonabstain_mean_|B|`	0.132367
`policy.mean_auc_trichotomy`	0.766119
`policy.mean_auc_always_adapt`	0.777145
`policy.mean_auc_always_freeze`	0.748056
`policy.mean_auc_oracle`	0.782816
`policy.regret_trichotomy_vs_oracle`	0.016697
`policy.regret_always_adapt_vs_oracle`	0.005671
`policy.regret_naive_vs_oracle`	0.004220
`policy.mean_auc_naive_val_rule`	0.778596
`policy.false_adapt_rate_naive_(B<0|naive ADAPT)`	0.179775
Non-identifiability witness (in-distribution)	
`nonidentifiability_witness.z_distance`	0.798586
`nonidentifiability_witness.task_i`	`cv_CIFAR10_7`
`nonidentifiability_witness.B_i`	+0.038255
`nonidentifiability_witness.task_j`	`cv_SVHN_3`
`nonidentifiability_witness.B_j`	−0.017858
Evidence feature names (9 features)	
	`val_max`, `val_gap`, `val_mean`
	`ks_mean`, `ks_max`, `disagree`
	`val_stack_margin`, `anomaly_rate`, `log_ntest`

Appendix C

Notation and Glossary

This appendix collects all mathematical notation used in the monograph in a single reference table, followed by a glossary of key technical terms. Symbol conventions are drawn directly from the paper [25] and are consistent throughout all chapters.

C.1 Notation table

Table C.1: Mathematical notation used throughout the monograph.

Symbol	Meaning
$P_S(X, Y)$	Source distribution. Labels Y are available at training and validation time.
$P_T(X, Y)$	Target distribution. At deployment only $P_T(X)$ is observable (no target labels).
f_0	Frozen (baseline) predictor. Deployed without modification unless ADAPT fires.
f_a	Adapted (candidate) predictor. The output of an adaptation mechanism (Tent, EATA, SAR, or any other) that is being evaluated.
ℓ	Per-sample loss function, e.g. 0/1 loss or cross-entropy.
$R_T(f)$	Target risk of predictor f: $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)]$.
Δ	Adaptation benefit: $\Delta = R_T(f_0) - R_T(f_a)$. Positive means adapting reduces risk (helps); negative means it hurts.
$\Delta(z)$	Conditional benefit at evidence value z: $\Delta(z) = \mathbb{E}[\ell(f_0(X), Y) - \ell(f_a(X), Y) \mid Z = z]$.

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Symbol	Meaning
$\widehat{\Delta}$	Label-free (point) estimator of Δ or $\Delta(z)$, built without target labels.
ε	Certificate radius at level α: with probability $\geq 1-\alpha$, $\lvert\Delta - \widehat{\Delta}\rvert \leq \varepsilon$ (or $\Delta \geq \widehat{\Delta} - \varepsilon$ for the one-sided form).
α	Miscoverage level in $(0, 1)$; bounds the false-adapt probability $\mathbb{P}(\textsc{adapt} \text{ and } \Delta \leq 0) \leq \alpha$.
Z	Label-free observable evidence: $Z = \phi(X_{1:m}, f_0, f_a)$, a vector of statistics computable from unlabelled test scores and fixed predictors.
$\phi(\cdot)$	Evidence map; extracts Z from the unlabelled batch and predictor pair.
$g(Z)$	Decision rule; maps evidence Z to one of $\{\textsc{adapt}, \textsc{freeze}, \textsc{abstain}\}$.
$g^\star$	Bayes-optimal gate: $g^\star(z) = \mathbf{1}[\Delta(z) > 0]$.
D	Observable disagreement region: $D = \{x : f_0(x) \neq f_a(x)\}$; the support of the sign-of-difference signal.
p_a	Probability mass of the target distribution on D: $p_a = P_T(X \in D)$.
p_0	Probability mass of the source distribution on D: $p_0 = P_S(X \in D)$.
$\mathrm{TV}(P, Q)$	Total variation distance between distributions P and Q: $\mathrm{TV}(P, Q) = \sup_A \lvert P(A) - Q(A)\rvert$.
$\mathrm{Law}(X \mid P)$	Distribution (law) of X under P.
$\mathbf{1}[E]$	Indicator of event E; equals 1 if E holds, 0 otherwise.
$\mathrm{sign}(t)$	Sign function: $\mathrm{sign}(t) = \mathbf{1}[t > 0]$ (as used in Theorem 4.3).
$\arg\max$, $\arg\min$	Arg-max and arg-min operators.
$\mathbb{R}$	The real numbers $(-\infty, +\infty)$.
$\mathbb{E}[\cdot]$	Expectation under the distribution clear from context.
$\mathbb{P}[\cdot]$	Probability under the distribution clear from context.
n	Sample size (number of paired benefits, calibration tasks, or stream steps depending on context).
m	Size of the unlabelled target batch.
R	Benefit range $R = b - a$ where the paired benefits $X_i \in [a, b]$.
$\widehat{V}$	Unbiased sample variance (denominator $n-1$) of the paired benefits X_i.
$\widehat{\mu}$	Empirical mean of the paired benefits; equals $\widehat{\Delta}$ in the batch estimator.

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Symbol	Meaning
E_t^+, E_t^-	Two one-sided betting e-processes (Theorem 5.2): E_t^+ tests $H_0^+ : \Delta \leq 0$; E_t^- tests $H_0^- : \Delta \geq 0$.
λ_i, ν_i	Predictable bet sizes for E_t^+ and E_t^- respectively (aGRAPA truncated bets, formed from past observations).
`ks_mean`, `ks_max`	Mean and maximum per-detector Kolmogorov–Smirnov drift between calibration and test score marginals (fields of `Evidence`).
`disagree`	$1 - \overline{\rho}$: one minus the mean pairwise rank correlation of test scores across detectors (field of `Evidence`).
`ess_frac`	$\mathrm{ESS}/n_{\text{test}}$: scale-free effective sample size of the Gaussian-ratio importance weights (field of `Evidence`).
δ	Detection threshold or evidence separation; in the quantitative non-identifiability theorem (Theorem 4.1) a location-shift parameter indexing $\mathrm{TV} = 2\Phi(\frac{1}{2}\sqrt{n}\,\delta) - 1$.
Regret-to-oracle	$R(g) - R(g^\star)$: excess risk of decision rule g over the per-condition oracle $g^\star = \mathbf{1}[\Delta > 0]$.

C.2 Glossary

Test-time adaptation (TTA). The practice of updating (adapting) a pre-trained model using only unlabelled data from a new target distribution at deployment time. Standard TTA mechanisms include entropy minimisation (Tent [52]), sample-filtered updates (EATA [36]), and sharpness-aware resets (SAR [37]).

Trichotomy. The three-way decision returned by KGA: ADAPT (adapting is certified beneficial), FREEZE (adapting is certified harmful), or ABSTAIN (the sign of the benefit is not knowable at level α). Contrasts with the binary adapt/freeze choice of all prior TTA methods.

Label-free evidence Z. The vector of statistics $Z = \phi(X_{1:m}, f_0, f_a)$ computable from unlabelled target data, the frozen model, and the candidate adaptation, without any target labels. In the `kga` package, Z is represented by the `Evidence` dataclass with fields `ks_mean`, `ks_max`, `disagree`, `entropy_shift`, `conf_shift`, `calib_conf`, `test_conf`, `ess`, and `ess_frac`.

Certificate. A finite-sample statement of the form $\Delta \geq \widehat{\Delta} - \varepsilon$ holding with probability at least $1 - \alpha$. Represented in the `kga` package as the `Certificate` dataclass with fields `delta_hat`, `epsilon`, `method`, `alpha`, `n`, and derived properties `lower` / `upper`.

False adapt. The event that KGA returns ADAPT but the true benefit is non-positive ($\Delta \leq 0$); i.e. deploying f_a actually hurts. Theorem 5.1 guarantees $\mathbb{P}(\text{false adapt}) \leq \alpha$.

False freeze. The symmetric event: KGA returns FREEZE but $\Delta > 0$ (keeping f_0 forgoes a genuine benefit). Also bounded at $\leq \alpha$ by the two-sided version of the certificate.

Abstention. The safe default action returned when the certificate brackets zero (i.e. $\widehat{\Delta} - \varepsilon \leq 0 \leq \widehat{\Delta} + \varepsilon$). KGA keeps the frozen model f_0 and logs the evidence for monitoring. Abstention is mandatory, not merely conservative, in the non-identifiable regime (Theorem 4.1, Corollary 5.1).

Empirical-Bernstein certificate. The batch certificate estimator of Maurer & Pontil (2009) [34], the default in KGA (`method='ebern'`). The radius is $\varepsilon = \sqrt{2\widehat{V}\ln(2/\alpha)/n} + 7R\ln(2/\alpha)/(3(n-1))$, which is strictly tighter than Hoeffding when the sample variance $\widehat{V} \ll R^2$.

Conformal certificate (split-conformal). A certificate whose radius is the $(1-\alpha)$-quantile of held-out calibration residuals $r_i = |\widehat{\Delta}_i - \Delta_i|$ [51, 1]. The estimator used in the main K-Bound cross-task experiments (`method='conformal'`).

E-value / anytime certificate. The anytime-valid certificate based on two one-sided testing-by-betting e-processes [50, 55]. Unlike the batch and conformal certificates, this certificate is valid simultaneously over all stream lengths with no multiplicity correction (`method='evalue'`).

Covariate shift. The setting where $P_T(X) \neq P_S(X)$ but $P_T(Y \mid X) = P_S(Y \mid X)$. Theorem 5.3 identifies this as a provably knowable regime under the support-overlap condition: large KS drift combined with adequate importance-weight ESS allows KGA to certify the correct action.

Disagreement region. The set $D = \{x : f_0(x) \neq f_a(x)\}$ of inputs on which the frozen and adapted predictors differ. Its probability mass under P_T governs the sign-of-difference identifiability (Theorem 5.4).

Score archive. The 123-task score cache at `experiments/elara_u/score_archive/`, the main empirical dataset for K-Bound. Each task is one `.npz` file with keys `Sval`, `yval`, `Stest`, `ytest`, `det_names`, `val_auc`, and `domain`. The archive covers ADBench tabular (47 tasks), credit-card fraud, UNSW-NB15 cyber, behaviour/HAR, NLP (Enron), time-series (NAB/SMD/TSB-AD), and image benchmarks.

Non-identifiability witness. A pair of target worlds (P_T^1, P_T^2) with $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$ and $\operatorname{sign}\Delta^1 = -\operatorname{sign}\Delta^2 \neq 0$, proving that no label-free rule can commit correctly in both worlds. The canonical witness (Theorem 4.1) uses $X \sim \mathcal{N}(0,1)$, $f_0 = \mathbf{1}[X > 0]$, $f_a = \mathbf{1}[X < 0]$, and flipped labels.

Regret-to-oracle. The excess risk $R(g) - R(g^\star)$ of a decision rule g over the per-condition oracle $g^\star = \mathbf{1}[\Delta > 0]$. By Theorem 4.3, regret decomposes as $\mathbb{E}[|\Delta(Z)|\mathbf{1}\{\operatorname{sign}\widehat{\Delta}(Z) \neq \operatorname{sign}\Delta(Z)\}]$; the experiments report this quantity as the primary performance metric alongside AUROC.

Risk-aligned evidence. Evidence Z is risk-aligned (Definition 3 of Chapter 3) if it identifies or brackets the sign of $\Delta(z)$; equivalently, Z suffices to determine whether adapting helps. Risk alignment is Assumption (iii) inherited by KGA from Theorem 5.1.

Mechanism-agnostic wrapper. The design pattern of KGA: the algorithm wraps any existing adaptation mechanism (Tent, EATA, SAR, or other) without modifying it. The adaptation mechanism supplies f_a; KGA decides whether to deploy it.

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